

RECORDS KEEPING

Stores Records & Procedures

● SYLLABUS COVERED BY THIS BOOK

- **Stores Records & Procedures** — complete coverage of the prescribed topic.
- **Clerical procedure in the goods inward section.**
- **Records and procedures in main stores.**
- **Classification and codification** of stores/drug items.
- **Keeping of stock books.**
- **Preparation of indents.**
- **Methods of storing drugs.**

Subject: Hospital & Clinical Pharmacy / Drug Store & Business Management

Exam: JKSSB Junior Pharmacist

Paper coverage: Records Keeping — Stores Records & Procedures
14 chapters • fully tabulated • high-yield boxes • quick-revision capsule

TABLE OF CONTENTS

Chapter-wise contents with page numbers

TOPIC	PAGE
PART I • FOUNDATIONS OF STORES & RECORDS KEEPING	
Chapter 1. Drug Store Management & Records Keeping — Foundation.....	5
1.1 Store, Stores and Drug Store — Basic Terminology.....	5
1.2 Records Keeping — Meaning, Purpose and Scope.....	5
1.3 Objectives and Functions of a Drug Store.....	6
1.4 Importance / Advantages of Systematic Records Keeping.....	7
1.5 Principles (Golden Rules) of Records Keeping.....	7
1.6 Types and Classification of Store Records.....	8
1.7 Types of Stores in a Hospital / Pharmaceutical Set-up.....	8
1.8 Organisation and Staffing of the Stores Department.....	9
1.9 Planning and Layout of the Main Drug Store.....	9
1.10 Legal & Regulatory Framework Governing Store Records.....	10
1.11 Materials Management — Where Records Keeping Fits.....	11
Chapter 2. The Purchase Cycle and Its Documents.....	12
2.1 Purchasing — Meaning, Objectives and the 5 R's.....	12
2.2 Steps in the Purchase Procedure.....	12
2.3 Methods of Purchase.....	13
2.4 The Tender System in Detail.....	13
2.5 Purchase Committee, PTC and the Hospital Formulary.....	14
2.6 Documents of the Purchase-Receipt-Payment Cycle.....	14
2.7 Purchase Order — Contents and Distribution of Copies.....	15
2.8 Records Maintained by the Purchase Section.....	15
PART II • GOODS INWARD SECTION & MAIN STORES	
Chapter 3. Clerical Procedure in the Goods Inward Section.....	17
3.1 Goods Inward Section — Meaning, Siting and Layout.....	17
3.2 Functions of the Goods Inward Section.....	17
3.3 Staff of the Goods Inward Section and Their Duties.....	18
3.4 Documents Accompanying / Supporting a Consignment.....	18
3.5 The Clerical Procedure — Step by Step.....	19
3.6 Checking of the Consignment in Detail.....	20
3.7 Goods Received Note (GRN) — The Pivotal Document.....	21
3.8 Goods Inward Register (Daily Receipt Register).....	22
3.9 Quarantine, Acceptance and Rejection.....	22
3.10 Handling of Discrepancies, Shortages, Damages and Claims.....	22
3.11 Bill Passing and Payment Procedure.....	23
3.12 Receipt of Special Categories of Items.....	24
3.13 Precautions, Do's and Don'ts in the Goods Inward Section.....	24
3.14 Summary — Records of the Goods Inward Section.....	25
Chapter 4. Records and Procedures in the Main Stores.....	26
4.1 Main (Central) Store — Meaning and Functions.....	26
4.2 Broad Groups of Main-Store Records.....	26
4.3 Master List of Registers and Records Maintained in the Main Store.....	27
4.4 Receipt Procedure in the Main Store.....	28
4.5 Storage Procedure in the Main Store.....	28
4.6 Issue Procedure in the Main Store.....	29
4.7 Transfers, Returns and the Imprest (Top-up) System.....	29
4.8 Handling of Expiry, Breakage, Loss and Condemnation.....	30
4.9 Stock Verification in the Main Store.....	30
4.10 Filing, Indexing and Documentation Systems.....	31
4.11 Security, Confidentiality and Computerisation of Records.....	31

4.12 Common Errors and Audit Objections in Store Records.....	32
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PART III • CLASSIFICATION & CODIFICATION OF STORES

Chapter 5. Classification of Stores and Drug Items.....	33
5.1 Meaning, Definition and Need.....	33
5.2 Bases of Classification.....	33
5.3 Classification of Hospital Store Items (Functional Groups).....	34
5.4 Classification of Drugs.....	34
5.5 Standardisation, Simplification and Variety Reduction.....	36
Chapter 6. Codification of Stores Items.....	37
6.1 Meaning and Definition.....	37
6.2 Objectives and Advantages of Codification.....	37
6.3 Essentials (Requirements) of a Good Coding System.....	37
6.4 Methods / Systems of Codification.....	38
6.5 Standard Codification Systems — Brisch and Kodak.....	39
6.6 Bar Coding and Modern Identification Technology.....	39
6.7 Colour Coding — A Special Form of Codification.....	40

PART IV • STOCK BOOKS & INVENTORY CONTROL

Chapter 7. Keeping of Stock Books.....	42
7.1 Stock Book — Meaning and Necessity.....	42
7.2 Bin Card (Stock Card / Bin Tag).....	42
7.3 Stock Ledger (Stores Ledger).....	43
7.4 Kardex, Two-Bin and Location Systems.....	43
7.5 Rules and Procedure for Posting Stock Books.....	44
7.6 Types of Stock Books Maintained in a Drug Store.....	44
7.7 Physical Verification of Stock.....	45
7.8 Valuation of Stock (Pricing of Issues).....	46
7.9 Performance Indicators Derived from Stock Books.....	46
7.10 Computerised Stock Keeping.....	46
Chapter 8. Inventory Control Techniques.....	48
8.1 Inventory — Meaning, Types and Costs.....	48
8.2 Selective Inventory Control Techniques.....	48
8.3 Economic Order Quantity (EOQ).....	50
8.4 Stock Levels and Lead Time.....	50
8.5 Systems of Inventory Review and Modern Approaches.....	51

PART V • INDENTS

Chapter 9. Preparation of Indents.....	53
9.1 Indent — Definition and Related Terms.....	53
9.2 Indenting Authority and Financial Powers.....	53
9.3 Types of Indents.....	53
9.4 Contents (Columns) of an Indent Form.....	54
9.5 Step-by-Step Procedure for Preparing an Indent.....	54
9.6 Calculation of the Indent Quantity.....	55
9.7 Factors to be Considered While Indenting.....	56
9.8 Scrutiny, Sanction and the Indent Register.....	56
9.9 Common Mistakes and Precautions in Indenting.....	57
9.10 Emergency and Local Purchase.....	57
9.11 From Indent to Issue — The Complete Internal Cycle.....	58

PART VI • STORAGE OF DRUGS

Chapter 10. Methods of Storing Drugs.....	59
10.1 Storage — Meaning and Objectives.....	59
10.2 Requirements of a Good Drug Store.....	59
10.3 Methods / Systems of Arranging Drugs in the Store.....	60
10.4 Storage Conditions and Temperature Terminology.....	61

10.5 The Cold Chain.....	62
10.6 Storage of Special Categories of Drugs.....	63
10.7 Deterioration of Drugs — Causes and Signs.....	64
10.8 Shelf Life, Expiry Date and Near-Expiry Management.....	65
10.9 Housekeeping, Pest Control, Fire and Security in the Store.....	66
10.10 Storage at the Ward / Sub-Store Level.....	67

PART VII • STATUTORY RECORDS & MODERN SYSTEMS

Chapter 11. Statutory and Legal Records of a Drug Store.....	68
11.1 Records under the Drugs and Cosmetics Act, 1940 and Rules, 1945.....	68
11.2 Schedule H, H1 and X — Record Requirements Compared.....	69
11.3 Records under the NDPS Act, 1985.....	69
11.4 Other Statutory Records Relevant to a Drug Store.....	70
11.5 Retention (Preservation) Periods of Records.....	70
11.6 Inspection by the Drugs Inspector — What is Examined.....	71
Chapter 12. Computerisation, Bar Coding and Modern Store Management.....	72
12.1 Why Computerise Store Records?.....	72
12.2 Typical Modules of Drug Store / Pharmacy Software.....	72
12.3 Safeguards for Electronic Records.....	73
12.4 Limitations and Precautions.....	73

PART VIII • REVISION

Chapter 13. Quick Revision Capsule.....	74
13.1 One-Line Definitions.....	74
13.2 All Formulae on One Page.....	74
13.3 Selective Inventory Control — Criterion Table (memorise).....	75
13.4 Storage Temperatures and Conditions (IP).....	75
13.5 Schedules, Forms and Retention Periods.....	76
13.6 Colour Codes.....	76
13.7 Differences You Must Be Able to Write.....	77
13.8 Sequences and Procedures in Order.....	77
13.9 Numbers, Percentages and Standards Worth Memorising.....	77
13.10 Mnemonic Collection.....	78
13.11 Fifty Facts — Final Rapid Fire.....	78
Chapter 14. Glossary and Abbreviations.....	80
14.1 Glossary of Terms.....	80
14.2 A Final Word on Method.....	81

How to use this book: read the chapter, then re-read only the coloured boxes and tables the night before the examination. Chapter 13 is a complete revision capsule.

PART I • FOUNDATIONS OF STORES & RECORDS KEEPING

CHAPTER 1 | DRUG STORE MANAGEMENT & RECORDS KEEPING — FOUNDATION

Terminology • objectives • principles of record keeping • types of stores • organisation • layout • legal framework

Every single record discussed in this book exists for only three reasons — to prove that a drug was received, to prove where it went, and to prove that it was safe and legal at every moment in between. Keep these three "proofs" in mind and the whole chapter of Stores Records becomes logical instead of a list to be memorised.

1.1 Store, Stores and Drug Store — Basic Terminology

◆ **STORE** — A store is a specified, secure place where materials (drugs, surgicals, chemicals, equipment, stationery) are **received, kept safely and issued** as and when required. The word "**stores**" (plural) is used for the department/section that performs this function, while "**stock**" means the material actually lying in that store.

- ▶ **Drug store:** A store that specifically receives, stores and issues drugs, medicines, surgical dressings and allied pharmaceutical items.
- ▶ **Store keeping:** The physical activity of receipt, custody, preservation, record-keeping and issue of materials.
- ▶ **Stores management:** The managerial function of planning, organising, staffing, directing and controlling all store activities economically.
- ▶ **Materials management:** The **umbrella function** covering purchasing, stores, inventory control, transportation, material handling and disposal of surplus. Stores + Records keeping is one arm of it.
- ▶ **Records keeping (record maintenance):** Systematic **writing, posting, preserving and retrieving** of all documents that evidence the movement (in-flow, holding and out-flow) of every item of stock.
- ▶ **Custodian:** The officer personally responsible and accountable for the stock — normally the **Store Officer / Pharmacist-in-charge of stores**.

1.1.1 Difference between Store Keeping and Stores Management

Point	Store Keeping	Stores Management
Nature	Operative / physical activity	Managerial / decision-making activity
Scope	Narrow — receipt, custody, issue, records	Wide — includes store keeping + planning, layout, inventory control, budgeting, staff
Performed by	Store keeper, store clerk, packer	Store officer / Chief Pharmacist / Materials Manager
Aim	Safe custody and correct records	Right material, right quantity, right time, right cost
Output	Registers, bin cards, vouchers	Policy, SOPs, budget, performance reports

1.2 Records Keeping — Meaning, Purpose and Scope

A **record** is any written, printed or electronic document that furnishes *permanent, retrievable evidence* of a transaction or an event. In a drug store, a transaction is either a **receipt**, a **transfer**, an **issue**, a **return**, a **loss** or a **disposal** — and each one must leave a paper (or electronic) trail.

1.2.1 Purposes of Records Keeping

1. **Legal / statutory compliance** — the Drugs and Cosmetics Act, 1940 & Rules, 1945, the NDPS Act, 1985 and the Pharmacy Act, 1948 make certain records *compulsory*; absence of a record is itself an offence.
2. **Accountability and custody** — fixes personal responsibility for every unit of stock on a named custodian.
3. **Financial control** — shows the value of stock held, consumed and wasted; feeds the annual accounts, budget and audit.
4. **Inventory control** — consumption data from records is the raw material for ABC/VED analysis, EOQ, re-order level and indenting.
5. **Traceability / recall** — batch-wise records allow instant location of every unit of a batch declared *Not of Standard Quality (NSQ)* or recalled.
6. **Prevention of pilferage, theft and misuse** — especially of narcotics, psychotropics and Schedule X drugs.

7. **Avoidance of expiry and over-stocking** — expiry registers and near-expiry reports.
8. **Information for management (MIS)** — consumption trends, supplier performance, lead-time analysis, price trends.
9. **Evidence in disputes** — with suppliers, carriers, insurers, and in courts of law.
10. **Continuity** — the store does not stop working when a store keeper is transferred; the records carry the memory of the organisation.

★ **HIGH-YIELD** — The single-line examination answer: "*Records keeping is the systematic maintenance of documents relating to receipt, storage, issue and disposal of stores, to ensure accountability, legal compliance, financial control and traceability.*" Remember the four italicised words — they are the usual four options of an MCQ.

1.2.2 Scope — What all is "Records Keeping" in a Drug Store?

Table 1.1 Records generated at each stage of the material cycle

Stage of material flow	Typical records generated
Demand / need identification	Indent, requisition slip, consumption statement, stock position statement
Procurement	Purchase requisition, tender/quotation file, comparative statement, purchase order, rate-contract register
Receipt (goods inward)	Gate entry register, delivery challan, invoice, packing list, Goods Inward Register, Goods Received Note (GRN), inspection/test report, discrepancy & damage report
Storage	Bin card, stock ledger / stock register, batch & expiry register, temperature-humidity log, location (bin) index, narcotic register
Issue / distribution	Issue voucher, issue register, daily consumption register, sub-store & ward ledger, gate pass
Return / rejection	Material Return Note (MRN), rejection memo, debit note, credit note, supplier-return register
Loss / deterioration	Breakage & loss register, expiry register, discrepancy report, write-off sanction
Disposal	Condemnation register, board proceedings, auction/destruction certificate, bio-medical waste record
Verification & audit	Stock verification sheet, shortage/excess statement, verification certificate, audit para file

1.3 Objectives and Functions of a Drug Store

1.3.1 Objectives

- To make available the **right drug, of right quality, in right quantity, at the right time, at right cost** to the user departments (the 5 R's).
- To **protect** the stock from deterioration, damage, pilferage, fire, moisture, heat, light, rodents and insects.
- To maintain **minimum investment** in inventory consistent with uninterrupted supply (*no stock-out, no over-stock*).
- To keep **accurate, up-to-date records** of receipt, balance and issue of every item.
- To provide information for **purchase planning and budgeting**.
- To ensure **statutory compliance** in storage and documentation.
- To achieve **economy** in space, manpower and handling cost.

1.3.2 Functions (the classic "receipt-to-disposal" list)

1. **Receiving** the materials and checking them against purchase order.
2. **Inspection** (quantitative + qualitative) and arranging quality testing where required.
3. **Storage / preservation** in an orderly, coded, retrievable and condition-appropriate manner.
4. **Record keeping** — bin cards, ledgers, registers, batch and expiry data.
5. **Issue** of materials against authorised indents, on FIFO/FEFO basis.
6. **Stock control** — maintaining levels, initiating indents at re-order level.
7. **Stock verification** — perpetual and periodic physical counting.

8. **Housekeeping and safety** — cleanliness, pest control, fire safety, security.
9. **Disposal** of expired, damaged, obsolete, surplus and condemned items.
10. **Information & liaison** — with purchase, accounts, user departments and audit.

🎵 **MNEMONIC — FUNCTIONS OF STORES** — "R-I-S-R-I-S-V-H-D-I" → Receive, Inspect, Store, Record, Issue, Stock-control, Verify, Housekeep, Dispose, Inform.

1.4 Importance / Advantages of Systematic Records Keeping

Beneficiary	How good records help
Patient	Correct, unexpired, potent drug; quick recall of a defective batch
Pharmacist / store keeper	Protection from blame; proof of due diligence; easy hand-over on transfer
Hospital / firm	Control on expenditure; reduced wastage; better negotiation with suppliers using purchase history
Purchase department	Reliable consumption and lead-time data for indenting
Accounts / audit	Verifiable link between money spent and material received
Drug regulatory authority	Evidence of legal storage, sale and distribution
Management	MIS reports — turnover ratio, stock-out rate, expiry loss %, supplier performance

⚠️ CONSEQUENCES OF POOR RECORD KEEPING

- Stock-outs of life-saving drugs and simultaneous over-stocking of slow movers.
- Heavy **expiry loss** and wastage of public money.
- Undetected **pilferage**, particularly of narcotics — invites criminal action under the NDPS Act, 1985.
- Failure of **recall** of a Not-of-Standard-Quality batch → patient harm.
- **Audit objections**, disallowance of bills, recovery from the custodian's salary.
- **Suspension or cancellation of the drug licence** for contravention of Rule 65 conditions of the Drugs and Cosmetics Rules, 1945.

1.5 Principles (Golden Rules) of Records Keeping

1. **Contemporaneous entry** — record the transaction *at the time it happens*, never from memory later.
2. **Accuracy & completeness** — every column of the format must be filled; "nil" is written, not left blank.
3. **Legibility & permanence** — written in **ink** (blue/black ball pen), never in pencil; no loose sheets — bound registers with printed serial page numbers.
4. **No erasure / no overwriting / no whitener** — a wrong entry is **scored out with a single line**, the correct entry written above/beside, and the correction **initialled and dated** by the authorised person.
5. **Chronological order** — entries strictly date-wise; each page totalled and carried forward.
6. **Authentication** — every entry and every voucher signed (with name, designation, date) by the person making it and countersigned by the controlling officer.
7. **Cross-referencing** — every entry quotes its source document number (GRN No., Invoice No., Indent No., Issue Voucher No.) so that any figure can be traced back.
8. **Double entry / two-point recording** — the same transaction is posted in two records (bin card **and** stock ledger) which are periodically tallied.
9. **Confidentiality and security** — records kept under lock; access restricted; narcotic registers under the personal custody of a named officer.
10. **Retention** — preserved for the statutory period (see Chapter 11) and destroyed only after proper sanction, with a destruction certificate.
11. **Uniformity & standardisation** — same format, same code numbers, same units of issue throughout the organisation.
12. **Simplicity & economy** — the minimum number of records that satisfy law and control; avoid duplicate registers.

13. Auditability — records must satisfy an outsider (auditor, drug inspector) without oral explanation. *"If it is not written, it was not done."*

✓ **FREQUENTLY ASKED** — A wrong entry in a stock register is corrected by **scoring out with a single line and attesting with signature and date** — *never* by erasing, overwriting, or using correcting fluid. Registers must be **bound with machine-numbered pages**, and a **certificate of number of pages** is recorded on the first page by the officer-in-charge.

1.6 Types and Classification of Store Records

1.6.1 On the basis of legal compulsion

Category	Meaning	Examples
Statutory (mandatory) records	Prescribed by law; must be produced on demand to an inspector	Schedule H / H1 / X sale registers, prescription records, narcotic (NDPS) register, purchase & sale bills, batch-wise records, blood bank records, bio-medical waste records
Non-statutory (managerial) records	Maintained for internal control and efficiency	Bin card, stock ledger, indent register, GRN, issue voucher, temperature log, supplier performance register

1.6.2 On the basis of function

- **Receipt records** — Goods Inward Register, GRN, inspection report, day book of receipts.
- **Custody / holding records** — bin card, stock ledger, batch-expiry register, bin index.
- **Issue records** — indent, issue voucher, issue register, consumption register, gate pass.
- **Financial records** — invoice file, bill register, payment voucher, priced stores ledger.
- **Control records** — re-order level statement, ABC/VED list, stock verification sheet.
- **Statutory records** — as listed above.
- **Miscellaneous records** — correspondence file, guard file of orders/circulars, complaint and drug-alert file.

1.6.3 Other useful classifications

Basis	Types
Form	Manual (registers, cards, vouchers) • Mechanised/Computerised (software, barcode, e-registers)
Period	Permanent (dead-stock register, licence file) • Temporary/periodic (daily receipt register, monthly consumption)
Origin	Internal (indent, GRN, issue voucher) • External (invoice, challan, test report, credit note)
Level of detail	Primary/original (voucher, challan) • Secondary/derived (ledger, register, statement)

1.7 Types of Stores in a Hospital / Pharmaceutical Set-up

Type of store	Purpose / features
Central (Main) store	Single large store receiving all supplies and feeding all sub-stores; keeps master records (stock ledger); usually located near the receiving dock with easy vehicular access
Sub-store / departmental store	Small store in a ward, OT, ICU, casualty, dispensary; receives from main store on indent; works on imprest / top-up system
Dispensary / pharmacy counter stock	Working stock for daily dispensing to patients
Bonded store	For goods on which duty/excise is not yet paid (e.g. rectified spirit, alcohol) — kept under excise bond and double lock
Narcotic store (DDA cupboard)	Separate locked steel cupboard/safe embedded in wall, double lock, key with a named officer; NDPS register maintained
Cold store / cold chain room	Walk-in cooler, ILR, deep freezer for vaccines, sera, insulin, blood products
Inflammable / hazardous store	Detached building for ether, spirit, LPG, cylinders; flame-proof fittings
Quarantine store	"Under test" area for material awaiting inspection/QC clearance
Transit / receiving store	Temporary holding of consignments in the goods inward section

Type of store	Purpose / features
Salvage / condemned store	Holds expired, damaged, obsolete and condemned items pending disposal — must be physically separate and labelled
Equipment / dead-stock store	Non-consumable capital items on permanent register
General / miscellaneous store	Stationery, linen, sanitary, POL, engineering spares

★ **HIGH-YIELD** — Expired, damaged and recalled drugs must be kept in a **separate, clearly marked and preferably locked area** (quarantine/salvage) to prevent accidental issue — this is a favourite one-liner MCQ.

1.8 Organisation and Staffing of the Stores Department

Medical Superintendent / MD ➤ Materials Manager / Store Officer ➤ Pharmacist (Stores) ➤ Store Keeper ➤ Store Clerk ➤ Packer / Helper

Post	Principal duties relating to records
Store Officer / Materials Manager	Overall custody & accountability; sanctions indents; countersigns GRN and issue vouchers; orders stock verification; replies to audit
Pharmacist (Stores) / Chief Pharmacist	Technical scrutiny of receipts (batch, expiry, storage condition); maintains narcotic, Schedule X and expiry registers; prepares indents
Store Keeper	Physical receipt, storage, issue; maintains bin cards and stock registers; bin location; FIFO/FEFO
Store Clerk / Ledger Clerk	Posting of stock ledger , preparation of GRN, issue vouchers, statements, filing
Receipt (Goods Inward) Clerk	Gate entry, checking of challan/invoice, unpacking, counting, drafting GRN and discrepancy reports
Packer / Helper / Loader	Material handling, stacking, packing for despatch
Stock Verifier (independent)	Physical verification; must not be the custodian — principle of <i>separation of duties</i>

✓ **PRINCIPLE OF SEPARATION OF DUTIES** — The person who **orders** (purchase), the person who **receives/keeps** (stores) and the person who **pays** (accounts) must be *different*; and the **stock verifier must be independent of the custodian**. This is the basic internal-check principle behind all store documentation.

1.9 Planning and Layout of the Main Drug Store

1.9.1 Sections (zones) of a well-planned store

1. Goods inward / receiving bay — unloading platform, weighing scale, unpacking table.
2. Quarantine ("under test") area — clearly demarcated, yellow-labelled.
3. Approved / main storage area — racks, bins, pallets, code-wise arrangement.
4. Cold storage area — refrigerators, ILR, deep freezer, walk-in cooler + temperature charts.
5. Narcotic & Schedule X vault — double-locked cupboard.
6. Inflammable / hazardous bay — separate, ventilated, fire-fighting equipment.
7. Rejected / expired / salvage area — red-labelled, locked.
8. Issue / despatch counter — with issue window, trolleys, packing table.
9. Records / office room — ledgers, registers, filing cabinets, computer.
10. Bulk & heavy item area — IV fluids, oxygen cylinders, near the entrance at ground level.

1.9.2 Points to remember in layout & storage practice

Parameter	Norm / good practice
Location	Ground floor, near the receiving gate, with easy access to lift/ramp and to the dispensary; away from wards for noise/traffic reasons
Flow of material	One-way (straight-line) flow : receive → inspect → store → issue; no back-tracking, no crossing of clean and rejected stock

Parameter	Norm / good practice
Aisle / gangway	Wide enough for trolleys (about 1–1.5 m); main gangway wider
Distance from wall	Stacks/pallets kept at least 30 cm (1 ft) away from walls to allow air circulation, cleaning, inspection and pest control
Height from floor	Nothing stored directly on the floor; use pallets/duck-boards (≈ 10 cm high) to protect from damp and flooding
Distance from ceiling / lights	Adequate gap below ceiling and light fittings; sprinkler clearance maintained
Racks & shelves	Metal/steel, adjustable, labelled with item name + code + bin no.
Lighting	Adequate, diffused; direct sunlight avoided ; windows painted/shaded; UV-filtered where needed
Ventilation & temperature	Cross-ventilation, exhaust fans, air-conditioning; store temperature normally kept below 25 °C with RH < 60 %
Floor	Smooth, non-absorbent, crack-free, easily washable; no crevices for pests
Security	Grilled windows, single controlled entry, locks, CCTV, fire alarm, restricted entry
Fire safety	CO ₂ / dry-powder extinguishers, sand buckets, hydrants, "NO SMOKING" boards, marked exits
Pest control	Rodent-proof, insect screens, periodic fumigation record
Space norm	Provide for present + future stock; commonly ~60–65 % of area for storage and the rest for aisles, receiving and office

♪ **MNEMONIC — LAYOUT OF A STORE — "R-Q-S-C-N-H-R-I-O-B"** → Receiving, Quarantine, Storage, Cold, Narcotic, Hazardous, Rejected, Issue, Office, Bulk.

1.10 Legal & Regulatory Framework Governing Store Records

Law / rule	What it demands of a drug store
Drugs and Cosmetics Act, 1940 and Rules, 1945	Licence to stock & sell (Form 20/20B/21/21B; Form 20F/20G for Schedule X); qualified person; storage conditions; records of purchase and sale; prescription and Schedule H/H1/X registers; labelling; no sale of expired drugs
Pharmacy Act, 1948	Drugs to be compounded/dispensed only by a registered pharmacist ; register of pharmacists; renewal records
NDPS Act, 1985 & Rules	Licence/permit for narcotics & psychotropics; separate register; safe custody under lock; consumption and balance accounted; destruction only with official sanction
Drugs (Prices Control) Order (DPCO) / NPPA	Ceiling price of scheduled formulations; price list display; records of price compliance
Bio-Medical Waste Management Rules, 2016	Segregation & recording of discarded/expired medicines and cytotoxic waste (yellow category); annual return
Legal Metrology Act & Packaged Commodities Rules	Correct declaration of net quantity, MRP, mfg. date; verified weighing scales in store
GST Act / financial rules	Tax invoice, HSN code, e-way bill; books of account preserved for the prescribed period
General Financial Rules (GFR) / State Store Purchase Rules	Government stores: purchase procedure, stock registers, physical verification, condemnation, write-off powers
Factories Act / Explosives & Petroleum Rules	Storage of inflammables, compressed gas cylinders, spirit
AERB regulations	Radiopharmaceuticals — licence, lead storage, radiation log-book
Schedule M / GMP (for manufacturers)	Warehousing area requirements, batch manufacturing & distribution records, retention periods

● **REMEMBER THE HIERARCHY — Act** (passed by Parliament) → **Rules / Order** (made by Government under the Act) → **Schedule** (a list appended to the Rules). Thus *Schedule H is a part of the Drugs and Cosmetics Rules, 1945***, not of the Act.

1.11 Materials Management — Where Records Keeping Fits

Demand forecast ➤ Indent ➤ Purchase ➤ Goods inward ➤ Storage & records ➤ Issue ➤ Consumption data

Notice that the cycle is **closed**: consumption data recorded at the issue stage becomes the forecast for the next indent. This is exactly why the syllabus links *records, classification, stock books, indents and storage* into one topic — they are five points on one circle.

Element of materials management	Objective
Materials planning & forecasting	How much will be needed
Purchasing	Buying at right price/quality/time
Receiving & inspection	Getting exactly what was ordered
Stores & warehousing	Safe custody and quick retrieval
Inventory control	Minimum investment, no stock-out
Material handling & transport	Least damage, least cost
Records, codification & standardisation	Identification, accountability, information
Disposal of surplus/scrap	Recovery of value, space release

★ **ECONOMIC IMPORTANCE** — In a typical hospital, **drugs and medical supplies consume roughly 30–40 % of the total recurring budget** (second only to salaries). Materials/stores management is therefore the biggest single area where a pharmacist can save money.

CHAPTER 2 | THE PURCHASE CYCLE AND ITS DOCUMENTS

Because every store record begins with a purchase • methods of purchase • tenders • purchase order • document trail

The goods inward section cannot check a consignment unless it holds the purchase order; the main store cannot post a ledger unless it holds the GRN; accounts cannot pay unless all three tally. Hence a working knowledge of the purchase cycle is compulsory before studying stores records.

2.1 Purchasing — Meaning, Objectives and the 5 R's

◆ **PURCHASING** — Purchasing is the **procurement of materials of the right quality and quantity, from the right source, at the right price, and delivered at the right time and place.**

2.1.1 The 5 R's (Six R's) of purchasing

R	Meaning	Record that proves it
Right Quality	Conforms to pharmacopoeial/tender specification	Specification in PO, analytical/test report, GRN remarks
Right Quantity	Neither excess (blocked capital, expiry) nor short (stock-out)	Indent computation sheet, EOQ working, PO quantity
Right Price	Lowest evaluated cost, not merely lowest quoted rate	Comparative statement, rate contract, invoice
Right Time	Supply before stock falls below minimum level	Delivery schedule in PO, GRN date, lead-time register
Right Source	Licensed, GMP-compliant, reliable manufacturer/dealer	Supplier registration file, licence copies, performance register
(Right Place)	Delivered at the specified store/consignee	Consignee address in PO, challan, gate entry

♪ **MNEMONIC** — "Q-Q-P-T-S" — Quality, Quantity, Price, Time, Source.

2.2 Steps in the Purchase Procedure

- Step 1 Recognition of need** — stock reaches re-order level / user department demand / new item approved by the Pharmacy & Therapeutics Committee.
- Step 2 Preparation of indent (purchase requisition)** by the store, showing item, specification, code, quantity required, stock in hand, average consumption and budget head.
- Step 3 Scrutiny & sanction** of the indent by the competent authority (technical scrutiny by pharmacist, financial sanction by the officer having powers).
- Step 4 Selection of method of purchase** — tender, rate contract, spot purchase, etc.
- Step 5 Issue of enquiry / tender notice (NIT)** with complete specifications, terms, earnest money and last date.
- Step 6 Receipt and opening of quotations/tenders** on the notified date and time before a committee, all members initialling each rate.
- Step 7 Preparation of comparative statement and technical evaluation** (licence, GMP, market standing, samples).
- Step 8 Negotiation** (only with the lowest evaluated/L-1 bidder, as a rule) and **approval of the Purchase Committee.**
- Step 9 Placing the Purchase Order (PO) / supply order** on the successful bidder; security deposit / performance guarantee obtained.
- Step 10 Follow-up** with the supplier for timely delivery (reminders, delivery-period extension).
- Step 11 Receipt of consignment in the goods inward section;** checking, inspection, testing, GRN.
- Step 12 Posting of stock records** (bin card, stock ledger, batch/expiry register).
- Step 13 Bill passing** — three-way matching of PO, GRN and invoice; deduction of penalties/liquidated damages if any.
- Step 14 Payment** through accounts and **release of security deposit** after the warranty period.
- Step 15 Recording of supplier performance** (delay, short supply, quality failure) for future vendor rating.

Indent ➤ Sanction ➤ Tender/Enquiry ➤ Comparative statement ➤ Purchase Order ➤ Receipt + GRN ➤ Stock records ➤ Bill passing ➤ Payment

2.3 Methods of Purchase

Method	When used / salient features
Open / global tender (advertised tender)	Widest publicity in newspapers & website; for large value purchases; most transparent but slow. "Global" when foreign firms may also bid
Limited / closed tender	Enquiry sent only to a short-list of registered, approved suppliers; used for medium value or urgent needs
Single tender / proprietary purchase	Only one source exists (patented or proprietary item, sole manufacturer); justification recorded in writing
Rate contract (RC)	Rates & terms fixed in advance for a period (usually 1–2 years) by a central agency; indenting unit simply places supply orders as needed. No tendering each time, no stock holding by the buyer — biggest advantage
Running contract / standing contract	Fixed quantity to be supplied over a period in instalments
Negotiated purchase	Price negotiated with supplier(s) after bids; normally only with L-1
Spot / cash purchase (local purchase)	Small, urgent, emergency requirement purchased locally within delegated financial powers, against cash memo
Direct purchase from manufacturer / government pool	e.g. from a public-sector drug company, Jan Aushadhi/BPPI, CMSS or state medical services corporation
GeM / e-procurement portal	Government e-Marketplace; online bidding, reverse auction, transparent audit trail
Group / centralised (pooled) purchase	Several hospitals pool demand to obtain quantity discount
Blanket order	One order for a period; deliveries as per call-off schedule
Consignment / VMI purchase	Supplier keeps stock in the hospital store; payment only on consumption

✓ **MOST COMMONLY ASKED** — **Rate Contract** = rate is fixed, quantity is not fixed. **Running contract** = quantity is also fixed. The **lowest tenderer** is called **L-1**; earnest money is also called **bid security**, and money kept after award of contract is the **security deposit / performance guarantee**.

2.4 The Tender System in Detail

2.4.1 Sequence

1. Notice Inviting Tender (NIT) published / uploaded.
2. Sale or download of **tender document** (specifications, terms & conditions, eligibility, penalty clause).
3. Submission in **two-bid system** — *Technical bid* and *Price (financial) bid* in separate sealed covers.
4. Deposit of **Earnest Money Deposit (EMD)** with the technical bid.
5. **Opening of technical bids** on the appointed date before the tender committee; evaluation of licences, GMP/WHO-GMP certificate, market standing, samples, turnover.
6. **Opening of price bids** of only technically qualified firms.
7. Preparation of **comparative statement**; identification of **L-1**.
8. Approval of the competent authority / Purchase Committee.
9. Issue of **Letter of Intent / Purchase Order**; EMD of unsuccessful bidders refunded.
10. Successful bidder furnishes **Security Deposit / Performance Bank Guarantee** and signs the agreement.

2.4.2 Important tender terminology

Term	Meaning
EMD (earnest money / bid security)	Token deposit with the bid to prevent frivolous bidding; forfeited if bidder withdraws
Security deposit / performance guarantee	Held during the contract to ensure performance; refunded after satisfactory completion

Term	Meaning
Liquidated damages (LD) / penalty	Deduction for late supply, commonly a % of value per week of delay subject to a maximum
Risk purchase	If supplier fails, the buyer purchases elsewhere and recovers the extra cost from the defaulter
Validity of offer	Period during which the quoted rate cannot be withdrawn
Fall-clause	Supplier must not sell the same item to anyone else at a lower rate during the contract
Escalation clause	Provision for price revision if raw material/statutory levies change
FOR destination	Freight paid by supplier up to the consignee's store
FOB / CIF	<i>Free On Board</i> (buyer bears freight & insurance from port) / <i>Cost, Insurance, Freight</i> (seller bears them up to destination port)
Pre-qualification	Screening of bidders before price bids are opened
Reverse auction	Online downward bidding by suppliers to reach the lowest price

2.5 Purchase Committee, PTC and the Hospital Formulary

- ▶ **Purchase / Tender Committee:** Opens and evaluates tenders, negotiates, recommends award. Usually includes the Medical Superintendent, Store Officer, Chief Pharmacist, Accounts Officer and a clinician.
- ▶ **Pharmacy and Therapeutics Committee (PTC):** Advisory body of doctors + pharmacists that selects which drugs the hospital will stock, prepares and revises the **hospital formulary**, frames drug-use policies, reviews ADRs and monitors drug utilisation.
- ▶ **Hospital formulary:** A continually revised compilation of the drugs **approved for use in that hospital**, with essential prescribing information. It *limits the variety of items* the store must stock — i.e. it is the first step of **standardisation and variety reduction**.
- ▶ **Essential Medicines List (NLEM):** National list of medicines that satisfy the priority health-care needs of the population, selected on efficacy, safety, cost-effectiveness; the basis of most government indenting and of DPCO price control.

★ **HIGH-YIELD** — The body that decides **which** drugs are stocked = **Pharmacy & Therapeutics Committee**. The body that decides **from whom and at what rate** they are bought = **Purchase / Tender Committee**. The document that lists approved drugs = **Hospital Formulary**.

2.6 Documents of the Purchase–Receipt–Payment Cycle

Document	Raised by	Purpose / contents
Indent / Purchase requisition	Store or user department	Formal demand: item, code, specification, quantity, stock in hand, consumption, urgency, budget head
Enquiry / RFQ / NIT	Purchase section	Invites rates with specification and terms
Quotation / tender / bid	Supplier	Offer of rate, terms, delivery period, validity
Comparative statement	Purchase section	Side-by-side rates of all bidders to identify L-1
Purchase Order (PO) / supply order	Purchase section	The contract : what, how much, at what rate, when and where to deliver
Acknowledgement / acceptance of order	Supplier	Confirms acceptance of PO terms
Delivery challan / despatch note	Supplier	Accompanies goods, lists what has physically been sent
Invoice / bill (tax invoice)	Supplier	Demand for money: rate, quantity, discount, GST, total; batch nos. and expiry dates for drugs
Packing list / case-wise list	Supplier	Contents of each case/carton
Lorry receipt (LR) / consignment note / railway receipt	Transporter	Proof of handing over goods to the carrier; needed for transit claims
Insurance policy / transit insurance	Supplier or buyer	Cover for transit loss
Gate entry (inward) register	Security / gate	Vehicle no., time, name of supplier, no. of packages

Document	Raised by	Purpose / contents
Goods Received Note (GRN) / Material Receipt Report	Goods inward section	Certifies what was actually received, in what condition, accepted/rejected quantity
Inspection / analytical test report	Inspection cell / QC laboratory	Quality clearance ("approved"/"rejected")
Discrepancy / shortage / damage report	Goods inward section	Basis of claim on supplier, carrier or insurer
Debit note / credit note	Buyer / supplier	Financial adjustment for rejected, short or over-charged goods
Bill passing order / payment voucher	Accounts	Sanction and record of payment after three-way matching
Material Return Note (MRN)	User dept. / store	Return of unused or rejected material
Gate pass (returnable / non-returnable)	Store	Authorises material to leave the premises

✓ **THREE-WAY MATCHING** — No bill is passed for payment until **Purchase Order = Goods Received Note = Invoice** agree on item, quantity and rate. Learn this triad — it is the most commonly examined internal-control concept in stores accounting.

2.7 Purchase Order — Contents and Distribution of Copies

2.7.1 Essential contents

- PO number and date (the unique reference quoted in every later document).
- Name and address of the **supplier**; reference to their quotation/tender & rate contract no.
- Name and address of the **consignee** (where to deliver) and of the **paying authority**.
- Full **description, specification, brand/generic name, strength, dosage form, pack size** and **store code number** of each item.
- **Quantity** ordered with unit of measurement; **rate** per unit; discount; GST/HSN; total value.
- **Delivery period / schedule** and place; mode of despatch; packing & marking instructions.
- **Shelf-life clause** — commonly "*not less than 5/6 of the total shelf life remaining at the time of delivery*" (a very frequently asked condition for drug purchase).
- Requirement of **batch-wise analytical report** and/or pre-despatch inspection.
- Terms of **payment**, penalty/liquidated damages, warranty/guarantee, replacement of NSQ or expired stock.
- Arbitration and jurisdiction clause; signature of the competent authority with seal.

2.7.2 Typical distribution of copies

Copy	Sent to	Why
Original	Supplier	The contract / authority to supply
2nd copy	Accounts / finance	To create liability and check the bill
3rd copy	Goods inward / receiving section	To check the consignment on arrival
4th copy	Main store / indenting department	To know dues-in and update records
5th (office) copy	Purchase section file	Record & follow-up

- **DUES-IN (ON-ORDER) REGISTER** — The quantity ordered but not yet received is called **dues-in / on-order quantity**. It is recorded in the *Purchase Order (dues-in) Register* and **must be deducted while preparing the next indent**, otherwise double ordering occurs — a classic audit objection.

2.8 Records Maintained by the Purchase Section

Register / file	Contents
Indent register	Serial no., date, indenting dept., item, quantity, action taken
Tender / quotation register	NIT no., date of opening, firms responded, L-1

Register / file	Contents
Purchase order (supply order) register	PO no., date, firm, item, quantity, rate, value, delivery due date
Dues-in / pending order register	Outstanding quantity against each PO, reminders sent
Rate contract register	Item, firm, rate, validity period
Supplier / vendor registration file	Licence, GMP certificate, PAN/GST, past performance
Vendor rating / performance register	Timeliness, quality failures, short supplies, blacklisting
Bill / invoice register	Bill no., date, amount, GRN no., date of passing, payment date
Price history / market rate register	Rate trend of each item over the years
Budget control register	Head-wise allotment, commitment, expenditure, balance

★ **LINK TO THE SYLLABUS** — The examiner's phrase "*clerical procedure in the goods inward section*" begins exactly where this chapter ends — at the moment the supplier's vehicle reaches the gate with the goods, the challan and the invoice, against the Purchase Order already lying with the receiving clerk.

PART II • GOODS INWARD SECTION & MAIN STORES

CHAPTER 3 | CLERICAL PROCEDURE IN THE GOODS INWARD SECTION

Receiving section • documents accompanying a consignment • checking • GRN • quarantine • discrepancies & claims • bill passing

This is the chapter the syllabus names first, and it is the most examined. Learn the procedure as a chain of documents: Gate entry → Challan/Invoice → GRN → Inspection report → Bin card & Ledger → Bill passing. If you can write that chain in order, you can answer almost any question set on it.

3.1 Goods Inward Section — Meaning, Siting and Layout

◆ **GOODS INWARD SECTION** — The **goods inward (receiving) section** is that part of the stores department where all incoming consignments are **received, unloaded, unpacked, counted, checked against the purchase order, inspected for quality, documented on a Goods Received Note and then either taken on charge or rejected**. It is also called the *receiving section, receiving bay, inward goods section or goods receipt section*.

- ▶ **Position in the flow of materials:** It is the **entry gate of the store** — the point at which the ownership, custody and accountability of material passes from the supplier (and carrier) to the organisation.
- ▶ **Why "clerical" procedure:** Because the physical work of unloading is only half the job; the other half is **paper work** — entering, verifying, certifying, posting and filing the documents, which is what fixes legal and financial responsibility.
- ▶ **Siting:** On the **ground floor**, adjoining the main store, with a *covered loading/unloading platform* at truck-bed height, wide doors, direct road access and a weighing machine. It must be **separate from the issue/despatch counter** so that incoming and outgoing material never mix.
- ▶ **Golden rule of layout: One-way (straight-line) flow** — unloading → unpacking → counting/checking → quarantine → approved storage; never allow back-tracking or cross-traffic.

3.1.2 Facilities and equipment needed in the receiving section

Facility	Purpose
Covered unloading dock / platform + ramp / trolleys / pallet truck	Safe, weather-protected unloading; prevents breakage of glass and IV fluids
Weighing scale (verified/stamped)	To verify weight-based consignments and detect pilferage in transit
Unpacking table with good light, cutters, crow-bar	Opening cases without damaging contents
Counting space / counting tray	Accurate quantitative check
Sampling table + sampling tools, sample bottles, labels	Drawing samples for quality testing
Quarantine ("Under Test") area — yellow marked	Holding the consignment till quality clearance; must be lockable
Approved (green) and Rejected (red) areas	Physical segregation prevents accidental issue of untested/rejected goods
Refrigerator / ILR near the dock	Immediate shifting of cold-chain items
Thermometer, data-logger, cold-chain monitor readers	Verifying that the cold chain was not broken in transit
Desk, computer/terminal, GRN books, rubber stamps, PO file	The clerical work itself
Fire extinguisher, first-aid box, spill kit	Safety during handling of chemicals and glass

3.2 Functions of the Goods Inward Section

1. To **receive** all incoming consignments and record their arrival (gate entry).
2. To verify that the goods are **meant for the organisation** and are covered by a valid **purchase order / rate contract**.
3. To **unload, unpack and count** the goods and carry out the **quantitative check**.

4. To carry out the **qualitative (technical) check** — name, strength, dosage form, pack, batch number, manufacturing and expiry date, storage condition, physical condition, labelling, MRP, manufacturer's licence number.
5. To **draw samples** and send them for analytical testing where the order or policy requires.
6. To place the consignment in **quarantine** pending inspection/test results.
7. To prepare the **Goods Received Note (GRN)** and the entry in the **Goods Inward Register**.
8. To record and report **discrepancies** — shortage, excess, breakage, damage, wrong supply, short expiry, unordered goods.
9. To arrange **acceptance** (transfer to main store with documents) or **rejection** (segregation, intimation to supplier, return).
10. To lodge **claims** on the supplier, the carrier (transporter/railway) or the insurance company.
11. To **certify the invoice/bill** for payment and forward it to accounts with the GRN.
12. To **file and preserve** all receipt documents for audit and legal inspection.

3.3 Staff of the Goods Inward Section and Their Duties

Personnel	Duties
Receiving / Goods Inward Clerk	Gate entry; checks challan & invoice against PO; supervises unpacking and counting; prepares GRN and discrepancy report; maintains Goods Inward Register; files documents
Pharmacist (receiving)	Technical check — generic & brand name, strength, dosage form, batch no., mfg./exp. date, remaining shelf life, storage requirement, labelling & pack integrity; draws samples; certifies fitness for taking on charge
Store Keeper	Takes accepted goods on charge, allots bin/rack location, posts bin card
Store Officer / Materials Manager	Countersigns GRN; decides on rejections, claims, penalties; liaises with purchase and accounts
Inspection committee / QC laboratory	Independent quality inspection and analytical report
Security / gate staff	Gate entry register, vehicle details, number of packages, seal condition
Helper / packer / loader	Unloading, unpacking, stacking, shifting to quarantine or store

✓ **INTERNAL CHECK** — The person who **receives** the goods should **not** be the person who **ordered** them, and the **inspecting** authority should preferably be independent of both. This is why the GRN is prepared by the receiving clerk but **countersigned** by the store officer and supported by a separate **inspection report**.

3.4 Documents Accompanying / Supporting a Consignment

Document	Issued by	What the receiving clerk checks in it
Purchase Order (PO) copy	Purchase section (already held)	The master reference — item, specification, quantity, rate, delivery date, shelf-life clause, consignee
Delivery challan / despatch note	Supplier	What has physically been sent; number of cases; PO number quoted
Invoice / tax invoice / bill	Supplier	Description, quantity, rate, discount, GST & HSN, total; batch no. and expiry date of each item ; supplier's drug licence no., GSTIN, and (for scheduled formulations) the DPCO ceiling price
Packing list / case-wise list	Supplier	Contents of each carton, so that a short-packed case can be pinpointed
Lorry receipt (LR) / consignment note / railway receipt (RR)	Transporter	Number of packages booked, weight, condition at booking, freight paid/to-pay — basis of any transit claim
Insurance policy / cover note	Supplier or buyer	Whether transit loss is insured and by whom
Test / analytical report (batch certificate of analysis)	Manufacturer's QC	Batch-wise compliance with pharmacopoeial standards
Gate entry (inward) slip	Security	Date & time of arrival, vehicle number, driver, number of packages, seal intact?

Document	Issued by	What the receiving clerk checks in it
Guarantee / warranty certificate, and (for equipment) manuals	Supplier	Warranty period, installation obligation
Cold-chain / temperature data-logger record	Transporter/supplier	Whether 2–8 °C was maintained throughout transit

★ **CHALLAN VS INVOICE — CLASSIC MCQ** — **Delivery challan** = document of *movement of goods* (says *what has been sent*); it travels with the goods and does **not** demand money. **Invoice (bill)** = document of *demand for payment* (says *what is payable*), showing rate, GST and total value. A **GRN** is prepared by the **buyer**, not the supplier.

3.5 The Clerical Procedure — Step by Step

This is the *heart of the syllabus*. Learn the 16 steps in order; every sub-question (GRN, discrepancy, quarantine, bill passing) is only a detail of one of these steps.

- Step 1 Arrival & gate entry.** The vehicle reports at the gate. Security records date, time, vehicle number, name of supplier/transporter and **number of packages** in the **Gate Entry (Inward) Register** and issues a gate entry slip. Condition of seals and of packages is noted *before* unloading.
- Step 2 Receipt of documents.** The driver/representative hands over the **delivery challan, invoice, packing list and LR**. The clerk stamps them with date of receipt and a serial *inward number*.
- Step 3 Verification of authority to supply.** The clerk retrieves the **office copy of the Purchase Order** (or the rate-contract supply order) and confirms: correct consignee, valid PO number, item actually ordered, **delivery period not expired**. *Unordered goods are not unloaded* — they are refused or kept unopened with intimation to purchase section.
- Step 4 Unloading and counting of packages.** Packages are counted against the challan/LR. Any shortage in the **number of packages**, or a torn/wet/re-stitched/open package, is recorded **on the carrier's copy of the LR itself** and the driver's signature obtained — this is essential to sustain a claim later. Weighment is done where relevant.
- Step 5 Unpacking.** Cases are opened carefully at the unpacking table in the presence of the receiving clerk (and the supplier's representative if present). Packing material is retained until checking is complete, as small items often remain inside.
- Step 6 Quantitative check.** Each item is counted/measured and compared with (a) the challan, (b) the invoice and (c) the purchase order. Units of measurement are converted where necessary (e.g. PO in "1,000 tablets", supply in "10 boxes of 10 × 10").
- Step 7 Qualitative / technical check by the pharmacist.** Name (generic + brand), strength, dosage form, pack size, **batch number, date of manufacture, date of expiry, remaining shelf-life**, storage instruction, label completeness, manufacturer's name & licence number, MRP/DPCO price, ISI/BIS or sterility markings, physical condition (colour, clarity, caking, leakage, broken seal, cracked ampoule), and whether the batch figures in any **drug alert / recall list**.
- Step 8 Sampling for analytical test.** Where the PO or hospital policy requires, a sample of each batch is drawn, labelled (item, batch, GRN no., date), entered in the **Sample Register** and sent to the approved laboratory with a covering memo. The consignment stays in quarantine meanwhile.
- Step 9 Movement to the quarantine ("Under Test") area.** The consignment is stacked in the quarantine area with a **yellow "QUARANTINE / UNDER TEST"** label showing item, batch, quantity and GRN number. Cold-chain items go straight into the refrigerator/ILR, still labelled as quarantined.
- Step 10 Entry in the Goods Inward Register (Daily Receipt Register).** A serial, date-wise entry of every consignment received on that day — the *primary chronological record* of receipts.
- Step 11 Preparation of the Goods Received Note (GRN).** The clerk prepares the GRN in the prescribed number of copies showing quantity ordered, quantity received, quantity accepted, quantity rejected/short, batch and expiry details and remarks. It is signed by the receiving clerk, the pharmacist and countersigned by the Store Officer. **The GRN is the pivotal document of the whole section.**
- Step 12 Inspection / test result and disposal of the consignment.** On receipt of the inspection or analytical report the quarantine label is replaced by a **green "APPROVED"** label (goods moved to main store) or a **red**

"REJECTED" label (goods moved to the rejected/salvage area, supplier informed, arranged for return/replacement).

Step 13 Taking on charge and posting of stock records. For accepted quantities the store keeper allots a bin/rack location and posts the **bin card**; the ledger clerk posts the **stock ledger / stock register** and the **batch-and-expiry register** — each quoting the **GRN number** as authority.

Step 14 Reporting of discrepancies and lodging of claims. A **Discrepancy / Shortage / Damage Report** is raised and copies sent to the purchase section, the supplier, the carrier and the insurer as applicable, followed by a **debit note** for the value involved.

Step 15 Certification of the bill and passing for payment. The invoice is stamped and certified "*Received in good condition and taken on stock vide GRN No. ____ dated ____*", the accepted quantity and rate verified against the PO (**three-way matching**), penalties/liquidated damages for late supply noted, and the bill forwarded to accounts.

Step 16 Filing, follow-up and MIS. GRN, challan, invoice copy, inspection report and claim correspondence are filed **PO-wise / GRN-number-wise**; the purchase section is informed so that the **dues-in (pending order) register** is closed or partly closed; supplier performance (delay, short supply, quality failure) is recorded for **vendor rating**.

Gate entry ➤ Documents vs PO ➤ Unload & count ➤ Quantity check ➤ Quality check ➤ Sample to QC ➤ Quarantine ➤ GRN ➤ Approved / Rejected ➤ Bin card + Ledger ➤ Bill passing ➤ Filing

🎵 **MNEMONIC FOR THE 16 STEPS** — "G-D-P-U-U-Q-Q-S-Q-G-I-T-D-B-F" — Gate entry, Documents, PO check, Unload, Unpack, Quantity, Quality, Sample, Quarantine, Goods Inward Register, GRN, Inspection result, Take on charge, Discrepancy report, Bill passing, Filing.

3.6 Checking of the Consignment in Detail

3.6.1 Quantitative check

- Count/measure **every item** — sample counting is allowed only for large homogeneous consignments, and then the method must be recorded.
- Compare with **three** documents: challan (what was sent), invoice (what is charged) and **purchase order (what was ordered)**. The **lowest** of the three governs what can be accepted and paid for.
- Convert units carefully (strips → tablets, vials → ml, bottles → litres).
- Watch for **excess supply**: excess quantity is **not** taken on charge without sanction; it is either returned or a supplementary order/regularisation is obtained.
- Record **shortages** immediately on the carrier's documents and on the GRN.

3.6.2 Qualitative (technical) check — the pharmacist's checklist

What to check	Why it matters / what to reject
Generic name, brand, strength, dosage form	Supply of a different salt, strength or dosage form = wrong supply; reject
Pack size and pack type	Affects rate per unit and dispensing convenience
Batch (Lot) number	Mandatory for traceability and recall; unbatched stock is not acceptable
Date of manufacture (Mfg.) & date of expiry (Exp.)	Expired or expiring stock must never be accepted
Remaining shelf life	Standard tender condition: at least 5/6 (83 %) of the total shelf life , or a minimum of ~75 % / not less than 2 years as specified in the PO; short-expiry stock is rejected or accepted only with written sanction
Storage condition on the label	Cold-chain, "protect from light", "do not freeze" — dictates immediate shifting
Condition of container & seal	Leakage, breakage, cracked ampoules, loose caps, swollen/punctured blister, wet cartons, damaged strip = reject
Physical appearance of the drug	Discolouration, caking, mottling of tablets, turbidity or precipitate in injections, separation of emulsion, mould growth, crystallisation, softening of suppositories
Sterility indicators for sterile products	Intact seal, clarity, absence of particulate matter, valid sterilisation lot

What to check	Why it matters / what to reject
Label particulars	Manufacturer's name & address, mfg. licence number , net content, " Schedule H / H1 / X " warning with Rx / ✕ symbol , storage directions, batch, mfg./exp. dates, MRP, "for hospital/government supply — not for sale" where applicable
MRP / DPCO ceiling price	Overcharging on price-controlled formulations must be objected to
Presence in drug alert / NSQ / recall list	Batch to be quarantined and reported at once
Accompanying documents	Certificate of analysis, warranty, manuals, accessories
Cold-chain evidence	Vaccine Vial Monitor stage, cold-chain monitor card, data-logger printout — breach = reject

⚠ NEVER ACCEPT

- Drugs **without batch number, manufacturing date or expiry date**.
- Drugs that are **expired or of short expiry** beyond the PO condition.
- Consignments with **broken seals, leakage, wet or re-stitched packages** (accept "under protest" only after recording the damage).
- Cold-chain items where the **cold chain has demonstrably broken** (VVM stage 3/4, frozen DPT/Hep-B, thawed insulin).
- **Unordered / excess** goods without written sanction.
- Physician's **samples** or "not for sale" stock for regular hospital use unless specifically permitted.

3.7 Goods Received Note (GRN) — The Pivotal Document

◆ **GRN** — A **Goods Received Note** (also *Goods Receipt Note, Material Receipt Report, Receiving Report, Store Receipt Voucher*) is the internal document prepared by the receiving section **certifying the quantity and condition of material actually received** against a purchase order. It is the **authority (a) for the store to take goods on charge, (b) for the ledger clerk to post the stock records and (c) for accounts to pay the bill**.

3.7.1 Contents of a GRN

Group	Columns / particulars
Identification	GRN serial number and date; inward/gate entry number
Supplier	Name & address of supplier; their challan no. & date; invoice no. & date; LR/RR no. & date; transporter
Order reference	Purchase order no. & date (or rate contract no.); indent no.
Item details	Sl. no.; store code number ; description/specification; unit of measurement
Quantities	Quantity ordered • quantity as per challan/invoice • quantity actually received • quantity accepted • quantity rejected/short/damaged
Drug-specific data	Batch no., date of manufacture, date of expiry , pack size, manufacturer
Value	Rate, discount, GST, total value of accepted quantity
Inspection	Sample sent (yes/no), test report no. & date, result: approved/rejected
Storage	Bin/rack/location allotted; storage condition required
Remarks	Shortage, breakage, short expiry, late delivery, damage in transit, "accepted under protest"
Signatures	Receiving clerk • Pharmacist/inspecting officer • Store keeper • Countersignature of Store Officer

3.7.2 Copies of the GRN and their distribution

Copy	Goes to	Use
1st (original)	Accounts / finance section	Attached to the supplier's bill; authority for payment
2nd	Purchase section	To close/partly close the order in the dues-in register and to record supplier performance
3rd	Main store (ledger clerk)	Authority to post the stock ledger and bin card

Copy	Goes to	Use
4th	Supplier (when required)	Acknowledgement of receipt of goods
5th (book copy)	Goods inward section file	Record, audit and reference

★ **REMEMBER** — GRN is prepared in the **goods inward/receiving section**, normally in **4–5 copies**, and is **numbered serially**. *No material may be taken on stock, and no bill may be paid, without a GRN.* Where the material fails inspection, the GRN still records the receipt but shows the quantity as **rejected**.

3.7.3 Specimen format — Goods Received Note

Format 3.1 GRN (upper part carries GRN No. & date, supplier, challan, invoice, LR and PO references; lower part carries signatures)

Sl.	Code	Description	Unit	Qty ord.	Qty recd.	Qty acc.	Qty rej.	Batch	Mfg.	Exp.	Rate	Value	Remarks
1													
2													
3													

3.8 Goods Inward Register (Daily Receipt Register)

While the GRN is *consignment-wise*, the Goods Inward Register is *date-wise* — it is the chronological diary of the receiving section and the first document an auditor or drug inspector asks for.

Format 3.2 Goods Inward / Daily Receipt Register

Sl. No.	Date of receipt	Supplier	Challan/Invoice No. & date	PO No. & date	Item & quantity	GRN No.	Remarks / disposal

3.9 Quarantine, Acceptance and Rejection

Status	Label colour (usual practice)	Action
Under test / quarantine	Yellow	Stacked separately in the quarantine area; no issue permitted; entry in quarantine register
Approved / released	Green	Moved to the main storage area; taken on charge; bin card and ledger posted
Rejected	Red	Moved to the rejected/salvage area under lock; supplier informed; returned within the stipulated period; debit note raised
Recalled / NSQ	Red "RECALLED — DO NOT ISSUE"	Batch withdrawn from all sub-stores, quarantined, reported to the licensing authority, returned/destroyed as directed

✓ **QUARANTINE** — Quarantine means **physical or administrative segregation of material awaiting a decision on its quality**. In a computerised store it may be "electronic quarantine" — the stock exists in the system but is *blocked for issue*. Colour convention: **yellow = under test, green = approved, red = rejected**.

3.10 Handling of Discrepancies, Shortages, Damages and Claims

Type of discrepancy	Meaning	Action by the goods inward section
Shortage in number of packages	Fewer cases received than booked	Record on the carrier's LR at the time of delivery, obtain driver's signature, take "short certificate" from the transporter, lodge claim on carrier

Type of discrepancy	Meaning	Action by the goods inward section
Short supply / short packing	Package intact but contents fewer than invoiced	Open in the presence of a witness, prepare shortage report, inform supplier, pay only for quantity received, raise debit note
Breakage / damage in transit	Broken ampoules, leaking bottles, crushed cartons	Preserve the damaged goods and packing as evidence, photograph, prepare damage report, claim on carrier and/or insurer, demand free replacement
Wrong supply	Different drug, strength, dosage form, brand or pack than ordered	Reject; do not take on charge; intimate purchase section; return at supplier's cost
Excess supply	More than ordered	Keep unopened/segregated; do not take on charge; obtain sanction for retention or return the excess
Short expiry / expired stock	Remaining shelf life below the PO condition	Reject; if unavoidable and urgently needed, accept only with written sanction of the competent authority and a supplier undertaking to replace unused stock
Not of Standard Quality (NSQ) on test	Analytical report fails pharmacopoeial limits	Quarantine the entire batch, inform supplier & licensing authority, recall issued stock, reject and claim replacement/refund; enter in NSQ register
Spurious / misbranded / adulterated suspicion	Doubtful label, price, source	Freeze the stock, inform the Drugs Control Organisation immediately, do not return to supplier without instructions
Unordered goods	No PO exists	Refuse delivery or keep unopened and inform purchase section
Late delivery	Received after the delivery period	Record the delay on the GRN so that liquidated damages / penalty can be deducted from the bill; extension of delivery period, if granted, must be on record
Cold-chain breach	Temperature excursion during transit	Segregate, do not use, obtain manufacturer's opinion, reject if stability is compromised; record data-logger printout

3.10.1 Debit note and credit note

- ▶ **Debit note:** Raised by the **buyer** on the supplier to reduce the amount payable — for short supply, rejected goods, breakage, over-charging, penalty. It *debts* the supplier's account.
- ▶ **Credit note:** Issued by the **supplier** to the buyer accepting that reduction, or on return of goods. It *credits* the buyer's account.
- ▶ **Material Return Note (MRN):** Prepared when material already taken on charge is returned — to the supplier (rejected/expired) or to the store by a user department (unused).
- ▶ **Non-returnable / returnable gate pass:** Authority for material to physically leave the premises (e.g. goods returned to supplier, equipment sent for repair).

3.11 Bill Passing and Payment Procedure

- Step 1** Invoice received (with the consignment or separately by post) is entered in the **Bill / Invoice Register** with date of receipt.
- Step 2** The invoice is linked with the **GRN** and the **Purchase Order** — the **three-way matching** of *item, quantity and rate*.
- Step 3** Arithmetical accuracy, discount, GST rate & HSN, freight and other charges are checked; DPCO ceiling price verified for scheduled formulations.
- Step 4** Deductions are computed: short/rejected quantity, breakage, **liquidated damages for late supply**, recovery of earlier advances, statutory deductions (TDS, etc.).
- Step 5** The receiving officer **certifies** on the invoice: *"Goods received in good condition, quantity and quality verified, taken on stock vide GRN No. __ dated __, entered at page __ of the stock register."*
- Step 6** The store officer **countersigns** and forwards the bill with GRN and inspection report to the accounts section.
- Step 7** Accounts pre-audit the bill, pass it for payment, and issue the cheque/e-payment; entry made in the payment register.
- Step 8** **Security deposit / performance guarantee** is released only after the warranty or contract period ends; EMD of unsuccessful bidders is refunded.

⚠️ AUDIT WILL OBJECT IF ...

- A bill is passed **without a GRN**, or the GRN quantity exceeds the PO quantity.
- The **stock register page number** is not quoted on the bill (no proof that the material was taken on charge).
- **Liquidated damages** for late supply are not deducted, or waiver is granted without sanction.
- Payment is made for **short-supplied or rejected** quantity.
- The **same GRN** is used to support two payments (double payment).
- Short-expiry stock is accepted without written sanction and later expires unused.

3.12 Receipt of Special Categories of Items

Category	Extra precautions at the goods inward stage
Narcotics & psychotropics (NDPS)	Received by a named officer ; counted ampoule/tablet-wise in the presence of a witness; entered the same day in the NDPS/DDA register; stored at once in the double-locked cupboard ; no quarantine outside the safe
Schedule X drugs	Separate register and separate locked cupboard; quantity verified against the special licence/permit
Cold-chain items (vaccines, sera, insulin, blood products)	Check VVM stage, cold-chain monitor card, data-logger ; shift to ILR/refrigerator within minutes; record receipt temperature; never leave on the dock
Blood and blood components	Verified against blood-bank records; stored at 2–6 °C (whole blood/PRBC), platelets at 20–24 °C with agitation , FFP below –30 °C
Cytotoxic / anticancer drugs	Check for leakage using intact outer packing; handle with gloves; keep spill kit ready; store separately with a "CYTOTOXIC — HANDLE WITH CARE" label
Radiopharmaceuticals	Received by the authorised radiation-safety officer; survey meter check; lead-lined storage; entry in the radiation log-book (AERB requirements)
Inflammables (ether, spirit, acetone, LPG)	Unloaded away from the main building; "No smoking"; stored in the detached inflammable store; spirit entered in the excise/bonded store register
Compressed gas cylinders	Verify colour code , valve condition, test date, cap in place; chain cylinders upright; never roll or drop
Glassware, IV fluids, ampoules	Open carefully; count broken units; claim on carrier; stack at low level
Equipment / capital items	Check accessories, manuals, warranty card, serial number; installation and demonstration certificate; entry in the dead-stock (non-consumable) register
Free samples, donations, drugs received from other institutions	Entered in a separate register ; checked for expiry and labelling; donated drugs must have adequate remaining shelf life
Bio-medical / hazardous chemicals	MSDS obtained; segregated storage; PPE used

3.13 Precautions, Do's and Don'ts in the Goods Inward Section**Do's**

- Always keep the **PO copy** ready before unloading; unload only what is covered by an order.
- Record damage **before** the carrier leaves and get his signature.
- Check **batch, mfg. and expiry dates of every item** — this is the pharmacist's personal responsibility.
- Shift **cold-chain items first**, other items later.
- Prepare the GRN **on the day of receipt**; delay is an audit objection.
- Use **serially numbered, bound** GRN books and registers.
- Segregate quarantine, approved and rejected stock physically.
- Retain packing material until checking is complete.
- File every document against its GRN number for instant retrieval.

Don'ts

- Do **not** sign "received in good condition" on the transporter's copy before counting.
- Do **not** take excess or unordered goods on charge.
- Do **not** mix a new batch with an existing batch in the same bin without separate batch identification.

- Do **not** issue any material lying in quarantine.
- Do **not** allow the supplier's representative to handle or count the stock unsupervised.
- Do **not** use pencil, whitener or erasers in any register.
- Do **not** keep narcotics even briefly outside the double-locked cupboard.

3.14 Summary — Records of the Goods Inward Section

Record	Nature	Purpose in one line
Gate entry (inward) register	Chronological	Proof of arrival of the vehicle and number of packages
Goods inward / daily receipt register	Chronological	Date-wise diary of all consignments received
Goods Received Note (GRN)	Transaction voucher	Certifies quantity & condition received; authority to take on charge and to pay
Inspection / test report file	Technical	Quality clearance of each batch
Sample register	Technical	Record of samples drawn and sent for analysis
Quarantine register	Control	What is under test, since when
Discrepancy / shortage / damage report file	Claim	Basis of claim on supplier, carrier, insurer
Rejection memo & supplier-return register	Claim	What was rejected and whether it went back
Debit note / credit note file	Financial	Adjustment of the supplier's account
Bill / invoice register	Financial	Receipt, passing and payment of bills
Freight / transport & insurance claim file	Financial	Recovery of transit losses
Supplier performance (vendor rating) register	MIS	Delay, short supply and quality record of each firm
Drug alert / NSQ / recall file	Statutory	Action taken on alerts and recalls
Gate pass (returnable & non-returnable) book	Control	Authority for material to leave the premises

★ ONE-LINE ANSWERS FROM THIS CHAPTER

- Document prepared **by the buyer** on receipt of goods — **GRN**.
- Document that **travels with the goods** — **delivery challan**.
- Document that **demand payment** — **invoice**.
- Document required for a **transit claim** — **lorry receipt / consignment note**.
- Area where goods await quality clearance — **quarantine (yellow label)**.
- Matching of PO + GRN + Invoice — **three-way matching**.
- Deduction for late supply — **liquidated damages**.
- Minimum remaining shelf life usually demanded in drug tenders — **5/6 of total shelf life**.

CHAPTER 4 | RECORDS AND PROCEDURES IN THE MAIN STORES

Functions of the main store • master list of registers • receipt, storage, issue, transfer & return • specimen formats • filing • verification

The main store is the central bank of materials. Everything it receives must be accounted for, everything it issues must be authorised, and at any moment the physical stock on the shelf must equal the balance shown in its books. The records described here are simply the tools that make that equality provable.

4.1 Main (Central) Store — Meaning and Functions

◆ **MAIN STORE** — The **main or central store** is the principal storage facility of the institution which **receives all approved material from the goods inward section, holds it in safe custody, maintains its complete accounts, and issues it to sub-stores, wards, departments and the dispensary against authorised indents.**

4.1.1 Functions

- 1. Custody** — safe keeping of all drugs and supplies under proper storage conditions.
- 2. Accounting** — maintenance of quantitative (and where required, priced) records of receipt, issue and balance of every item, batch-wise.
- 3. Issue and distribution** — supply to sub-stores/wards on **FIFO/FEFO** basis against indents.
- 4. Stock control** — watching minimum, maximum, re-order and danger levels; initiating indents in time.
- 5. Preservation** — protection from heat, light, moisture, pests, fire, theft; cold-chain maintenance.
- 6. Codification and location control** — every item has a code and a fixed bin address.
- 7. Verification** — perpetual and annual physical verification, reconciliation of book balance with ground balance.
- 8. Expiry and quality surveillance** — near-expiry reporting, action on drug alerts and recalls.
- 9. Disposal** — segregation and condemnation of expired, damaged, obsolete and surplus stock.
- 10. Information** — consumption statements, stock position reports, expenditure statements for management, purchase and audit.

4.1.2 Centralised vs decentralised stores

Point	Centralised store	Decentralised (sub-)stores
Investment in stock	Lower (pooled buffer)	Higher (buffer at every point)
Control & records	Strong, uniform, single set of books	Weaker, duplicated records
Bulk purchase economy	High — quantity discounts possible	Low
Storage conditions & security	Easier to provide (cold room, vault)	Difficult and costly to replicate
Expiry / obsolescence loss	Lower — stock can be redistributed	Higher
Speed of supply to user	Slower; needs transport & indenting	Immediate at the point of use
Handling & transport cost	Higher internal movement	Lower
Best practice	Centralised main store for bulk custody + small imprest sub-stores for daily use	

4.2 Broad Groups of Main-Store Records

Receipt records ► Custody records ► Issue records ► Control records ► Financial records ► Statutory records

- **Receipt records** — GRN file, receipt register/day book, batch & expiry register.
- **Custody (holding) records** — **bin card, stock ledger / stock register**, bin-location index, kardex, temperature & humidity log.
- **Issue records** — indent file, issue voucher, issue register, daily consumption register, sub-store/ward ledger, gate pass book.
- **Control records** — re-order level statement, ABC/VED classification list, dues-in statement, stock verification sheet, near-expiry statement.

- **Financial records** — priced stores ledger, bill register, expenditure/budget control register, loss & write-off register.
- **Statutory records** — Schedule H/H1/X registers, NDPS register, prescription records, licences & renewals file, bio-medical waste records, DPCO price list.

4.3 Master List of Registers and Records Maintained in the Main Store

No.	Register / record	What it contains / why it is kept
1	Stock register / stock ledger (item-wise, page per item)	The master account: opening balance, receipts, issues, closing balance of every item, with source document references
2	Bin card (stock card)	Card kept <i>at the bin</i> : quantity received, issued and balance only — posted by the store keeper at the time of each transaction
3	Batch and expiry register	Batch-wise quantity with date of expiry; basis of FEFO issue and near-expiry reporting
4	Goods Received Note (GRN) file / receipt register	All receipts with supplier, PO and inspection references
5	Indent register	Serial record of indents received from wards/departments and of indents raised on the purchase section
6	Issue voucher / issue register	Every issue with date, indent no., indenting department, item, batch, quantity, receiver's signature
7	Daily / monthly consumption register	Consumption per item per period — the basis of all forecasting
8	Sub-store & ward ledger (departmental account)	What each ward/department has drawn; controls over-drawal
9	Dead stock / non-consumable (permanent) register	Equipment, furniture, instruments — never "consumed", verified annually, written off only on condemnation
10	Consumable (expendable) stores register	Drugs, dressings, chemicals, stationery
11	Narcotic / NDPS (DDA) register	Drug-wise, date-wise receipt, issue and balance of narcotics & psychotropics; countersigned; kept under lock
12	Schedule X register	Purchase and sale/issue of Schedule X (habit-forming) drugs
13	Schedule H / H1 register	Sale/issue records of prescription drugs; H1 register for the specified antibiotics, anti-TB and psychotropic drugs
14	Prescription file / register	Prescriptions retained as required by the Drugs and Cosmetics Rules
15	Spirit / alcohol (bonded store) register	Excise-controlled receipt, issue and wastage of rectified spirit
16	Expiry register / near-expiry statement	Items expiring in the next 3–6 months, for redistribution or return to supplier
17	Breakage, damage and loss register	Every breakage/loss with cause, value, sanction of write-off
18	Condemnation / disposal register	Items condemned, board proceedings, mode and date of disposal
19	Stock verification register & discrepancy statement	Date of verification, book balance, ground balance, shortage/excess, action taken
20	Temperature and humidity record (log)	Twice-daily readings of store, refrigerator, ILR and deep freezer
21	Refrigerator/ILR maintenance & breakdown register	Defrosting, servicing, power failure, action taken
22	Bin location index / location register	Code number → rack/shelf/bin address, for instant retrieval
23	Gate pass book (returnable & non-returnable)	Authority for material to leave the premises
24	Material Return Note (MRN) file	Unused material returned by departments; rejected material returned to suppliers
25	Transfer voucher file	Inter-store / inter-institutional transfers
26	Priced stores ledger / valuation register	Value of receipts, issues and closing stock for accounts
27	Bill register & payment register	Bills received, passed and paid
28	Budget control (allotment vs expenditure) register	Head-wise funds, commitment and balance
29	Licence and statutory documents file	Drug licences, renewals, pharmacist registration, NDPS permits, AERB licence
30	Drug alert / NSQ / recall register	Alerts received, batches held, action taken, report sent
31	ADR (adverse drug reaction) & complaint file	Quality complaints from wards, forwarded to the manufacturer/authority

No.	Register / record	What it contains / why it is kept
32	Bio-medical waste register (yellow category)	Expired/discarded medicine and cytotoxic waste handed over, with dates & quantities
33	Free supply / donation / sample register	Drugs received free, from donations or as samples
34	ABC / VED analysis file & re-order level statement	Inventory control working papers
35	Guard file of orders, circulars and SOPs	Standing instructions, delegation of financial powers, price lists
36	Visitor / inspection book	Inspections by drug inspector, audit, accreditation team and their remarks

✓ MOST ASKED PAIRS

- **Bin card** — kept *with the material*, shows **quantity only**, posted by the **store keeper**, *at the time of each transaction*.
- **Stock ledger** — kept *in the office*, may show **quantity and value**, posted by the **ledger clerk / accounts**, from vouchers.
- **Dead stock register** — for **non-consumable** items (equipment, furniture); **consumable register** — for drugs and dressings.

4.4 Receipt Procedure in the Main Store

- Step 1** Accepted material is transferred from the goods inward section **with the GRN** (and the inspection/test report).
- Step 2** The store keeper **re-counts** the quantity handed over and signs the GRN as having taken the goods on charge.
- Step 3** **Bin/rack location** is allotted as per the code number; the item is placed with the **label facing outward**.
- Step 4** **Bin card** is posted at once: date, GRN no., quantity received, new balance, batch and expiry.
- Step 5** **Stock ledger** is posted by the ledger clerk from the GRN, quoting GRN number and date.
- Step 6** **Batch & expiry register** is posted; the expiry date is also written on the bin card and on the carton in bold marker.
- Step 7** For narcotics, Schedule X and spirit — entry in the **respective statutory register the same day** and immediate storage under double lock.
- Step 8** Cold-chain items are placed in the ILR/refrigerator and the **temperature log** noted.
- Step 9** Where the new receipt is of a **different batch**, it is stacked **behind** the existing stock so that the **earlier-expiring stock is issued first (FEFO)**.
- Step 10** The GRN is filed; the purchase section is informed to close the dues-in entry.

★ **GOLDEN RULE** — No material enters the store without a GRN, and no material leaves the store without an issue voucher. Every figure in the stock ledger must be traceable to one of these two documents (or to a transfer voucher, MRN or write-off sanction).

4.5 Storage Procedure in the Main Store

- Store **code-wise / group-wise** as per the classification adopted (see Chapters 5, 6 and 10).
- Observe the **storage condition on the label** — cold place, cool place, room temperature, protect from light/moisture.
- Follow **FIFO** (first-in-first-out) and more correctly **FEFO** (first-expiry-first-out).
- Keep stock **30 cm away from walls** and on **pallets ~10 cm above the floor**; never on the floor directly.
- Heavy and bulky items (IV fluids, cylinders) at **floor/lower shelf level**, light items above.
- Fast-moving items **near the issue counter**; slow-moving items at the far end.
- Keep **narcotics, Schedule X, spirit, inflammables, cytotoxics and radiopharmaceuticals in their designated secure areas**.
- Never store **look-alike / sound-alike (LASA)** drugs adjacent to each other; use LASA labels and separation.
- Record **temperature and humidity twice daily**; maintain the cold-chain log for every refrigerator and ILR.

- Keep the **expiry date visible**; mark near-expiry stock with a coloured sticker (traffic-light system).
- Maintain cleanliness, pest control and fire-safety equipment; keep aisles clear.

4.6 Issue Procedure in the Main Store

- Step 1 Receipt of indent** from the ward/department/sub-store on the prescribed form, signed by the **authorised indenting officer** within his delegated powers.
- Step 2 Scrutiny of the indent** — is the item stocked? is the quantity reasonable in relation to the previous issue and the patient load? is the indent within the imprest scale? is stock available?
- Step 3 Entry in the indent register** with a serial number and date.
- Step 4 Approval / sanction** by the store officer; quantity may be reduced (“cut”) and the reason recorded.
- Step 5 Preparation of the issue voucher** (issue slip / material issue note) showing item, code, batch, expiry, quantity issued and balance.
- Step 6 Physical picking of stock on FEFO basis** — the batch expiring earliest is issued first; batch and expiry are written on the voucher.
- Step 7 Handing over** to the authorised messenger/representative, who **signs the issue voucher/register in token of receipt**.
- Step 8 Posting of records:** bin card (quantity issued and new balance), stock ledger, batch & expiry register, departmental/ward ledger, daily consumption register; narcotics also in the NDPS register.
- Step 9 Closing balance checked** against the bin card; if the balance has fallen to the **re-order level**, an indent is initiated on the purchase section.
- Step 10 Filing** of the indent and issue voucher; preparation of the monthly consumption and expenditure statement.

Indent received ➤ Scrutiny & sanction ➤ Issue voucher ➤ FEFO picking ➤ Receiver signs ➤ Bin card + Ledger ➤ Consumption register

4.6.1 Specimen format — Issue Voucher / Material Issue Note

Format 4.1 Issue voucher (header: voucher no. & date, indent no. & date, indenting department; footer: signatures of store keeper, receiver and store officer)

Sl.	Code No.	Item & specification	Unit	Qty indented	Qty issued	Batch No.	Expiry	Ledger folio	Remarks

⚠ ISSUE DISCIPLINE

- Never issue against a **verbal order**; if unavoidable in an emergency, it must be **regularised in writing the same day**.
- Never issue to a person who is **not authorised**; obtain signature with name and designation.
- Never issue **expired, near-expiry (without a plan of use), quarantined or recalled** stock.
- Never issue **without posting the records** — unposted issues are the commonest cause of shortage at verification.

4.7 Transfers, Returns and the Imprest (Top-up) System

- **Transfer voucher:** Used when stock moves from one store/institution to another; the transferring store shows it as an **issue**, the receiving store as a **receipt**, both quoting the same voucher number.
- **Material Return Note (MRN):** Unused or excess material returned by a ward to the main store; it is taken back on charge with a receipt entry, and the ward ledger is credited.
- **Return to supplier:** Rejected, damaged, expired or recalled stock returned under an **MRN + non-returnable gate pass + debit note**; the reduction is posted in the stock records.
- **Imprest system:** A ward/department is allotted a **fixed scale (imprest) of each item**. It periodically indents **only the quantity consumed**, so that the stock is topped up back to the imprest level. Also called the **top-up or replenishment system**.

- ▶ **Advantages of imprest:** Ward never runs dry; indenting is simple (consumption = indent); over-stocking at ward level is prevented; audit is easy because the closing stock should always equal the imprest scale.
- ▶ **Floor/ward stock check:** The pharmacist inspects ward stock periodically for expiry, storage condition, unauthorised accumulation and correctness of the ward register.

4.8 Handling of Expiry, Breakage, Loss and Condemnation

4.8.1 Expiry management

1. A **near-expiry statement** is generated **monthly**, listing stock expiring within the next 3–6 months.
2. Colour-coded stickers (**traffic-light system**) are pasted: e.g. *red* = expires within 3 months, *yellow* = 3–6 months, *green* = more than 6 months.
3. Near-expiry stock is **redistributed** to high-consumption wards/institutions or **returned to the supplier** if the purchase order has a buy-back/replacement clause.
4. On expiry the stock is **immediately removed from the shelf**, entered in the **expiry register**, stored in the **separate locked expired-drug area** with a red label, and its quantity written off from the stock register after sanction.
5. Disposal is done as **bio-medical waste (yellow category)** through the authorised common facility, or returned to the manufacturer, and the certificate of destruction is filed.
6. The **value of expiry loss** is reported to management and analysed (wrong forecasting, over-indenting, prescriber change, short-expiry acceptance).

4.8.2 Breakage, pilferage and loss

- Every breakage/loss is recorded **immediately** in the **Breakage and Loss Register** with item, batch, quantity, value, date, cause and name of the person responsible.
- Losses are classified as **normal/unavoidable** (handling breakage, evaporation, leakage — within permissible limits) or **abnormal/avoidable** (negligence, theft).
- Write-off requires **sanction of the competent authority**; abnormal losses may involve **recovery from the person at fault** and disciplinary/criminal action.
- Loss of **narcotics** must be reported to the authority prescribed under the NDPS Rules without delay.
- After sanction, the quantity is deducted from the stock register quoting the sanction number — *never* quietly adjusted.

4.8.3 Condemnation and disposal of unserviceable stores

- Step 1** Items that are **expired, damaged beyond use, obsolete, surplus or beyond economical repair** are listed by the store keeper.
- Step 2** A **Condemnation Board / Survey Committee** (usually 3 members, including a technical officer and an officer not connected with the store) inspects and certifies them.
- Step 3** The board records the **reason, residual value and recommended mode of disposal**.
- Step 4** Sanction of the **competent authority** is obtained as per delegated financial powers.
- Step 5** Disposal is by **auction/sale of scrap, transfer, return to supplier, or destruction** (incineration/deep burial for drugs, as per the Bio-Medical Waste Rules).
- Step 6** The **Condemnation Register** is completed, the stock register credited/deducted, sale proceeds deposited, and the destruction certificate filed.

4.9 Stock Verification in the Main Store

Type	How it is done	Remarks
Perpetual / continuous verification	A few items verified every day by rotation, so that every item is covered at least once (high-value items more often) during the year	Best method; no closure of the store; errors detected early; suits ABC-based frequency (A items monthly, B quarterly, C annually)
Periodic / annual (physical) stock taking	The whole stock counted on a fixed date, usually at the close of the financial year; store may be closed for issues	Statutory/accounting requirement; heavy work load; disrupts supply

Type	How it is done	Remarks
Spot / surprise verification	Sudden check of selected items by a superior officer or a flying squad	Deterrent against pilferage; compulsory for narcotics and high-value items
Verification by an independent verifier / audit party	External stock verifier not connected with custody	Satisfies the principle of internal check

4.9.1 Procedure and treatment of discrepancies

1. The verifier counts the **ground (physical) balance** and records it on the verification sheet **before** looking at the book balance (blind counting is the ideal).
2. The **book balance** is then taken from the bin card and stock ledger and the two compared.
3. Differences are listed in the **shortage/excess (discrepancy) statement**.
4. Obvious causes are investigated first: unposted vouchers, wrong unit of measurement, issue not recorded, receipt posted twice, transfer not accounted, breakage not written off.
5. Genuine **shortages** are regularised by write-off sanction, or recovered from the custodian if negligence is established; **excesses** are taken on charge to prevent their misuse.
6. The custodian and the verifier both **sign the verification certificate**, and a note of verification with date is made in the stock register.
7. A report is placed before management; repeated discrepancies attract change of custodian or a special audit.

f BASIC STOCK EQUATION — **Closing balance = Opening balance + Receipts – Issues – Losses/Write-offs + Returns.**
If the physical count does not satisfy this equation, either a voucher is unposted or stock is missing. This equation is the logic of every stock register page.

4.10 Filing, Indexing and Documentation Systems

System of filing	Basis of arrangement	Typical use in a store
Alphabetical	By name of item or supplier	Supplier files, drug information files
Numerical	By serial number of the document	GRN files, issue voucher bundles, PO files
Code / classification-wise (decimal)	By store code number of the item	Stock ledger folios, bin index
Chronological (date-wise)	By date	Daily receipt register, gate passes
Subject-wise	By topic	Circulars, licences, audit paras
Geographical	By place/institution	Sub-store and peripheral institution files
Vertical / horizontal / lateral filing	Physical method of keeping files	Filing cabinets, box files, guard files
Electronic / digital	Scanned and indexed soft copies	Back-up of GRNs, invoices, test reports

- Each file has a **file number, subject and index of contents**; pages are serially numbered.
- A **guard file** keeps original standing orders, circulars and delegation of powers for ready reference.
- Registers are **bound with machine-numbered pages**, and the first page carries a **certificate of the number of pages** signed by the officer-in-charge.
- A **movement/transit register** is kept for files and registers taken out of the store.
- **Hand-over / take-over (charge) certificate** is prepared whenever the custodian changes, listing verified balances of all items and all registers handed over.

4.11 Security, Confidentiality and Computerisation of Records

- Registers kept under **lock in a fire-resistant almirah**; narcotic and Schedule X registers in the personal custody of a named officer.
- Access restricted on a **need-to-know basis**; no register leaves the store without entry in the movement register.
- Patient-identifiable data in prescription/Schedule H1 registers is **confidential**.

- In computerised stores: **individual user IDs and passwords, role-based rights, no shared logins, automatic audit trail** of every entry and modification, daily **back-up** kept off-site, and **periodic printing and binding** of statutory registers where law requires a physical record.
- The **manual register is still legally required** for narcotics and certain statutory records unless the rules expressly permit electronic records.

4.12 Common Errors and Audit Objections in Store Records

Error / objection	Correct practice
Entries made in pencil, or erased/whitened	Ink only; single-line scoring with attestation
Loose sheets used as stock register	Bound, page-numbered register with page certificate
Bin card balance ≠ ledger balance	Post both at the time of the transaction; reconcile weekly
Issues made without vouchers / verbal issues	Written indent and signed issue voucher for every issue
Batch and expiry not recorded	Batch-wise posting; expiry register maintained
Expired stock lying on the shelf with live stock	Immediate segregation in a locked, labelled area
Goods taken on charge without a GRN / without inspection	GRN + inspection report compulsory
Bill paid for short-supplied quantity; LD not deducted	Three-way matching and penalty computation
Dues-in not deducted while indenting → double ordering	Maintain and consult the pending-order register
Physical verification not done / no verification certificate	Perpetual verification plus annual stock taking with certificate
Narcotic register not countersigned, balance not tallied daily	Daily tally and countersignature by the designated officer
Condemned items lying for years without disposal	Timely board, sanction and disposal
Sub-store stock not accounted after issue from main store	Ward ledgers and periodic ward stock checks
Free/donated drugs not recorded	Separate register; taken on charge like purchases

★ CHAPTER SUMMARY IN FIVE LINES

- Main store = custody + accounting + issue + control of all materials.
- Receipt is authorised by the **GRN**; issue is authorised by the **indent + issue voucher**.
- Quantity records = **bin card (at the bin)** and **stock ledger (in the office)**.
- Issue is always on **FEFO/FIFO**; batch and expiry are recorded at every step.
- Book balance must equal ground balance — proved by **perpetual + annual verification**.

PART III • CLASSIFICATION & CODIFICATION OF STORES

CHAPTER 5 | CLASSIFICATION OF STORES AND DRUG ITEMS

Meaning • need • bases of classification • hospital store groups • classification of drugs • schedules • standardisation

Classification answers the question "what kind of thing is this?"; codification then answers "what shall we call it in our books?". Classification always comes first — you cannot give a rational code number to an item whose group you have not yet decided.

5.1 Meaning, Definition and Need

◆ **CLASSIFICATION** — Classification of stores is the **systematic grouping (sorting) of all items of stores into classes, groups and sub-groups** on the basis of some common characteristic — nature, use, source, value, movement, criticality or storage requirement — so that similar items lie together and can be identified, stored, accounted and controlled uniformly.

5.1.1 Need / objectives of classification

1. To **identify** each item unambiguously and to eliminate confusion between similar items.
2. To make **codification possible** — a code is meaningful only when built on a classification.
3. To group items that need **similar storage conditions** (cold chain, inflammables, narcotics).
4. To permit **selective control** — different inventory-control rules for costly, vital and fast-moving items.
5. To simplify **location, stacking and retrieval** in the store.
6. To standardise **specifications and units of issue**, and thereby reduce the **variety** of items stocked.
7. To help **purchase** by grouping items of the same trade/supplier for combined tendering.
8. To make **accounting, costing and budgeting** group-wise (drugs, surgicals, X-ray, laboratory, stationery).
9. To simplify **stock verification** and **statistical analysis** (consumption of a group over the years).
10. To meet **legal requirements** that certain classes (Schedule H, H1, X, narcotics) be stored and recorded separately.

★ **ONE LINE** — Classification = grouping of similar items. Codification = giving a symbol/number to each item. Standardisation = fixing a specification. Variety reduction (simplification) = reducing the number of varieties of the same item. These four are always examined together as a set of definitions.

5.2 Bases of Classification

Basis	Classes formed	Example
Nature / physical character	Solids, liquids, semi-solids, gases; glass, rubber, metal, plastic, textile	Tablets • syrups • ointments • oxygen
End use / function	Drugs, surgical dressings, laboratory reagents, X-ray, linen, stationery, equipment, engineering spares, POL, diet	The standard hospital store groups
Source of supply	Indigenous, imported, government pool, rate contract, local	Basis of GOLF analysis
Value / cost	High, medium, low unit cost; high, medium, low annual consumption value	Basis of HML and ABC analysis
Criticality / importance	Vital, essential, desirable	Basis of VED analysis
Movement / rate of consumption	Fast, slow, non-moving	Basis of FSN analysis
Availability	Scarce, difficult, easy to obtain	Basis of SDE analysis
Shelf life	Long shelf life, short shelf life, perishable	Vaccines, sera, blood products
Storage requirement	Room temperature, cool, cold chain, frozen, light-sensitive, inflammable, hazardous	Determines the storage area

Basis	Classes formed	Example
Legal status	Schedule H, H1, X, C, C(1), G, narcotics & psychotropics, OTC/general sale	Determines the record and the lock
Therapeutic / pharmacological action	Antibiotics, analgesics, antihypertensives, anti-diabetics ...	Common arrangement in hospital pharmacies
Dosage form / route	Oral solids, orals liquids, injections, externals, ophthalmics, inhalations, suppositories	Commonest physical arrangement in a store
Consumable vs non-consumable	Expendable (drugs, dressings) vs permanent/dead stock (equipment, furniture)	Determines which register is used
Accounting classification	Capital vs revenue; budget head-wise	For financial reporting

5.3 Classification of Hospital Store Items (Functional Groups)

Group	Typical items
Drugs & pharmaceuticals	Tablets, capsules, injections, IV fluids, syrups, ointments, eye/ear drops, inhalers, vaccines & sera
Surgical dressings & disposables	Gauze, bandages, cotton, plaster of Paris, sutures, syringes, needles, catheters, IV sets, gloves, masks
Chemicals & laboratory reagents	Stains, indicators, solvents, diagnostic kits, glassware
X-ray & imaging materials	Films, contrast media, developer, fixer, cassettes
Instruments & equipment (dead stock)	Surgical instruments, BP apparatus, autoclaves, monitors, refrigerators
Linen & clothing	Bed sheets, draw sheets, OT gowns, uniforms, mackintosh
Furniture & fixtures	Beds, lockers, trolleys, almirahs, chairs
Stationery & printing	Registers, case sheets, forms, prescription pads, computer consumables
Sanitary & housekeeping	Disinfectants, phenyl, bleaching powder, brooms, detergents, waste bags
Engineering / maintenance spares & POL	Electrical fittings, plumbing spares, oils, lubricants, generator fuel
Medical gases	Oxygen, nitrous oxide, carbon dioxide, medical air, nitrogen
Diet / provisions	Rations for the hospital kitchen
Narcotics & psychotropics	Morphine, pethidine, fentanyl, diazepam, pentazocine, buprenorphine
Cold-chain items	Vaccines, sera, insulin, oxytocin, blood products, some antibiotic eye drops
Hazardous items	Inflammables (ether, spirit), corrosives (acids), cytotoxics, radiopharmaceuticals

5.4 Classification of Drugs

5.4.1 On the basis of therapeutic / pharmacological action

e.g. analgesics & antipyretics, NSAIDs, antibiotics (further sub-grouped as penicillins, cephalosporins, aminoglycosides, macrolides, quinolones...), antitubercular, antimalarial, antifungal, antiviral, antihypertensives, diuretics, cardiac drugs, antidiabetics, hormones, vitamins & minerals, antihistamines, corticosteroids, anaesthetics, psychotropics, anticonvulsants, gastro-intestinal drugs, respiratory drugs, oxytocics, vaccines & sera, antiseptics & disinfectants, IV fluids and electrolytes, anticancer drugs.

5.4.2 On the basis of dosage form

Class	Forms
Oral solids	Tablets, capsules, powders, granules, lozenges
Oral liquids	Syrups, elixirs, suspensions, emulsions, drops, linctus
Parenterals (injections)	Ampoules, vials, pre-filled syringes, large-volume IV fluids
Externals / topicals	Ointments, creams, pastes, gels, lotions, liniments, powders
Ophthalmic / otic / nasal	Eye drops & ointments, ear drops, nasal sprays
Rectal & vaginal	Suppositories, pessaries, enemas
Inhalations / respiratory	MDI, DPI, nebuliser solutions, medical gases

Class	Forms
Transdermal & implants	Patches, implants

5.4.3 On the basis of legal status — Schedules of the Drugs and Cosmetics Rules, 1945

Schedule	Subject matter (as relevant to a store)
Schedule C and C(1)	Biological and special products — sera, vaccines, toxoids, antitoxins, insulin, pituitary/other hormones, sterile products and other special drugs; require licence in Form 21/21B and generally cold storage
Schedule D	Classes of drugs exempted from import provisions
Schedule F / F(1)	Requirements for blood banks, blood components and for vaccines & sera
Schedule F(II)	Standards for surgical dressings
Schedule F(III)	Standards for umbilical tapes
Schedule G	Drugs to be used only under medical supervision — label carries the caution " <i>Caution: it is dangerous to take this preparation except under medical supervision</i> " (e.g. many antineoplastics)
Schedule H	Prescription drugs — sold only on the prescription of a registered medical practitioner; label bears " Rx " and the warning " <i>Schedule H drug — Warning: to be sold by retail on the prescription of a Registered Medical Practitioner only</i> "
Schedule H1	Specified 3rd/4th-generation antibiotics, anti-TB drugs and certain psychotropics ; label bears a red vertical line on the left border with the H1 warning; a separate register of sales must be kept
Schedule J	Diseases & conditions for which no drug may claim to cure (advertisement prohibition)
Schedule K	Classes of drugs exempted from certain provisions (e.g. household remedies, supply by registered practitioners, drugs supplied by a hospital to its own patients)
Schedule M	GMP — requirements of premises, plant, equipment and of records & documentation for manufacturers
Schedule M(III)	Requirements for medical devices
Schedule N	Minimum equipment for the efficient running of a pharmacy
Schedule P	Life periods (shelf life) and storage conditions of drugs
Schedule P(1)	Pack sizes of drugs
Schedule Q	Permitted colours/dyes in drugs & cosmetics
Schedule V	Standards for patent and proprietary medicines
Schedule W	Drugs marketable only under generic names
Schedule X	Habit-forming / narcotic & psychotropic type drugs (e.g. barbiturates, amphetamine, methaqualone-type substances) — require special licence (Forms 20F/20G on application in Form 19C) , storage in a separate locked cupboard , prescription in duplicate , and a separate register
Schedule Y	Requirements for clinical trials and import/manufacture of new drugs

★ **SCHEDULE P — REMEMBER THE NAME** — Schedule P of the Drugs and Cosmetics Rules prescribes the ***life period (shelf life)*** and the ***conditions of storage*** of drugs — the single most directly relevant schedule for the topic "methods of storing drugs". **Schedule N** = minimum equipment for a pharmacy; **Schedule M** = GMP and records.

5.4.4 Other classifications of drugs

- **By source** — natural (plant, animal, mineral, microbial), synthetic, semi-synthetic, biotechnological.
- **By system of medicine** — allopathic (modern), ayurvedic, siddha, unani, homoeopathic (stored **separately**, with separate licences and registers).
- **By essentiality** — NLEM/EML drugs vs non-essential; hospital formulary vs non-formulary.
- **By price control** — scheduled formulations (ceiling price under DPCO) vs non-scheduled.
- **By risk in handling** — LASA (look-alike sound-alike), high-alert/high-risk drugs (insulin, heparin, concentrated electrolytes, narcotics, chemotherapy), ordinary drugs.
- **Alphabetical** — simple A-Z arrangement of generic (or brand) names; commonest in small pharmacies.

5.5 Standardisation, Simplification and Variety Reduction

- ▶ **Standardisation:** Fixing a **definite specification, size, quality and unit** for each item to be stocked (e.g. only 5 ml disposable syringes of one make and specification).
- ▶ **Simplification / variety reduction:** Reducing the **number of varieties, brands, strengths and pack sizes** of the same item — the work of the hospital formulary and the Pharmacy & Therapeutics Committee.
- ▶ **Benefits:** Fewer items to store and record • larger quantity per item → better price • less capital blocked • less obsolescence • easier codification and stock verification • fewer dispensing errors.
- ▶ **Link with codification:** Standardisation and variety reduction must be done **before** codification; otherwise the code list is bloated with duplicate items under different brand names.

✓ **SEQUENCE TO REMEMBER** — **Classification → Standardisation / variety reduction → Codification → Location (bin) allotment → Stock records.** Examiners often ask which step comes first: *classification*.

CHAPTER 6 | CODIFICATION OF STORES ITEMS

Meaning • objectives • essentials of a good code • methods • Brisch & Kodak systems • bar codes • colour codes

A code is the name of an item in the language of the store. It replaces a long, ambiguous description ("Injection Ceftriaxone 1 g vial, 10's pack") with a short, unique symbol that a clerk, a computer and an auditor all understand identically.

6.1 Meaning and Definition

◆ **CODIFICATION** — Codification (coding) is the **process of assigning a systematic, unique symbol — numbers, letters or a combination of both — to every item of stores**, so that the item can be identified, described, located, recorded and communicated briefly and without ambiguity.

- ▶ **Code / code number:** The symbol itself, e.g. 02-14-0357.
- ▶ **Vocabulary of stores / stores catalogue:** The master printed or electronic list showing every code number with its full description, unit of issue and location. Also called the **item master**.
- ▶ **Nomenclature:** The standard *naming* of the item; codification is the standard *numbering* of it.

6.2 Objectives and Advantages of Codification

1. **Unique identification** of every item — eliminates confusion between similar items (different strengths, pack sizes, brands).
2. **Brevity and accuracy** in all documents — indents, POs, GRNs, vouchers, ledgers.
3. **Avoidance of duplication** — the same item cannot be stocked twice under two descriptions, so duplicate purchase is prevented.
4. **Secrecy** — the nature/value of the item is not obvious to outsiders.
5. **Ease of location** — the code can be linked to the rack/bin address.
6. **Mechanisation / computerisation** — a computer needs a unique key field; the code is that key.
7. **Standardisation and variety reduction** are exposed by coding (duplicates become visible).
8. **Simplified pricing, costing and accounting** — group codes roll up into budget heads.
9. **Faster stock verification** and posting; fewer clerical errors.
10. **Better inventory analysis** — group-wise consumption and ABC/VED analysis become easy.
11. **Language independence** — useful where staff speak different languages.

6.2.1 Limitations / disadvantages

- Initial **cost, time and expertise** required to design the system and code thousands of items.
- Staff need **training**; a wrong digit means a wrong item (**transposition errors**).
- A badly designed (rigid) code becomes **inadequate when new items appear**.
- Purely numeric codes are **difficult to memorise**; very long codes are error-prone.
- Requires **continuous maintenance** of the item master (additions, deletions, mergers).

6.3 Essentials (Requirements) of a Good Coding System

Essential	Meaning
Uniqueness	One code for one item and one item for one code — never two codes for the same item
Exhaustiveness	Every item in the store must be codable; no item left out
Mutual exclusiveness	An item must fall in one group only, with no overlap
Brevity / concise	As short as possible while remaining unique (usually 6–10 characters)
Flexibility / expansibility	Room for new items, new groups and new technologies without redesigning the whole system
Simplicity / ease of use	Easy to read, write, remember and dictate; avoid confusing characters (I/1, O/0)

Essential	Meaning
Mnemonic value	The code itself should suggest the item wherever possible (e.g. TAB for tablet, INJ for injection)
Logical / hierarchical structure	Digits should progress from the general group to the specific item
Uniformity	Same length and same structure for all codes; same code used by every department
Compatibility	Should fit the accounting heads, the computer system and, ideally, national standards (e.g. GS1 bar codes)
Documented & centrally controlled	Only one authority may allot codes; the item master is the sole authentic list

♪ **MNEMONIC — A GOOD CODE IS "U-B-F-S-M-E"** — **U**nique, **B**rief, **F**lexible, **S**imple, **M**nemonic, **E**xhaustive.

6.4 Methods / Systems of Codification

Method	How it works	Example	Comment
Alphabetical (mnemonic) system	Letters, usually the initials of the item name, form the code	PCM = paracetamol; CTX = ceftriaxone; COT = cotton	Easy to remember; but limited combinations, confusing for similar names
Numerical (numeric) system	Plain serial numbers allotted to items, either sequentially or in blocks reserved for groups	0001, 0002 ... or block 1000–1999 = tablets	Simple and unlimited; but the number carries no meaning
Block / group numerical system	Blocks of numbers reserved for each group and sub-group	10–19 = drugs, 20–29 = surgicals	Adds group meaning to a numeric code; widely used
Alpha-numerical (combined) system	Letters denote the group and numbers the individual item	TAB-0125, INJ-0341, SUR-0088	Most popular — combines mnemonic value with unlimited numbering
Decimal system (Dewey decimal)	Each digit after the decimal point denotes a finer sub-division; based on the library classification of Melvil Dewey	61.24.031	Highly logical and expandable; codes may become long
Universal Decimal Classification (UDC)	International extension of the Dewey system	615.4	Used in technical libraries and large depots
Brisch system	Seven-digit, monocode, applied in three stages	See 6.5	Hierarchical, British origin
Kodak system	Ten-digit numeric code, source-of-supply oriented	See 6.5	Developed by Eastman Kodak Co., USA
Colour coding	Colours (or coloured labels/stickers) identify a group, hazard, year of receipt or expiry status	Gas cylinders; BMW bins; red/yellow/green expiry stickers	Instantly visible; limited number of distinguishable colours
Bar coding / QR / 2-D DataMatrix	Machine-readable representation of the code printed on the pack	GS1 GTIN + batch + expiry	Fast, error-free data capture; the modern standard
RFID tagging	Radio-frequency tag read without line of sight	Tagged cold-chain boxes, equipment	Enables real-time tracking and temperature logging
Arbitrary / random system	Numbers allotted arbitrarily as items arrive	—	Simple but unstructured; not recommended for large stores

✓ **MONOCODE VS POLYCODE** — **Monocode (hierarchical code)** — each digit's meaning **depends on the digits before it** (like Dewey, Brisch); compact and good for retrieval of groups. **Polycode (attribute/multi-code)** — each digit position denotes an **independent attribute** (size, material, colour); better for describing items fully. Many practical systems are **hybrid**.

6.5 Standard Codification Systems — Brisch and Kodak

Feature	Brisch system	Kodak system
Origin	Developed in the United Kingdom by E. G. Brisch (a consulting engineer)	Developed by Eastman Kodak Company, New York, USA
Number of digits	7 digits (all numeric)	10 digits (all numeric)
Structure	A monocode — hierarchical; applied in 3 stages : (1) items grouped into broad preliminary categories (assemblies, sub-assemblies, components, off-the-shelf/standard items, raw material); (2) grouping within each category on the basis of function, material, type; (3) allotment of the final code number to the individual item	Digits are grouped 3 + 4 + 3 ; the first digits indicate the source/mode of procurement , the rest indicate the classification by material, function and use; materials are spread over 100 main classifications
Chief emphasis	Logical, hierarchical grouping of similar items so that every item has a meaningful and unique identity	Procurement orientation — how and from where the item is bought
Suitability	Engineering stores and large organisations with many components	Large organisations with wide, varied purchasing

6.5.1 Steps in developing a codification system

- Step 1** Prepare a **complete inventory list** of all items actually stocked.
- Step 2** Carry out **standardisation and variety reduction** — delete duplicates and obsolete varieties.
- Step 3** Decide the **classification** (groups and sub-groups) and the number of levels required.
- Step 4** Decide the **type of code** (alpha-numeric, decimal, etc.) and the **number of digits**, keeping room for expansion.
- Step 5** Allot **group and sub-group digits**, then **serial numbers** to individual items.
- Step 6** Prepare the **stores vocabulary / item master** with code, full description, unit of issue, storage condition and bin location.
- Step 7** Mark the code on **bins, racks, bin cards, ledger folios and all forms**.
- Step 8** Train the staff, and centralise the authority to **create or amend a code**.
- Step 9** **Review periodically** — add new items, delete obsolete codes, never re-use a deleted code.

6.5.2 A worked example of building a drug code

Table 6.1 Structure of a hierarchical (monocode) drug code

Digit position	Meaning	Example value
1-2 (Group)	Main group of stores	01 = drugs, 02 = surgicals, 03 = laboratory, 04 = X-ray
3-4 (Sub-group)	Dosage form / category within the group	01 = tablets, 02 = capsules, 03 = injections, 04 = orals liquids
5-6 (Therapeutic class)	Pharmacological class	07 = antibiotics, 12 = analgesics
7-9 (Item serial)	Individual drug	045 = ceftriaxone
10 (Variant)	Strength / pack variant	2 = 1 g vial

- **READING THE CODE** — The code **`01-03-07-045-2`** therefore reads: *drug* → *injection* → *antibiotic* → *ceftriaxone* → *1 g vial*. Notice that the code tells the store keeper where to put it, the pharmacist what it is, and the accountant which budget head to debit.

6.6 Bar Coding and Modern Identification Technology

Technology	Key points
Linear (1-D) bar code	Parallel bars of varying width read by a laser scanner; holds a single number such as the GS1 GTIN (Global Trade Item Number, commonly a 13-digit EAN); printed on retail medicine packs

Technology	Key points
2-D codes (DataMatrix, QR)	Store much more data in a small area — GTIN plus batch number, expiry date and serial number ; the basis of track-and-trace / anti-counterfeiting systems
GS1 application identifiers	(01) GTIN, (10) batch/lot, (17) expiry date, (21) serial number — the standard used for pharmaceutical traceability
RFID	Radio-frequency identification tag; no line of sight needed; can be read in bulk and can log temperature — used for cold-chain boxes, high-value implants and equipment
Advantages of bar coding in a store	Very fast data entry • near-zero transcription error • instant batch and expiry capture • automatic FEFO enforcement • instant recall of a batch • real-time stock position • easy physical verification with hand-held scanners
Limitations	Cost of scanners & printers • damaged/dirty labels do not scan • needs power & software • staff training • loose/repacked items must be re-labelled
Indian context	Bar coding/track-and-trace is mandatory on exported pharmaceutical packs, and QR codes are required on the packs of specified formulations to let a buyer verify authenticity

6.7 Colour Coding — A Special Form of Codification

6.7.1 Medical gas cylinders

Table 6.2 Colour coding of medical gas cylinders (Indian practice differs from the US code — read the question carefully)

Gas	Indian standard (IS) colour	US (compressed gas) colour
Oxygen	Black body with white shoulder	Green
Nitrous oxide	Blue	Blue
Carbon dioxide	Grey	Grey
Medical air	Grey body, white & black quartered shoulders	Yellow
Nitrogen	Grey body, black shoulder	Black
Helium	Brown	Brown
Entonox (50 % O ₂ + 50 % N ₂ O)	Blue body, blue & white quartered shoulders	Blue/green quartered
Cyclopropane	Orange	Orange
Ethylene	Violet	Red
Vacuum (suction) pipeline	Yellow	White

⚠ EXAM TRAP — In India (IS 3933 / hospital practice) oxygen is **BLACK** with a **WHITE** shoulder, whereas in the **USA** oxygen cylinders are **GREEN**. If the question says "as per Indian standard", answer *black body, white shoulder*.

6.7.2 Bio-medical waste colour coding (BMW Rules, 2016)

Colour of container/bag	What goes in it	Treatment & disposal
Yellow	Human & animal anatomical waste; soiled waste; expired, discarded and contaminated medicines ; chemical waste; cytotoxic waste (in a separate yellow cytotoxic bag); microbiology & laboratory waste	Incineration / plasma pyrolysis / deep burial; expired medicines may also be returned to the manufacturer; cytotoxic waste incinerated at high temperature
Red	Contaminated recyclable plastic — tubing, bottles, IV sets, catheters, urine bags, syringes without needles , gloves	Autoclaving / microwaving / hydroclaving, then shredding and recycling
White (translucent, puncture-proof)	Sharps — needles, syringes with fixed needles, scalpels, blades	Autoclave / dry heat sterilisation, then shredding or encapsulation
Blue (puncture-proof box/bag)	Broken or discarded glassware including medicine vials & ampoules; metallic body implants	Disinfection or autoclaving, then recycling

6.7.3 Other colour conventions used in stores

■ Quarantine yellow / approved green / rejected red labels on consignments.

- **Expiry traffic-light stickers** — red (expires ≤ 3 months), yellow (3–6 months), green (> 6 months).
- **LASA (look-alike/sound-alike)** drugs given contrasting coloured labels and shelf separation.
- **High-alert drug labels** (e.g. bright orange for concentrated potassium chloride).
- **Year-of-receipt colour dots** — a quick visual FIFO aid on cartons.
- **Fire-safety and hazard labels** — inflammable, corrosive, oxidising, cytotoxic, radioactive (trefoil) symbols.

★ **CHAPTER 6 IN ONE BOX**

- Codification = allotting a **unique symbol** to each item.
- **Brisch = 7 digits, 3 stages (UK); Kodak = 10 digits, 3+4+3, source-of-supply oriented (USA); Decimal = Dewey.**
- Most widely used practical system = **alpha-numerical**.
- A good code is unique, brief, flexible, simple, mnemonic and exhaustive.
- The master list of all codes = **stores vocabulary / item master**.
- India: oxygen cylinder = **black body, white shoulder**; expired medicines = **yellow** BMW bag; sharps = **white** container.

PART IV • STOCK BOOKS & INVENTORY CONTROL

CHAPTER 7 | KEEPING OF STOCK BOOKS

Stock register • bin card • stock ledger • kardex • posting rules • types of stock books • verification • valuation

"Keeping of stock books" is the examiner's way of asking: which books, what columns, who posts them, how are they posted, and how are they proved correct? Answer in that order and nothing can be missed.

7.1 Stock Book — Meaning and Necessity

◆ **STOCK BOOK** — A stock book (stock register) is the permanent, bound book of account of materials in which, for every item separately, the opening balance, all receipts, all issues and the resulting balance are recorded in chronological order, with a reference to the voucher authorising each transaction.

- It is the **quantitative account** of stores — the equivalent of a cash book for money.
- It is maintained **item-wise (one folio/page per item)**, not date-wise.
- For drugs it must also carry **batch number and expiry date**, because two units of the same drug are not interchangeable if their batches differ.
- It is a **legally inspectable document** — a Drug Inspector, auditor or accreditation assessor may demand it at any time.
- The **closing balance of the stock book must equal the physical stock on the shelf** at every moment; that equality is the whole purpose of the book.

f **THE STOCK BOOK EQUATION** — Closing balance = Opening balance + Total receipts – Total issues – Losses/write-offs + Returns received back

7.2 Bin Card (Stock Card / Bin Tag)

◆ **BIN CARD** — A bin card is a card (or tag) attached to the bin, rack or shelf in which the material is kept, on which the store keeper records, at the time of every transaction, the quantity received, the quantity issued and the resulting balance of that item.

7.2.1 Contents of a bin card

Part of the card	Particulars
Heading (fixed information)	Name & full description of the item • store code number • unit of issue • bin/rack/shelf number • storage condition • minimum, maximum and re-order levels • lead time
Body (transaction columns)	Date • reference/voucher number (GRN no. or issue voucher no.) • quantity received • quantity issued • balance • batch number • expiry date • initials of the store keeper
Foot / remarks	Date of physical verification and verifier's initials; remarks on shortage, breakage, near expiry

Format 7.1 Bin card (heading carries item, code, unit, bin no., min/max/re-order levels)

Date	GRN / Issue Voucher No.	Batch No.	Expiry	Received	Issued	Balance	Initials

7.2.2 Advantages of a bin card

- The **balance is available at the point of storage**, without going to the office.
- Acts as a **ready check at the time of issue** — physical count can be compared instantly.
- Shows **minimum, maximum and re-order levels**, so the need to indent is noticed automatically.

- Provides an **independent second record** which, when tallied with the stock ledger, detects errors and fraud (**two-point recording**).
- Helps the stock verifier to compare book and ground balance on the spot.
- Simple, cheap and needs no accounting knowledge.

7.2.3 Limitations

- Shows **quantity only** (no value), so it cannot be used for costing.
- Cards can be **soiled, torn, lost or manipulated** as they lie in the open store.
- Many cards mean **repetitive writing**; errors of posting are possible.
- Does not show the **outstanding orders** unless a special column is provided.

7.3 Stock Ledger (Stores Ledger)

◆ **STORES LEDGER** — The **stores (stock) ledger** is the book maintained in the **store office / accounts section**, one folio per item, in which every receipt and issue is posted **from the vouchers (GRN, issue voucher, transfer voucher)**, showing quantity and — in a **priced stores ledger** — also rate and value of receipts, issues and closing balance.

Group of columns	Particulars
Item identification	Name, code number, unit, folio number, storage condition, min/max/re-order level
Receipts	Date • GRN no. • supplier • PO no. • batch • expiry • quantity • rate • value
Issues	Date • issue voucher no. • indenting department • batch • quantity • rate • value
Balance	Quantity • rate • value • (and in some formats, quantity on order = dues-in)
Verification	Date of physical verification, ground balance found, discrepancy, signature

7.3.1 Bin card vs Stores ledger — the classic comparison

Point of difference	Bin card	Stores ledger
Where kept	With the material, on/near the bin	In the store office / accounts section
Who posts it	Store keeper	Ledger clerk / accounts clerk
When posted	At the time of the transaction (immediately)	Later, from the vouchers (may be a day or two behind)
What it records	Quantity only	Quantity and value (priced ledger)
Basis of entry	The physical movement of material itself	The document/voucher evidencing the movement
Form	A loose card or tag	A bound register or a set of ledger folios/computer records
Individual transactions	Each transaction posted separately	Transactions may be posted in summary
Main use	Physical control, ready reference at the bin, re-ordering	Accounting, costing, valuation, MIS
Legal/audit standing	Working record	Primary book of account for audit

✓ **REMEMBER THE THREE-WORD ANSWERS** — *Bin card* → **store keeper, quantity, at the bin**. *Stores ledger* → **ledger clerk, quantity + value, in the office**. Both must be **reconciled periodically**; a difference means an unposted or wrongly posted voucher.

7.4 Kardex, Two-Bin and Location Systems

- ▶ **Kardex / visible index card system**: Cards for all items kept in a **shallow tray or vertical cabinet so that the top line (item name, code and balance) of every card is visible at a glance**; coloured signal clips are attached when stock reaches the re-order level. Excellent for quick review of many items.
- ▶ **Two-bin system**: The stock of an item is kept in **two bins**. Issues are made from the first bin; **the moment the first bin is empty, an order is placed**, and the second bin (which holds the re-order quantity + safety stock)

carries the store through the lead time. Simple, visual, needs little paper work; suited to **low-value C-class items**.

- ▶ **Fixed location system:** Every item has a **permanent bin address**; easy to memorise and to locate, but space lies idle when the item is out of stock.
- ▶ **Random / floating location system:** An item is placed **wherever space is available** and the location is recorded in the **location index / computer**; gives maximum space utilisation but depends totally on accurate records.
- ▶ **Zoned / semi-fixed location:** A **zone** is fixed for a group (e.g. all injections in zone C) but the exact bin within the zone is flexible — the practical compromise used in most hospital stores.
- ▶ **Bin/location index:** The register (or computer table) that translates **code number → rack/shelf/bin address**; without it a random-location store becomes unusable.

7.5 Rules and Procedure for Posting Stock Books

1. Use a **bound register with machine-numbered pages**; record on the first page a **certificate of the number of pages**, signed and dated by the officer-in-charge.
2. Allot a **separate folio (page) to each item**, writing its full description, code number and unit of issue at the top; maintain an **index of items with folio numbers** at the beginning of the register.
3. Post **in ink, on the day of the transaction**, in chronological order.
4. Every entry must quote its **authority** — GRN number for a receipt, issue voucher number for an issue, sanction number for a write-off.
5. Post the **batch number and expiry date** for every drug receipt and issue.
6. Work out the **balance after every transaction**; do not leave the balance column blank.
7. **Total and carry forward** at the end of each page and each month; write "Carried over to page and "Brought forward from page ".
8. A wrong entry is **scored out with a single horizontal line**, the correct figure written above it, and the correction **initialled with date** by the authorised officer. **No erasing, no overwriting, no whitener, no page torn out.**
9. Leave **no blank lines or blank pages** between entries; cancel unused space by a diagonal line.
10. Record **physical verification** on the folio itself: date, ground balance, discrepancy and the verifier's signature.
11. **Reconcile bin card with ledger** at fixed intervals (weekly/monthly) and record the reconciliation.
12. Close the register at the end of the financial year, carry the closing balances forward as the opening balances of the new register, and get them **certified**.

⚠ WHAT MAKES A STOCK BOOK WORTHLESS IN AUDIT

- Pencil entries, erasures, overwriting, whitener.
- Loose sheets, missing pages, un-numbered pages.
- Entries without voucher references.
- Balances not struck after each transaction.
- Corrections not attested.
- No record of physical verification.
- Batch and expiry not recorded for drugs.

7.6 Types of Stock Books Maintained in a Drug Store

Stock book	What it covers / special features
Drug (consumable) stock register	All medicines; item-wise folios with batch and expiry columns
Surgical & dressing stock register	Gauze, bandages, sutures, disposables
Chemical / laboratory reagent register	With hazard and shelf-life remarks
Dead stock (non-consumable / permanent) register	Equipment, instruments, furniture; columns for identification/serial number, date of purchase, cost, warranty, location, condition and date of annual verification ; items are never "consumed" — they leave the register only on transfer, loss or condemnation

Stock book	What it covers / special features
Narcotic / NDPS (Dangerous Drugs) register	Separate page per drug; daily balance; kept under lock and countersigned ; entries in ink, no corrections without attestation; stock physically tallied daily/at every shift change
Schedule X register	Purchase and issue of habit-forming drugs; kept for the prescribed period; prescriptions retained
Schedule H / H1 register	Sale/issue of prescription drugs; the H1 register records patient name & address, prescriber, drug, quantity and date
Spirit / alcohol (bonded) register	Receipt, issue and permissible wastage of rectified spirit under excise control
Batch and expiry register	Batch-wise quantity and expiry date; drives FEFO and the near-expiry statement
Expiry / near-expiry register	Items expiring within 3–6 months and action taken (redistribution, return, destruction)
Breakage, damage and loss register	Quantity, value, cause, responsibility, write-off sanction
Condemnation / disposal register	Board proceedings, mode of disposal, sale proceeds, destruction certificate
Free supply / donation / sample register	Drugs received without payment
Temperature & humidity log	Store, refrigerator, ILR and deep-freezer readings twice daily
Equipment maintenance & breakdown register	For refrigerators, ILRs, generators, autoclaves
Ward / sub-store (departmental) stock register	Imprest scale, receipts from main store, consumption, balance
Stock verification register	Date, items verified, book vs ground balance, discrepancy, certificate

7.6.1 Consumable vs non-consumable (dead) stock

Point	Consumable (expendable) stock	Non-consumable (dead/permanent) stock
Nature	Used up in a single use or a short period	Used repeatedly over years
Examples	Drugs, dressings, chemicals, stationery, X-ray film	Autoclave, refrigerator, BP apparatus, furniture, surgical instruments
Register used	Consumable stores register (with batch/expiry)	Dead stock register (with identification number, cost, warranty)
Accounting head	Revenue expenditure	Capital expenditure
Removal from books	By issue/consumption	Only by transfer, loss or condemnation with sanction
Verification	Perpetual + annual quantity verification	Annual physical verification of every item with condition noted

7.7 Physical Verification of Stock

(The types — perpetual, periodic, spot, independent — were introduced in Chapter 4. Here are the points most often asked in the context of stock books.)

Aspect	Key points
Purpose	To prove that the book balance equals the ground balance , to detect pilferage, deterioration, expiry, wrong posting and obsolete stock
Who should verify	A person independent of the custodian ; the custodian must be present during counting
Frequency (ABC-linked)	A items — monthly/quarterly; B items — half-yearly; C items — annually. Narcotics — daily/shift-wise tally and surprise checks
Method	Count/weigh/measure physically; do not accept the custodian's figure; "blind" verification (ground balance recorded before seeing the book balance) is ideal
Record	Verification sheet, discrepancy (shortage/excess) statement , and a verification certificate signed by verifier and custodian; note the date of verification on the bin card and ledger folio
Discrepancy — first check	Unposted vouchers, wrong unit of measurement, transfers not accounted, breakage not written off, double posting, arithmetical error
Shortage	Regularised by write-off sanction of the competent authority; recovery from the custodian if negligence is proved; criminal action for theft, and immediate reporting for narcotics
Excess	Taken on charge (credited to stock) so that it cannot be misused

Aspect	Key points
Hand-over of charge	On transfer of the custodian a full physical verification is done and a charge (hand-over/take-over) certificate signed by both officers

7.8 Valuation of Stock (Pricing of Issues)

Method	How the issue is priced	Effect / remark
FIFO (First In First Out)	Issues priced at the rate of the oldest stock	Closing stock valued at latest (current) prices; matches the physical FIFO practice in a drug store
LIFO (Last In First Out)	Issues priced at the rate of the latest purchase	Closing stock valued at old prices; not suitable for drugs because physically the oldest stock must go out first
Simple average	Average of the rates of the lots in stock	Ignores quantity; rarely used
Weighted average	(Total value of stock) ÷ (total quantity in stock), recomputed after every receipt	Smooths out price fluctuations; widely used in computerised systems
Standard price	A pre-determined rate fixed for the year	Simple, good for budgeting; differences go to a price-variance account
Market / replacement price	Current market rate	Used for insurance and for valuing condemned stock
Last purchase price	Rate of the most recent purchase	Simplest; commonly used in government hospital stores for valuing closing stock

★ **FIFO VS FEFO — DO NOT CONFUSE** — **FIFO = First In, First Out** (the *oldest received* stock is issued first). **FEFO = First Expiry, First Out** (the stock with the *earliest expiry date* is issued first). For drugs, **FEFO is superior and is the recommended practice**, because a later purchase may carry an earlier expiry date.

7.9 Performance Indicators Derived from Stock Books

Indicator	Formula / definition	What it tells
Inventory (stock) turnover ratio	Annual consumption value ÷ Average inventory value held	How many times the stock is used up in a year; higher = more efficient . Average inventory = (opening + closing) ÷ 2
Months / days of stock in hand	Closing stock ÷ Average monthly (or daily) consumption	How long the present stock will last
Stock-out rate	(No. of items out of stock ÷ total items) × 100, or stock-out days per item	Service level of the store; should be near zero for vital items
Expiry / wastage rate	(Value of drugs expired or damaged ÷ total value of drugs consumed) × 100	Efficiency of forecasting and FEFO discipline
Inventory carrying (holding) cost	Interest on capital + storage + insurance + handling + obsolescence + expiry; commonly taken as about 20–25 % of the average inventory value per year	Why over-stocking is expensive
Lead time	Time from raising the indent to the material being available for issue	Determines the re-order level
Order frequency	Annual consumption ÷ EOQ	Number of orders per year
Fill rate	(Quantity issued ÷ quantity indented) × 100	Ability of the store to satisfy demand
Percentage of non-moving items	(No. of items with no issue in 12 months ÷ total items) × 100	Dead stock and obsolescence

7.10 Computerised Stock Keeping

Aspect	Manual system	Computerised system
Speed & effort	Slow; heavy writing work	Instant posting and retrieval
Accuracy	Prone to arithmetical & posting errors	Balances computed automatically

Aspect	Manual system	Computerised system
Batch/expiry control	Depends on the clerk's vigilance	Automatic near-expiry alerts and FEFO enforcement
Re-ordering	Manual watch on re-order level	Automatic re-order level alert and draft indent generation
Reports/MIS	Prepared laboriously by hand	ABC/VED analysis, consumption trends, expiry loss at a keystroke
Security	Physical lock	User ID, password, role-based rights, audit trail
Risk	Loss/damage of registers	Data loss, power/software failure → needs daily back-up and a manual fallback
Legal position	Accepted everywhere	Acceptable where rules permit; statutory registers (e.g. narcotics) may still have to be maintained/printed in physical form

- **DATA INTEGRITY — "ALCOA"** — Records (paper or electronic) must be **Attributable, Legible, Contemporaneous, Original and Accurate**. Modern additions ("ALCOA+") are **Complete, Consistent, Enduring and Available**. This principle is increasingly examined in pharmacy papers.

CHAPTER 8 | INVENTORY CONTROL TECHNIQUES

Because an indent cannot be prepared without them — ABC, VED, FSN, HML, XYZ, SDE, GOLF, SOS • EOQ • stock levels • lead time

Inventory control is the science of deciding **how much to order and when to order**. Chapter 9 (preparation of indents) is nothing but the application of the formulae in this chapter, so learn the formulae here thoroughly — they are pure-mark questions.

8.1 Inventory — Meaning, Types and Costs

◆ **INVENTORY** — Inventory means the **stock of any item or resource held to meet a future demand** — in a hospital, the drugs, surgicals, reagents, spares and provisions lying in the main store, sub-stores and wards. **Inventory control** is the technique of maintaining stock at the **desired level** — enough to serve patients, small enough not to waste money.

8.1.1 Objectives of inventory control

- Uninterrupted availability — **no stock-out of vital drugs**.
- Minimum **investment** locked up in stock.
- Minimum **expiry, deterioration and obsolescence** loss.
- Lowest total of **ordering cost + carrying cost**.
- Reduction in **emergency/local purchases** (which are always costlier).
- Optimum use of **storage space and manpower**.
- Reliable information for **budgeting and purchase planning**.

8.1.2 Costs associated with inventory

Cost	Components	Behaviour
Ordering (procurement) cost	Tendering, correspondence, inspection, transport, bill processing — <i>cost per order</i>	Falls as the order quantity rises (fewer orders per year)
Carrying (holding) cost	Interest on blocked capital, rent, electricity & refrigeration, insurance, salaries, pilferage, expiry & obsolescence	Rises as the order quantity rises; usually 20–25 % of average inventory value per annum
Stock-out (shortage) cost	Emergency purchase at a higher rate, patient harm, loss of reputation, loss of revenue	Rises as stock is cut too fine
Capital / opportunity cost	Return foregone on the money tied up in stock	Proportional to inventory value

✓ **EOQ IS THE BALANCE POINT** — The **Economic Order Quantity** is that order size at which the **total ordering cost equals the total carrying cost**, and therefore the **total inventory cost is minimum**.

8.2 Selective Inventory Control Techniques

No store can control 10,000 items with equal attention. **Selective control** means applying tight control to the few items that matter most. Each technique classifies items on a *different criterion* — that criterion is what the MCQ tests.

Technique	Criterion of classification	Classes and their meaning
ABC analysis	Annual consumption VALUE ($\approx$ Pareto / 80:20 principle)	A $\approx$ 10 % of items $\rightarrow \approx$ 70 % of value; B $\approx$ 20 % of items $\rightarrow \approx$ 20 % of value; C $\approx$ 70 % of items $\rightarrow \approx$ 10 % of value
VED analysis	Criticality of the item for patient care / functioning	V = Vital (life-saving; stock-out not permissible), E = Essential (serious inconvenience if out of stock), D = Desirable (minor effect)
VEN analysis	Same idea, WHO terminology	Vital, Essential, Non-essential
HML analysis	Unit price of the item (not annual value)	High, Medium, Low cost per unit — used to control pilferage-prone costly items

Technique	Criterion of classification	Classes and their meaning
XYZ analysis	Value of the stock actually lying in store (closing inventory value)	X = high inventory value, Y = moderate, Z = low. Used at the time of physical verification to attack over-stocking
FSN / FSND analysis	Rate of movement / consumption	F = Fast moving, S = Slow moving, N = Non-moving (D = Dead). Non-moving stock indicates obsolescence
SDE analysis	Availability / difficulty of procurement	S = Scarce (long lead time, imported, short supply), D = Difficult, E = Easily available
GOLF analysis	Source of supply	G = Government (e.g. from a government undertaking), O = Ordinary/open market, L = Local purchase, F = Foreign/imported
SOS analysis	Seasonality of availability	S = Seasonal, OS = Off-seasonal. Applies to items whose supply or demand is seasonal (ORS, antimalarials, herbs)
MUSIC-3D	Multi-unit selective inventory control — three dimensions	Combines cost, criticality and availability in one framework
Two-bin technique	Simplicity of control for cheap items	Order when the first bin empties

🎵 MNEMONICS FOR THE CRITERIA

- ABC — Annual consumption value ("A for Amount").
- VED — Vitality / criticality.
- HML — High unit price ("price per piece").
- XYZ — value of stock in the store ("X for eXisting stock").
- FSN — Frequency of movement.
- SDE — Scarcity / ease of purchase.
- GOLF — Geographical source of supply.
- SOS — Season.

8.2.1 ABC analysis — steps and use

Step 1 List every item with its **annual consumption quantity** and **unit price**.

Step 2 Compute **annual consumption value** = **quantity × unit price** for each item.

Step 3 Arrange items in **descending order of annual consumption value**.

Step 4 Compute the **cumulative value and cumulative percentage** of items and of value.

Step 5 Draw the cut-offs: roughly the first **70 %** of cumulative value = **A**, the next **20 %** = **B**, the last **10 %** = **C**.

Step 6 Apply **differential control**: A items — tight control, low safety stock, frequent small orders, monthly verification, senior-level approval; C items — loose control, bulk annual purchase, two-bin system, annual verification.

Control parameter	A items	B items	C items
Share of items / of value	≈ 10 % / 70 %	≈ 20 % / 20 %	≈ 70 % / 10 %
Degree of control	Very tight	Moderate	Loose
Safety stock	Low (but no stock-out of vital A items)	Moderate	High
Ordering frequency	Frequent, small quantities	Moderate	Few large orders (once or twice a year)
Physical verification	Monthly / quarterly	Half-yearly	Annual
Value analysis & negotiation	Rigorous	Moderate	Minimal
Level of approval	Highest authority	Middle level	Store keeper level

8.2.2 Coupling ABC with VED — the practical matrix

Table 8.1 ABC-VED matrix. Category I items get the tightest control and the top management's attention; Category II moderate; Category III routine control.

	V (Vital)	E (Essential)	D (Desirable)
A (costly)	Category I	Category I	Category II

	V (Vital)	E (Essential)	D (Desirable)
B	Category I	Category II	Category III
C (cheap)	Category I	Category III	Category III

★ **WHY COUPLE THEM?** — A life-saving injection may cost only a few rupees, so pure ABC would put it in the careless "C" class. VED rescues it by calling it **Vital**. Hence all V items, whatever their cost, are placed in Category I and are never allowed to go out of stock.

8.3 Economic Order Quantity (EOQ)

f **EOQ (WILSON'S FORMULA)** — $EOQ = \sqrt{2 \times A \times O \div C}$ where

A = Annual consumption (demand) in units; O = Ordering cost per order; C = Carrying (holding) cost per unit per year. If the carrying cost is expressed as a fraction *i* of the unit price *p*, then $C = i \times p$ and $EOQ = \sqrt{2AO \div ip}$.

- At the EOQ, **total ordering cost = total carrying cost**, and the total inventory cost curve is at its **minimum** (the curve is flat near the minimum, so a small error in EOQ costs little).
- **Number of orders per year = $A \div EOQ$; interval between orders = $12 \div (A/EOQ)$ months.**
- **Average inventory = $EOQ \div 2$** (plus safety stock).
- EOQ **increases** if annual demand or ordering cost rises; it **decreases** if carrying cost or unit price rises. EOQ varies as the **square root** of demand — doubling the demand increases EOQ only $\sqrt{2}$ (≈ 1.41) times.

8.3.1 Assumptions and limitations of EOQ

Assumption	Why it may fail in a drug store
Demand is known, uniform and continuous	Epidemics, seasonal disease, change of prescriber preference
Lead time is known and constant	Tender delays, supplier failure, transport strikes
Price is constant (no quantity discount)	Bulk discounts and rate contracts change the economics
No stock-out is permitted	In practice a calculated stock-out of D items is accepted
Ordering and carrying costs are measurable and constant	Difficult to quantify in a hospital
The item has an unlimited shelf life	Drugs expire — so EOQ must be capped by the shelf life; this is the biggest limitation for pharmaceuticals
Money and space are unlimited	Budget is released in instalments; cold-chain space is limited

● **SOLVED EXAMPLE** — Annual consumption **A = 10,000** vials; ordering cost **O = ₹ 400** per order; unit price **p = ₹ 20**; carrying cost **i = 20 % = 0.20** → $C = 0.20 \times 20 = ₹ 4$ per vial per year.

$EOQ = \sqrt{2 \times 10,000 \times 400 \div 4} = \sqrt{2,000,000} \approx 1,414$ vials.

Number of orders per year = $10,000 \div 1,414 \approx 7$ orders, i.e. about one order every 7–8 weeks.

8.4 Stock Levels and Lead Time

8.4.1 Lead time

- ▶ **Lead time (procurement time):** The **total time between recognising the need (raising the indent) and the material becoming available for issue.**
- ▶ **Components:** **Administrative lead time** (preparing and sanctioning the indent, tendering, comparative statement, approval, placing the PO) + **supplier/delivery lead time** (manufacturing and despatch) + **transit time** + **inspection/testing and taking-on-charge time.**
- ▶ **Why it matters:** The whole purpose of the **re-order level** is to cover the consumption during the lead time. Longer or more erratic the lead time, larger the safety stock required.

8.4.2 The stock levels — definitions and formulae

Level	Meaning	Formula
Safety / buffer / reserve stock	Cushion held to absorb unexpected demand or delay in supply	Safety stock = (Maximum consumption rate – Average consumption rate) × Lead time or = Average consumption × (Maximum lead time – Average lead time) or, in practice, consumption of a fixed number of weeks/months
Re-order level (ROL) / ordering level	The balance at which a fresh order must be placed	ROL = (Average consumption per unit time × Lead time) + Safety stock (Simplified: ROL = Maximum consumption × Maximum lead time)
Minimum level	The level below which stock should not normally fall (= the safety stock)	Minimum = ROL – (Average consumption × Average lead time)
Maximum level	The highest stock that should be held, to avoid over-stocking	Maximum = ROL + Re-order (order) quantity – (Minimum consumption × Minimum lead time)
Danger level	Level at which normal issues are stopped and emergency purchase is resorted to	Danger level = Average consumption × Emergency (minimum) lead time
Average stock level	For valuation and carrying-cost calculation	Average stock = Minimum level + $\frac{1}{2}$ × Re-order quantity or = (Maximum + Minimum) ÷ 2
Re-order (order) quantity	How much to order each time	Usually the EOQ , adjusted for shelf life, pack size, budget and discounts

● **SOLVED EXAMPLE — RE-ORDER LEVEL** — Average consumption **500 strips/month**; lead time **2 months**; safety stock kept for **1 month** of consumption.

ROL = (500 × 2) + 500 = 1,500 strips. So when the balance on the bin card falls to 1,500 strips, the indent must be raised.

If EOQ = 3,000 strips and minimum consumption × minimum lead time = 300 × 1 = 300, then **Maximum level = 1,500 + 3,000 – 300 = 4,200 strips.**

8.5 Systems of Inventory Review and Modern Approaches

System	How it works	Suitability
Perpetual / continuous review (Q-system, fixed order quantity)	Stock is watched continuously; when it touches the re-order level , a fixed quantity (EOQ) is ordered. Order <i>quantity</i> fixed, order <i>time</i> variable	A and B items, vital drugs
Periodic review (P-system, fixed order interval)	Stock is reviewed at fixed intervals (say every quarter) and ordered up to a predetermined maximum. Order <i>time</i> fixed, <i>quantity</i> variable	C items, items from the same supplier ordered together
Two-bin system	Order placed when the first bin empties	Cheap C items
Min-max system	Stock kept between a fixed minimum and maximum	Ward/sub-store stocks
Imprest / top-up system	Fixed scale replenished to full at each cycle	Wards, OT, emergency trolley
Just-in-Time (JIT)	Material received exactly when needed; near-zero inventory	Possible only with utterly reliable local suppliers; risky for life-saving drugs
Vendor Managed Inventory (VMI) / consignment stock	Supplier owns and replenishes the stock lying in the hospital; payment on consumption	High-value implants, reagents, some costly injections
Rate contract + call-off orders	No stock held; ordered as and when needed at pre-fixed rates	Most government hospital purchasing
ERP / e-procurement (DVDMS, e-Aushadhi, GeM)	Computerised demand, purchase, receipt, issue and expiry management across a state	Large public health systems

★ FORMULA SHEET — MEMORISE THESE SEVEN

- $EOQ = \sqrt{2AO/C}$
- $ROL = (\text{Average consumption} \times \text{Lead time}) + \text{Safety stock}$
- $\text{Safety stock} = (\text{Max. consumption} - \text{Avg. consumption}) \times \text{Lead time}$
- $\text{Maximum level} = ROL + \text{Order quantity} - (\text{Min. consumption} \times \text{Min. lead time})$
- $\text{Danger level} = \text{Avg. consumption} \times \text{Emergency lead time}$
- $\text{Average stock} = \text{Minimum level} + \frac{1}{2} \text{ Order quantity}$
- $\text{Inventory turnover ratio} = \text{Annual consumption value} \div \text{Average inventory value}$

PART V • INDENTS

CHAPTER 9 | PREPARATION OF INDENTS

Definition • types • indenting authority • contents of the form • step-by-step preparation • quantity calculation • emergency purchase

An indent is the written demand that starts every purchase and every issue. The examiner's question "preparation of indents" has three parts: what an indent is, what it must contain, and how the quantity is worked out. The third part is where marks are won.

9.1 Indent — Definition and Related Terms

◆ **INDENT** — An indent is a **formal written requisition** by which a store or a user department **demand the supply of specified materials, in specified quantities, from a specified source** — either from the purchase section (*external indent*, leading to a purchase) or from the main store (*internal indent*, leading to an issue).

Term	Meaning	Who raises it	On whom
Indent / requisition (internal)	Demand for issue of material already in stock	Ward, OT, dispensary, sub-store	Main store
Indent / purchase requisition (external)	Demand to buy material	Main store / user department	Purchase section
Purchase Order (PO)	The actual order/contract to supply	Purchase section	The supplier
Issue voucher	Document by which the store hands over the material	Main store	(given to the indenting department)
Bill of materials	Pre-printed list of all items needed for one procedure/kit	User department	Store

★ **INDENT VS PURCHASE ORDER — SURE MCQ** — An indent is an **internal demand** (no legal obligation on anyone outside the organisation). A **purchase order is an external contract** with the supplier. The indent **precedes** the purchase order.

9.2 Indenting Authority and Financial Powers

- Only the **authorised indenting officer** (e.g. Head of the department, Ward Sister/Nursing In-charge, Chief Pharmacist, Store Officer) may sign an indent; a **specimen signature list** is kept in the store.
- The store must **refuse** an indent signed by an unauthorised person or exceeding the delegated **financial powers**.
- Indents involving purchase need **technical scrutiny** (is the item in the formulary? is the specification correct?) and **financial sanction** (is budget available under the correct head?).
- Indents for **narcotics, Schedule X drugs, spirit and costly items** require the countersignature of a senior/designated officer.
- A **budget certificate** ("funds are available under head ____") is usually recorded on the indent before it is processed.

9.3 Types of Indents

Type of indent	Purpose / features
Regular / routine / periodic indent	Raised at fixed intervals (weekly, monthly, quarterly) for items in regular use; the backbone of supply
Annual / bulk indent	Consolidated yearly demand used for annual tendering or rate contracts; based on last year's consumption plus growth
Emergency / urgent indent	For a life-saving or suddenly needed item; processed out of turn, often leading to local/spot purchase ; must state the reason for urgency and be signed by a senior officer

Type of indent	Purpose / features
Special / non-stock indent	For an item not normally stocked (a new drug, a special implant, a named-patient drug); needs approval of the PTC/competent authority
Standing indent	One indent covering supplies for a whole period, drawn in instalments as needed
Imprest / top-up indent	Sub-store/ward indent for only the quantity consumed, to restore the fixed imprest scale
Capital / equipment indent	For non-consumable (dead stock) items; needs capital budget sanction and technical specification committee approval
Transfer indent	Demand on another institution/store for transfer of surplus or urgently needed stock
Supplementary / additional indent	Raised when the quantity already indented proves insufficient before the next cycle
Local purchase (LP) indent	Purchase from the local market within delegated powers, when the item is out of stock/rate contract and urgently needed

9.4 Contents (Columns) of an Indent Form

Group	Columns / particulars
Identification	Indent number and date ; name of the indenting department/institution; period for which the demand is made; type (routine/emergency)
Item description	Sl. no.; store code number ; full description with generic name, strength, dosage form and pack size ; specification/standard (IP/BP/USP); unit of issue
Requirement data	Quantity required ; average monthly/annual consumption ; stock in hand (balance as per bin card) ; quantity already on order (dues-in) ; period of cover required
Justification	Purpose/patient load; reason for urgency; reason for any deviation from the previous pattern; whether the item is in the hospital formulary/NLEM
Financial data	Last purchase rate; estimated value; budget head and balance of funds ; rate contract number if any
Office use	Quantity sanctioned ("passed for"); reduction with reasons; ledger folio/bin card reference; issue voucher no. or PO no.; date of supply
Signatures	Indenting officer (with name, designation, date) • verifying/scrutinising officer • sanctioning authority • store keeper who issues • receiver

9.4.1 Specimen format — Indent / Requisition

Format 9.1 Indent form (header: indent no. & date, department, period; footer: signatures of indenting officer, scrutinising officer and sanctioning authority)

Sl.	Code No.	Item, strength & dosage form	Unit	Avg. monthly consumption	Stock in hand	On order	Qty indented	Qty sanctioned	Remarks

9.5 Step-by-Step Procedure for Preparing an Indent

- Step 1 Identify the need.** Either the balance on the bin card has touched the **re-order level**, or the periodic indenting date has arrived, or a user department has demanded an item, or a new drug has been approved by the PTC.
- Step 2 Take out the consumption data.** From the stock ledger/consumption register, extract the issues of the last 6–12 months and compute the **average monthly consumption (AMC)**. Ignore abnormal months (epidemic, camp) or note them separately.
- Step 3 Ascertain the present stock in hand** — physically verify it; do not rely on the ledger alone. Deduct any quantity that is **near expiry, quarantined, damaged or reserved**, because it is not really usable.
- Step 4 Ascertain the quantity already on order (dues-in)** from the pending-order register. *Failure to deduct dues-in is the commonest cause of double ordering.*

- Step 5 Fix the period of cover** (how many months the supply should last) in the light of the **lead time**, budget release pattern, shelf life, storage space and cold-chain capacity.
- Step 6 Compute the quantity to be indented** using the formula in 9.6, then **round it off to a convenient pack/multiple** (you cannot indent 1,013 tablets if the pack is 10×10).
- Step 7 Cap the quantity by shelf life.** Never indent more than can be consumed well before expiry; for short-shelf-life items (vaccines, sera, some antibiotics) indent small quantities more often.
- Step 8 Check the specification and unit.** Write the **generic name, strength, dosage form, pack size and pharmacopoeial standard**; never indent by brand name alone in government purchase; state the unit of issue unambiguously.
- Step 9 Check the formulary/NLEM status and the code number**; indent only approved items, and quote the correct store code.
- Step 10 Check budget availability** under the correct head and record the funds certificate; obtain the rate contract reference and the last purchase rate for costing.
- Step 11 Fill the indent form**, number it serially and enter it in the **indent register**.
- Step 12 Obtain signatures** — the indenting officer signs, the scrutinising officer verifies consumption and stock figures, and the competent authority **sanctions** the quantity.
- Step 13 Despatch** the indent to the purchase section (external) or to the main store (internal) and retain the office copy.
- Step 14 Follow up** — watch the dues-in register, send reminders, and on receipt verify that the supply matches the indent; record any short supply for the next indent.

Re-order level reached ➤ Consumption data ➤ Stock in hand ➤ Less dues-in ➤ Compute quantity ➤ Round off to pack ➤ Sanction ➤ Indent register ➤ Despatch ➤ Follow-up

9.6 Calculation of the Indent Quantity

f THE WORKING FORMULA — Quantity to be indented = (Average monthly consumption $\times$ [Period of cover + Lead time in months]) + Safety (buffer) stock – Stock in hand – Quantity already on order

Some manuals write the same thing as: **Indent quantity = Maximum stock level – (Stock in hand + Dues-in)**, where the maximum level already contains the lead-time consumption and the buffer.

9.6.1 Solved example 1 — a routine drug

Data	Value
Average monthly consumption (AMC)	2,000 tablets
Period of cover required	6 months
Lead time	2 months
Safety stock policy	1 month's consumption
Stock in hand (usable)	3,500 tablets
Quantity already on order (dues-in)	1,000 tablets

- Requirement for cover + lead time = $2,000 \times (6 + 2) = 16,000$ tablets.
- Add safety stock = 2,000 $\rightarrow$ total requirement = **18,000** tablets.
- Deduct stock in hand and dues-in = $18,000 - 3,500 - 1,000 = 13,500$ tablets.
- Round off to the pack ($10 \times 10 = 100$ tablets per box) $\rightarrow$ **indent 13,500 tablets (135 boxes)**.

9.6.2 Solved example 2 — annual indent with growth factor

- Last year's consumption = 48,000 vials; expected increase in patient load = 10 %.
- Estimated annual requirement = $48,000 \times 1.10 = 52,800$ vials.
- Add buffer of 2 months = $52,800 \times 2/12 = 8,800 \rightarrow 61,600$ vials.
- Deduct closing stock 6,000 and dues-in 4,000 $\rightarrow$ **annual indent = 51,600 vials**.
- If the shelf life is only 18 months, this is acceptable; if it were 9 months, the indent would be split into **two or more staggered deliveries**.

9.6.3 Methods of estimating requirement

Method	Basis	Remarks
Consumption (past usage) method	Actual issues of the past 6–12 months, adjusted for stock-out days, growth and seasonality	Most commonly used and most reliable where records are good
Morbidity (disease pattern) method	Number of expected cases of each disease × standard treatment schedule × course duration	Best for a new hospital/programme with no consumption history; needs reliable epidemiological data
Bed-occupancy / service-level method	Consumption per bed per day × number of beds × occupancy × days	Quick estimate for hospital planning
Budget (adjusted) method	Requirement trimmed to fit the money available, prioritised by VED/NLEM	Reality in most government institutions
Proxy / extrapolation method	Data of a comparable institution scaled to the size of this one	Rough; used when no other data exist

✓ **REMEMBER** — The **consumption method** uses *past issues*; the **morbidity method** uses *disease incidence and standard treatment guidelines*. For a brand-new facility with no past data, the **morbidity method** is the correct choice.

9.7 Factors to be Considered While Indenting

Factor	Effect on the indent
Average consumption & trend	The base figure of the calculation
Lead time and its reliability	Longer/erratic lead time → larger buffer
Stock in hand and dues-in	Deducted from the requirement
Shelf life / expiry	Caps the maximum quantity; short-life items indented frequently in small lots
Storage space & cold-chain capacity	You cannot indent what you cannot store
Budget provision and its release pattern	Determines both quantity and timing
Pack size and unit of issue	Quantity rounded to full packs
ABC / VED status of the item	Vital items get generous buffer; C items bulk-ordered
Seasonality and epidemics	ORS, antimalarials, antivenom, vaccines
Change in prescribing pattern / formulary revision	A discarded drug must not be re-indented
Quantity discounts and rate contracts	May justify a larger order
Statutory restrictions	Narcotics/Schedule X quantities limited by permit
New schemes, camps, programmes	Additional demand to be added separately
Supplier reliability / past short supply	Larger buffer or split ordering
Price trend	Rising prices may justify earlier/bulk purchase (within shelf life)

9.8 Scrutiny, Sanction and the Indent Register

9.8.1 Points checked while scrutinising an indent

- Is the indent on the **prescribed form**, serially numbered, dated and **signed by an authorised officer**?
- Is the item **in the formulary/NLEM** and is the **code number** correct?
- Is the **specification complete** (generic name, strength, dosage form, pack, pharmacopoeial standard)?
- Are the **consumption, stock-in-hand and dues-in figures correct** (verified against the ledger and bin card)?
- Is the **quantity reasonable** in relation to consumption, shelf life, storage space and budget?
- Is the **budget head correct** and are funds available?
- Is there any **duplication** with an earlier indent still pending?
- For an emergency indent — is the **reason for urgency recorded** and approved?
- For a non-stock/new item — is the **PTC/competent authority approval** attached?

9.8.2 Indent register

Format 9.2 Indent register

Sl. No.	Date	Indenting dept.	Item(s)	Qty indented	Qty sanctioned	Issue voucher / PO No.	Date of supply	Remarks

9.9 Common Mistakes and Precautions in Indenting

Mistake	Consequence	Precaution
Dues-in not deducted	Double supply, over-stocking, expiry	Consult the pending-order register every time
Stock in hand taken from the ledger without physical check	Wrong quantity indented	Verify the shelf physically
Near-expiry stock counted as usable stock	Stock-out despite "stock in hand"	Deduct near-expiry, quarantined and damaged stock
Indenting by brand name only	Restricts competition; audit objection	Indent by generic name + strength + dosage form + IP standard
Unit of issue not stated / ambiguous ("1 bottle")	Wrong quantity supplied (10 ml vs 100 ml)	State the exact unit and pack size
Over-indenting to be "safe"	Blocked capital, expiry loss, shortage of space	Base the quantity on AMC, lead time and shelf life
Under-indenting / indenting late	Stock-out, costly emergency purchase	Indent when the re-order level is reached, not when stock is exhausted
Ignoring shelf life	Expiry of large quantities	Cap by shelf life; stagger deliveries
Ignoring seasonality/epidemic	Shortage at the peak	Study last year's month-wise consumption
All emergency indents	Loss of tender discount, audit objection	Plan routine indents properly; emergency route only for genuine emergencies
Indent not signed / signed by unauthorised person	Indent invalid; supply irregular	Maintain a specimen-signature list

9.10 Emergency and Local Purchase

Step 1 The item is found **out of stock or insufficient**, and the need is **immediate and life-saving**.

Step 2 The pharmacist records the **non-availability** (from the stock register) and certifies that the item is **not available on rate contract / from the main store**.

Step 3 An **emergency indent** is prepared stating the reason for urgency and the quantity for the shortest reasonable period (usually a few days to a month).

Step 4 Sanction of the officer having **delegated financial powers for local purchase** is obtained.

Step 5 Quotations are obtained from **local licensed dealers** (at least 2–3 wherever possible) or purchase is made from the nearest approved chemist.

Step 6 The material is received, **checked for batch, expiry and MRP**, entered in the stock register and issued.

Step 7 The **cash memo/bill**, sanction order and certificate of urgency are filed together for audit.

Step 8 The event is analysed: *why did the stock-out happen?* — and the routine indent is corrected so that it does not recur.

⚠ WHY EMERGENCY PURCHASE IS DISCOURAGED

- Price is higher (no competitive tender, no bulk discount).
- Quality assurance is weaker (no pre-despatch testing).
- Heavy paper work and audit scrutiny.
- It is evidence of **failure of indenting and inventory control** — a high proportion of emergency purchases is a recognised indicator of poor store management.

9.11 From Indent to Issue — The Complete Internal Cycle

Ward indent ➤ Scrutiny in main store ➤ Sanction ➤ Issue voucher ➤ FEFO picking ➤ Receiver signs ➤ Bin card
& ledger posted ➤ Ward register posted ➤ Consumption data ➤ Next indent

Stage	Document created	Record posted
Ward raises demand	Indent (2 copies)	Indent register
Store sanctions	Sanction noted on the indent	Indent register
Store issues	Issue voucher	Bin card, stock ledger, batch/expiry register
Ward receives	Signature on the issue voucher	Ward/sub-store stock register (imprest)
Ward consumes	Ward consumption sheet / patient records	Ward register, patient charge sheet
Unused material returned	Material Return Note	Stock ledger (receipt entry)
Month end	Consumption statement	Monthly consumption register, MIS report

★ CHAPTER 9 IN ONE BOX

- Indent = **internal written demand**; PO = **external contract**.
- Types: routine, annual/bulk, emergency, special/non-stock, standing, imprest, capital, transfer, supplementary, local purchase.
- **Quantity** = $\text{AMC} \times (\text{cover} + \text{lead time}) + \text{buffer} - \text{stock in hand} - \text{dues-in}$, rounded to pack size and capped by shelf life.
- Estimation methods: **consumption** (past issues) and **morbidity** (disease pattern + standard treatment).
- Indent when the **re-order level** is reached — never after the stock is exhausted.
- Always deduct **dues-in**; always verify **stock physically**; always indent by **generic name with strength, dosage form and pack**.

PART VI • STORAGE OF DRUGS

CHAPTER 10 | METHODS OF STORING DRUGS

Good storage practice • systems of arrangement • storage conditions & temperature terms • cold chain • special categories • deterioration • expiry & disposal

Storage has two halves: **where you put the drug (the method of arrangement)** and **under what conditions you keep it (the storage environment)**. Examiners ask both, and they love the temperature definitions, the cold chain, and the special categories. This is the longest chapter for good reason.

10.1 Storage — Meaning and Objectives

◆ **STORAGE / GOOD STORAGE PRACTICE** — Storage is the holding of drugs and related materials **in such a manner and under such conditions that their identity, purity, potency, safety and quality are preserved until they are issued to the patient**. The set of rules that achieves this is called **Good Storage Practice (GSP)**, described by the WHO as part of Good Distribution Practice (GDP).

10.1.1 Objectives of proper storage

1. To **preserve the quality, potency and stability** of every drug up to its labelled expiry date.
2. To prevent **physical damage** (breakage, crushing, leakage) and **chemical deterioration** (oxidation, hydrolysis, photolysis).
3. To prevent **microbial contamination** of sterile and non-sterile products.
4. To enable **quick and correct retrieval** of any item (right drug to the right patient).
5. To prevent **expiry loss** by enforcing FEFO and near-expiry monitoring.
6. To prevent **theft, pilferage and misuse**, especially of narcotics and costly drugs.
7. To ensure **legal compliance** with the labelled storage conditions and with Schedule P.
8. To ensure **safety of staff and premises** (inflammables, cytotoxics, radioactives, fire).
9. To make the best use of **space, manpower and equipment**.

10.2 Requirements of a Good Drug Store

Requirement	Standard / good practice
Location & building	Ground floor, near the receiving gate and the dispensary; dry, well-drained site; pucca, leak-proof, rodent-proof construction; separate entry and exit
Area / space	Sufficient for present stock + expansion; separate zones for receiving, quarantine, bulk, cold chain, narcotics, inflammables, rejected stock, issue and office
Temperature	Store generally maintained below 25 °C (air-conditioned or well-ventilated); cold rooms/refrigerators at 2–8 °C ; deep freezer as required; temperature recorded twice daily with a calibrated thermometer
Humidity	Relative humidity preferably below 60 % ; dehumidifiers, silica gel, good ventilation; hygroscopic items in tightly closed containers
Light	Adequate diffused illumination for reading labels; direct sunlight excluded (shaded/painted windows, blinds); light-sensitive drugs in amber containers/cartons
Ventilation	Cross-ventilation and exhaust fans to prevent heat and vapour accumulation
Floor, walls, ceiling	Smooth, crack-free, non-absorbent, easily washable; no crevices for insects; no seepage
Racks, shelves, pallets	Metal/steel, adjustable, labelled with item name, code and bin number; stock ≥ 30 cm from walls and on pallets ≈ 10 cm above the floor ; nothing directly on the floor
Aisles & gangways	Wide enough for trolleys and for stock-taking; kept clear
Security	Grilled windows, strong doors, single controlled entry, restricted access, locks/CCTV; double-locked cupboard for narcotics and a separate locked cupboard for Schedule X
Fire safety	CO ₂ and dry-chemical extinguishers, sand buckets, hydrants, smoke detectors, marked exits, "NO SMOKING"; inflammables in a detached store
Pest & rodent control	Screened windows, rat traps/guards, periodic fumigation/pest control with records; no food or waste inside the store

Requirement	Standard / good practice
Cleanliness & housekeeping	Daily cleaning, no dust, no cobwebs, periodic defrosting of refrigerators, waste removed daily
Equipment	Refrigerator(s), ILR and deep freezer with voltage stabiliser, thermometers/data loggers, weighing scale, trolleys, ladders, fire extinguishers, generator/UPS back-up for the cold chain
Records area	Separate, secure space for registers, ledgers and the computer
Water & drainage	Wash basin, no water lines passing over the stock, no chance of flooding

★ NUMBERS EXAMINERS LIKE

- Stock kept **at least 30 cm (1 foot) away from the wall**.
- Stock kept on pallets **≈ 10 cm above the floor** — never directly on the floor.
- General drug store temperature: **below 25 °C**; RH **below 60 %**.
- Refrigerator for drugs/vaccines: **+2 to +8 °C**.
- Temperature recorded **twice a day** (morning and evening) in the log.

10.3 Methods / Systems of Arranging Drugs in the Store

This is the *core* of the syllabus phrase "methods of storing drugs". Learn all the systems, their advantages and their drawbacks, and remember which is commonest.

Method of arrangement	How it works	Advantages	Disadvantages
1. Alphabetical	Drugs arranged A–Z by generic (or brand) name	Simplest; anyone can locate an item; no training needed; ideal for small pharmacies and retail shops	Similar-sounding names come together → LASA dispensing errors ; a name change/brand change displaces the item; no relation to storage conditions
2. Therapeutic / pharmacological category	Grouped by action — antibiotics, analgesics, antihypertensives... and alphabetically within each group	Clinically logical; helps substitution of an equivalent drug; convenient for ward stock and formulary work	Needs pharmacological knowledge; a drug with several actions is hard to place
3. Dosage form / route of administration	Separate racks for oral solids, orals liquids, injections, externals, ophthalmics, inhalations, suppositories	Keeps liquids away from powders; heavy bottles on low shelves; suits the different storage needs of each form; prevents the serious error of issuing an external preparation for internal use	A drug available in several forms is stored in several places
4. Alphabetical within dosage form (the commonest hospital practice)	First divide by dosage form, then arrange alphabetically inside each group	Combines the advantages of methods 1 and 3; fast retrieval; widely recommended	Still needs LASA precautions
5. Code number / classification-wise (systematic)	Items arranged in the serial order of their store code numbers	Perfect link with bin cards, ledgers and computer records; ideal for large stores	Needs a codification system and a location index; a new staff member cannot find anything without the index
6. Storage-condition-wise	Separate areas for room temperature, cool, cold chain, frozen, light-sensitive, inflammable, hazardous	Protects quality — the single most important consideration for thermolabile drugs	Same drug group gets scattered
7. Frequency of use (popularity / movement) system	Fast-moving items nearest to the issue counter, slow movers at the far end	Minimises walking and handling time; based on FSN analysis	Locations must be revised as consumption changes
8. Fixed location system	Every item has a permanent bin address	Easy to memorise; no index needed	Space lies idle when the item is out of stock
9. Random / floating location system	Item placed wherever space exists, address recorded in the index/computer	Best space utilisation	Totally dependent on accurate records

Method of arrangement	How it works	Advantages	Disadvantages
10. Size / weight / volume-wise	Bulky and heavy items (IV fluids, cylinders, cartons) on the floor/lowest shelf near the entrance; small items in bins above	Safety, ease of handling, prevents breakage	Not a complete system by itself
11. Manufacturer / supplier-wise	All items of one firm kept together	Convenient for returns and for rate-contract checking	Poor clinical logic; not recommended for drugs
12. Legal-category-wise	Narcotics, Schedule X, Schedule H/H1 kept in their prescribed separate, locked places	Legally compulsory	Applies only to those categories
13. FIFO / FEFO arrangement (within any of the above)	Older/earlier-expiring stock in front, newer stock behind	Prevents expiry loss — the most important day-to-day storage discipline	Requires discipline at every receipt

✓ MOST LIKELY MCQ ANSWERS

- Commonest/most convenient system in a hospital drug store — **dosage-form-wise, alphabetically within each form.**
- System best suited to a large store with codification — **code-number (systematic) arrangement.**
- System that gives the best space utilisation — **random/floating location.**
- System based on rate of movement — **frequency-of-use (FSN) system.**
- The principle that prevents expiry — **FEFO (first expiry first out).**

10.3.1 FIFO and FEFO in practice

- New stock is **always placed behind (or below)** the existing stock; issue is from the front.
- The **expiry date is written boldly** on the carton/shelf label and on the bin card.
- When a new receipt has an **earlier expiry** than the existing stock, it is placed **in front** — that is the difference between FIFO and FEFO.
- A **monthly near-expiry list** is generated and colour stickers applied (traffic-light system).
- Bar-code/software systems can enforce FEFO automatically by prompting the batch to be picked.

10.4 Storage Conditions and Temperature Terminology

The label of every medicine states the condition under which it must be kept. The words used on the label have **precise, defined meanings** — these definitions are among the most frequently asked objective questions in pharmacy examinations.

Table 10.1 Defined storage terms (the IP/WHO column is the one to quote in an Indian examination)

Term on the label	Indian Pharmacopoeia (IP) / WHO	USP (for comparison)
Freezer / store frozen	Below – 18 to –20 °C (deep freezer)	–25 °C to –10 °C
Cold place	A temperature not exceeding 8 °C	Any temperature not exceeding 8 °C
Refrigerator	2 °C to 8 °C	2 °C to 8 °C
Cool place	8 °C to 25 °C	8 °C to 15 °C
Room temperature	The temperature prevailing in a working area; commonly taken as 15–30 °C (WHO: 15–25 °C)	The temperature prevailing in a working area
Controlled room temperature	Usually 20–25 °C, excursions permitted between 15 and 30 °C	20–25 °C, allowed excursions 15–30 °C
Warm	30 °C to 40 °C	30 °C to 40 °C
Excessive heat	Above 40 °C	Above 40 °C
Protect from freezing	Must not be frozen — freezing damages the product (e.g. DPT, Hep-B, Td, insulin) even though it is stored cold	Same
Store below 30 °C / below 25 °C	The stated maximum must never be exceeded	Same

Term on the label	Indian Pharmacopoeia (IP) / WHO	USP (for comparison)
Protect from light	Keep in an amber/opaque container or in its carton, away from sunlight and UV	Same
Protect from moisture	Tightly closed container, desiccant, low humidity	Same
Dry place	Relative humidity preferably below 60 %	Same

⚠ **THE CLASSIC TRAP** — "Cool place" is NOT the refrigerator. In the Indian Pharmacopoeia a cool place is 8–25 °C, whereas a cold place is not exceeding 8 °C and a refrigerator is 2–8 °C. (In the USP, "cool" is the narrower 8–15 °C.) Also remember: Schedule P of the Drugs and Cosmetics Rules prescribes the life period and storage conditions of drugs.

10.4.1 Types of containers and closures (storage-related)

Container	Meaning / use
Well-closed container	Protects the contents from extraneous solids and from loss during handling
Tightly closed container	Protects from extraneous solids, liquids and vapours (moisture) , and from loss, efflorescence, deliquescence or evaporation
Hermetically sealed container	Impervious to air or any other gas (e.g. sealed glass ampoule) — used for sterile products
Light-resistant container	Amber/opaque glass or an opaque outer wrapper, protecting light-sensitive drugs
Single-dose / multi-dose container	Ampoule vs multi-dose vial (which needs a preservative and a limited in-use shelf life)
Tamper-evident / child-resistant closure	Safety and evidence of interference

10.5 The Cold Chain

◆ **COLD CHAIN** — The cold chain is the system of people, equipment and procedures that keeps thermolabile products (vaccines, sera, insulin, oxytocin, blood products) continuously within their prescribed temperature range — usually +2 to +8 °C — from the manufacturer right up to the patient.

Equipment	Purpose / temperature
Walk-in cooler (WIC)	Bulk storage at +2 to +8 °C at regional/state level
Walk-in freezer (WIF)	Bulk storage of frozen items / ice packs at -15 to -25 °C
ILR (Ice-Lined Refrigerator)	Keeps vaccines at +2 to +8 °C and holds the temperature for many hours during power failure because of the lining of ice packs; top-opening design retains cold air
Deep freezer (DF)	For freezing ice packs and storing OPV and other freeze-safe vaccines at -15 to -25 °C
Domestic refrigerator	Acceptable only in small units; vaccines kept in the body of the fridge, never in the door shelves or freezer compartment
Cold box	Insulated box with ice packs for transporting vaccines over longer distances (holds cold for days)
Vaccine carrier	Small insulated carrier with 4 ice packs for daily field sessions (holds cold for several hours)
Ice packs / conditioned ice packs	Frozen water packs; conditioning (leaving them out until water begins to appear) prevents freezing of freeze-sensitive vaccines
Thermometer / dial thermometer / data logger	Continuous record of temperature; readings noted twice daily on the log chart
Vaccine Vial Monitor (VVM)	Heat-sensitive square on the vial label: as heat exposure accumulates the inner square darkens . Usable while the inner square is lighter than the outer ring (stages 1–2); discard at stage 3–4 when it matches or is darker than the ring
Cold Chain Monitor (CCM) card / freeze indicator (Freeze-tag)	Shows whether the consignment has been exposed to heat or to freezing during transit
Voltage stabiliser, generator/solar back-up	Protects the equipment and the chain during power fluctuation and failure

10.5.1 Freeze-sensitive vs heat-sensitive vaccines

Group	Examples	Rule
Most freeze-sensitive (damaged by freezing — never put in the freezer or against ice packs)	DPT, DT, Td, TT, Hepatitis B, Pentavalent, IPV, Rabies, Typhoid, JE (killed), Insulin	Store at +2 to +8 °C ; " do not freeze "; discard if frozen (shake test for adsorbed vaccines)
Most heat-sensitive (damaged by heat, tolerate freezing)	OPV (most heat-sensitive of all), Measles, MR/MMR, BCG	May be stored in the freezer/coldest part ; protect from heat and light
Diluents	Diluents of BCG, Measles, MMR	Never frozen; must be cooled to +2 to +8 °C before reconstitution ; use only the diluent supplied by the same manufacturer

10.5.2 Cold-chain do's and don'ts

- **Do** record temperature **twice daily** (morning & evening) on the log chart, including holidays; plot it.
- **Do** keep the ILR/refrigerator **away from direct sunlight and at least 10 cm from the wall**, on a level base, in a ventilated room.
- **Do** keep vaccines in the **basket/body of the ILR**, arranged with space for air circulation, earliest expiry on top.
- **Do** defrost regularly (frost > 5 mm reduces efficiency) and keep a **spare set of ice packs frozen** for emergencies.
- **Do** use the **shake test** if a freeze-sensitive vaccine is suspected to have frozen — a frozen-and-thawed vial shows rapid sedimentation and flakes; it must be discarded.
- **Don't** store food, drinks or laboratory specimens in the vaccine refrigerator.
- **Don't** keep vaccines in the **door shelves** of a domestic refrigerator (temperature fluctuates most there).
- **Don't** open the ILR unnecessarily, and never leave it open.
- **Don't** use a vaccine whose **VVM has reached stage 3 or 4**, or which has passed its expiry date, or a reconstituted vaccine beyond the permitted hours.
- **Don't** refreeze thawed ice packs together with freeze-sensitive vaccine without **conditioning** them first.
- **Do** have a written **power-failure/breakdown SOP**: do not open the ILR, arrange a generator, shift stock to a cold box, record the excursion and take a decision on usability in consultation with the manufacturer/authority.

10.6 Storage of Special Categories of Drugs

Category	Method of storage — key points
Narcotics & psychotropics (NDPS Act)	In a substantial, locked steel cupboard or safe, embedded in a wall or fixed to the floor, with a double lock ; keys in the personal custody of a named officer ; stock physically tallied daily/at each shift change ; entries in the NDPS/DDA register the same day , countersigned; no other article kept in the cupboard ; loss/theft reported immediately; destruction only in the presence of the authorised officers with a certificate
Schedule X (habit-forming) drugs	Separate locked cupboard (apart from other stock), special licence (Form 20F retail / 20G wholesale, applied for in Form 19C), prescription retained in duplicate , separate register, records preserved for the prescribed period (2 years)
Schedule H / H1 drugs	Stored with the general stock but sold/issued only against a prescription ; the H1 register (patient, prescriber, drug, quantity, date) maintained and preserved for 3 years
Thermolabile drugs (vaccines, sera, insulin, oxytocin, some eye drops, suppositories, probiotics)	+2 to +8 °C in a refrigerator/ILR with a twice-daily temperature log; never frozen unless the label permits; transported in a cold box/vaccine carrier
Blood and blood components	Whole blood & packed red cells +2 to +6 °C in a blood-bank refrigerator with an alarm; platelets +20 to +24 °C with continuous gentle agitation ; fresh frozen plasma at -30 °C or below ; never stored in an ordinary refrigerator
Light-sensitive drugs (e.g. vitamin A & D, riboflavin, chlorpromazine, nitroprusside, furosemide, adrenaline, silver salts, phenothiazines)	Amber/opaque containers , kept inside the outer carton, away from sunlight and fluorescent UV; do not transfer to clear containers

Category	Method of storage — key points
Moisture-sensitive / hygroscopic & deliquescent drugs (e.g. effervescent granules, aspirin, dispersible tablets, calcium chloride, KOH, NaOH, glycerin)	Tightly closed containers with a desiccant (silica gel) , in a dry place with RH < 60 %; strips/blisters not broken until issue
Volatile & inflammable liquids (ether, spirit, acetone, alcohol, chloroform, petrol)	Detached, well-ventilated, fire-resistant store away from the main building; cool place, away from heat and flame ; flame-proof electrical fittings; "NO SMOKING"; CO ₂ /dry-powder extinguishers, sand buckets; small quantities only in the working area; rectified spirit also in the excise-bonded store with its register
Corrosives and hazardous chemicals (strong acids, alkalis, phenol)	On low shelves in trays , never above eye level; acids away from alkalis and from oxidising agents; PPE, eye-wash and spill kit available; MSDS on file
Cytotoxic / anticancer drugs	Segregated, clearly labelled " CYTOTOXIC — HANDLE WITH CARE " area or cupboard; stored at the labelled temperature; kept in a tray/secondary container to contain spillage; spill kit and PPE nearby; reconstitution only in a biological safety cabinet; waste in yellow cytotoxic bags
Radiopharmaceuticals	Lead-lined containers inside a lead-shielded room/safe in a restricted area with the radiation trefoil sign; handled only by authorised persons; radiation survey and log-book maintained as per AERB requirements; decay storage before disposal
Medical gas cylinders	Stored upright, chained/secured , in a cool, dry, well-ventilated, fire-free area; full and empty cylinders kept separately and labelled ; valve caps in place; never oiled or greased (oxygen + oil = fire risk); not rolled or dropped; colour code verified; kept away from inflammables
LASA (look-alike, sound-alike) drugs	Never stored side by side ; separated physically, with bold/contrasting labels, "tall-man lettering" and LASA stickers; a LASA list is displayed
High-alert / high-risk drugs (concentrated KCl, heparin, insulin, neuromuscular blockers, concentrated electrolytes)	Stored in a separate, clearly marked location with a warning label ; quantity in ward stock restricted; double-check at issue
Antiseptics & disinfectants	Stored away from internal medicines (a favourite examination point) in a cool place; original labelled containers only; never decanted into medicine bottles
Sutures & sterile disposables	Dust-free, dry, cool place; cartons not crushed ; sterility date and pack integrity checked; FEFO strictly
X-ray films	Cool, dry place, away from radiation, chemicals and heat ; boxes stored vertically (on edge) to avoid pressure marks; expiry monitored
Diagnostic reagents & test kits	Mostly 2–8 °C ; protected from light; in-use stability noted
IV fluids & large-volume parenterals	On pallets on the lowest shelf/floor level (heavy); protected from freezing and from direct sun; checked for clarity, leakage and particulate matter before issue
Ayurvedic / homoeopathic medicines	Stored separately from allopathic drugs; homoeopathic preparations kept away from strong odours, camphor and direct sunlight
Expired, recalled and rejected stock	Separate, locked, red-labelled area , physically apart from live stock, pending return or destruction

♪ **MNEMONIC FOR SPECIAL STORAGE — "N-X-T-B-L-M-V-C-C-R-G-L-H"** — Narcotics, Schedule X, Thermolabile, Blood, Light-sensitive, Moisture-sensitive, Volatile/inflammable, Corrosives, Cytotoxics, Radiopharmaceuticals, Gases, LASA, High-alert.

10.7 Deterioration of Drugs — Causes and Signs

10.7.1 Factors causing deterioration

Factor	Mechanism	Examples / prevention
Heat / temperature	Accelerates hydrolysis, oxidation and loss of potency (rate of reaction roughly doubles for every 10 °C rise)	Vaccines, insulin, suppositories, antibiotics in syrup; prevent by cold chain and a cool store
Moisture / humidity	Hydrolysis, caking, softening, microbial growth	Aspirin, effervescent salts, dispersible tablets, capsules; tightly closed containers + desiccant
Light (especially UV)	Photolysis and oxidation	Vitamin A, riboflavin, chlorpromazine, adrenaline, sodium nitroprusside; amber containers, cartons

Factor	Mechanism	Examples / prevention
Air / oxygen	Oxidation, rancidity of oils	Oils, fats, vitamins, adrenaline; hermetic or tightly closed containers, antioxidants, nitrogen filling
Microbial contamination	Spoilage, loss of sterility	Syrups, creams, eye drops; preservatives, sterile technique, intact seals
pH and chemical interaction	Degradation, incompatibility	Buffered formulations; do not store acids with alkalis
Packaging failure / leaching	Interaction between drug and container, leakage	Use the specified container; reject leaking packs
Time (age)	Natural degradation up to and beyond the expiry date	FEFO, near-expiry monitoring
Physical stress / rough handling	Breakage, powdering, crushing of tablets	Careful handling, no over-stacking
Pests, rodents and insects	Contamination and physical destruction	Pest control, rodent-proofing, screens

10.7.2 Signs of deterioration by dosage form

Dosage form	Signs that the product must not be used
Tablets	Discolouration, mottling, spots, chips or cracks, swelling, softening, stickiness, unusual smell (e.g. vinegar smell of degraded aspirin), loss of coating, crumbling
Capsules	Softening, sticking together, hardening/brittleness, leakage, swelling, discolouration of the shell
Powders / granules	Caking, hardening, clumping, discolouration, damp lumps
Syrups & elixirs	Cloudiness, crystallisation, mould growth, change of colour or taste, gas formation, broken seal
Suspensions	Caking that does not redisperse on shaking, large crystals, colour change
Emulsions	Separation/cracking of phases that does not re-emulsify, rancid smell
Injections / ampoules & vials	Any turbidity, precipitate, particulate matter, discolouration, leakage, cracked ampoule, loose/bulging rubber closure, broken tamper seal — discard
IV fluids	Cloudiness, floating particles, leakage from the bag/bottle, damaged port
Ointments & creams	Separation of oil, change of colour or smell, granular or gritty texture, drying/hardening
Eye/ear drops	Turbidity, particles, discolouration, broken seal, exceeding the in-use period (usually 28 days after opening for a multi-dose preservative containing eye drop)
Suppositories & pessaries	Softening, melting, deformation, mould
Vaccines	VVM stage 3/4, frozen appearance of an adsorbed vaccine, flakes on shake test, expiry crossed
Effervescent tablets/powders	Loss of effervescence, swollen strip/tube, dampness

10.8 Shelf Life, Expiry Date and Near-Expiry Management

- ▶ **Shelf life (life period):** The period during which a drug, if stored under the labelled conditions, retains at least the specified percentage (generally **not less than 90 %**) of its labelled potency and remains within all other specifications. Prescribed for various classes in **Schedule P** of the Drugs and Cosmetics Rules.
- ▶ **Expiry date:** The date up to and including which the product is expected to comply with specifications **if stored correctly**. If only month and year are given, the drug is usable up to the **last day of that month**.
- ▶ **Date of manufacture:** The date on which the manufacturing process was completed; the shelf life is counted from it.
- ▶ **Retest date:** Used mainly for active raw materials — the date on which the material must be re-analysed to confirm it is still fit for use.
- ▶ **Beyond-use date (BUD) / in-use shelf life:** The date after which a product **that has been opened, reconstituted or diluted** must not be used (e.g. reconstituted antibiotic syrup 7–14 days refrigerated; opened multi-dose eye drop 28 days; reconstituted vaccine a few hours).

- **Near-expiry:** Conventionally stock expiring within the next **3–6 months**; it is listed monthly, colour-tagged and redistributed or returned.

10.8.1 Managing near-expiry and expired stock

- Step 1** Generate the **near-expiry statement monthly** from the batch/expiry register or software.
- Step 2** Apply **traffic-light stickers** (red ≤ 3 months, yellow 3–6 months, green > 6 months).
- Step 3** **Redistribute** to high-consumption wards or other institutions, or **return to the supplier** under the buy-back/replacement clause of the purchase order.
- Step 4** On expiry, **remove from the shelf the same day**, enter in the **expiry register**, and move to the **locked, red-labelled expired-stock area**.
- Step 5** Obtain **write-off sanction** and deduct from the stock register quoting the sanction.
- Step 6** Dispose of as **bio-medical waste (yellow category)** through the authorised common treatment facility, or return to the manufacturer; obtain and file the **destruction certificate**.
- Step 7** Analyse the **cause of expiry** (over-indenting, short-expiry acceptance, change of prescribing, poor FEFO) and correct it in the next indent.

Method of disposal of expired/unwanted drugs	Suitability
Return to the manufacturer/supplier	The best option where the contract permits
High-temperature incineration (≥ 1,200 °C)	Cytotoxics, controlled substances, most solid dosage forms
Encapsulation	Sealing drugs in a steel drum with cement/plastic foam before landfill
Inertisation	Grinding and mixing with cement, lime and water, then landfill
Engineered/sanitary landfill	Small quantities of non-hazardous solids
Sewer / dilution and flushing	Only small quantities of harmless liquids (e.g. dilute syrups, IV fluids) with plenty of water; never for antibiotics or cytotoxics
Waste immobilisation / deep burial	Permitted in specified situations under the Bio-Medical Waste Rules
Destruction before the prescribed authority	Compulsory for narcotics and psychotropics — in the presence of the authorised officer(s), with a certificate of destruction

⚠ ABSOLUTE RULES

- **Expired drugs must never be dispensed, sold, donated or kept with live stock** — it is an offence under the Drugs and Cosmetics Act.
- **Cytotoxic waste must not be landfilled or discharged into a drain** — incineration at high temperature only.
- **Narcotics cannot be destroyed by the store** — only by/before the authority prescribed under the NDPS Rules, with a certificate.
- Antibiotics must **not** be flushed into the sewer (antimicrobial resistance).

10.9 Housekeeping, Pest Control, Fire and Security in the Store

- **Daily** sweeping/mopping, dusting of racks, removal of waste and empty cartons; no food or personal articles in the store.
- **Weekly/monthly** deep cleaning, defrosting of refrigerators, checking of expiry stickers and of the near-expiry list.
- **Pest control** — screened windows and doors, rodent guards and traps, periodic professional pest control with a **record of date, agent used and area treated**; insecticides stored away from drugs.
- **Fire safety** — appropriate extinguishers (CO₂/dry powder for electrical and inflammable fires), sand buckets, water hydrant, smoke detectors, marked and unobstructed exits, a displayed **evacuation plan**, staff trained in drills, "NO SMOKING" boards.
- **Electrical safety** — no overloading, earthed equipment, voltage stabilisers, periodic inspection; flame-proof fittings in the inflammable store.
- **Security** — single controlled entry, restricted access, key register, locks on narcotic and Schedule X cupboards, CCTV, night watchman, gate pass for material leaving the premises.
- **Documentation of housekeeping** — cleaning checklist, pest-control record, temperature and humidity log, equipment maintenance register: an accreditation team looks for exactly these papers.

10.10 Storage at the Ward / Sub-Store Level

- Ward stock kept in a **locked medicine cupboard/trolley**, with the ward register and the imprest scale displayed.
- A **refrigerator with a temperature log** for thermolabile drugs; **no food** in it.
- **Narcotics** in a double-locked box within the ward cupboard, with the ward narcotic register and shift-wise tally.
- **Emergency (crash) trolley** — a fixed list of drugs, sealed, checked and signed **every shift/daily** for completeness and expiry, re-sealed after use.
- **LASA and high-alert drugs** separated and labelled; concentrated electrolytes not kept loose in the ward.
- **FEFO** followed; near-expiry items returned to the main store in time.
- Periodic **ward stock inspection by the pharmacist** with a written report: expiry, storage condition, unauthorised accumulation, register accuracy.

★ CHAPTER 10 — RAPID RECALL

- Commonest arrangement in a hospital store: **dosage-form-wise, then alphabetical**.
- **IP: cold place $\leq 8^{\circ}\text{C}$ • refrigerator $2-8^{\circ}\text{C}$ • cool place $8-25^{\circ}\text{C}$ • warm $30-40^{\circ}\text{C}$ • excessive heat $> 40^{\circ}\text{C}$.**
- **Schedule P** = life period & storage conditions; **Schedule N** = minimum equipment for a pharmacy.
- Cold chain = **$+2$ to $+8^{\circ}\text{C}$; ILR for vaccines; deep freezer for ice packs & OPV; VVM shows heat exposure.**
- Freeze-sensitive: **DPT, TT/Td, Hep-B, IPV, Pentavalent, insulin**. Heat-sensitive: **OPV (most), measles, BCG**.
- Blood **$2-6^{\circ}\text{C}$; platelets $20-24^{\circ}\text{C}$ with agitation; FFP below -30°C .**
- Stock **30 cm from the wall, 10 cm off the floor**; store below **25°C , RH below 60 %**.
- Narcotics: **double-locked cupboard, daily tally, NDPS register**; Schedule X: **separate locked cupboard, duplicate prescription**.
- Expired medicines → **yellow BMW category**; cytotoxic waste → **incineration**.

PART VII • STATUTORY RECORDS & MODERN SYSTEMS

CHAPTER 11 | STATUTORY AND LEGAL RECORDS OF A DRUG STORE

Drugs & Cosmetics Rules • Schedules H, H1 & X • NDPS records • licence forms • retention periods • inspection by the Drug Inspector

Managerial records can be simplified; statutory records cannot. They must exist in the prescribed form, contain the prescribed particulars, be preserved for the prescribed period, and be produced to the inspector on demand. This chapter collects them in one place — it is the densest source of one-mark questions in the whole topic.

11.1 Records under the Drugs and Cosmetics Act, 1940 and Rules, 1945

Requirement	Substance of the provision
Licence to stock and sell	A drug store must hold a valid licence; it must be displayed on the premises, and renewed in time
Qualified person	Sale/dispensing of drugs must be under the personal supervision of a registered pharmacist (retail) or a competent person (wholesale); registration certificate kept on record
Premises & storage	Adequate space, proper storage including refrigeration where required ; storage conditions as per the label and Schedule P
Records of purchase	Purchase bills/invoices showing the name of the manufacturer, batch number and quantity, preserved and available for inspection
Records of sale	Cash/credit memo (bill) for every sale showing the name & address of the seller, licence number, date, name of the drug, quantity, batch number and price; counterfoils preserved
Prescription records	Sales of Schedule H, H1 and X drugs only against a prescription; the prescription/register entry preserved for the prescribed period
No sale of expired drugs	Stocking or selling a drug after its expiry date is an offence; expired stock must be segregated
Inspection book / records available	All registers, bills and records must be produced to a Drugs Inspector on demand
Price display	Price list of scheduled formulations displayed as required under the DPCO

11.1.1 Principal licence forms (frequently asked)

Form	Purpose
Form 19	Application for a licence to sell, stock, exhibit or distribute drugs (retail/wholesale)
Form 19A	Application for a restricted licence (sale by a dealer without a qualified person)
Form 19C	Application for a licence to sell drugs specified in Schedule X
Form 20	Licence for retail sale of drugs other than those in Schedules C, C(1) and X
Form 20A	Restricted licence for retail sale
Form 20B	Licence for wholesale of drugs other than those in Schedules C, C(1) and X
Form 21	Licence for retail sale of drugs specified in Schedule C and C(1) (other than Schedule X)
Form 21A	Restricted licence for retail sale of Schedule C/C(1) drugs
Form 21B	Licence for wholesale of Schedule C and C(1) drugs
Form 20F	Licence to sell Schedule X drugs by retail
Form 20G	Licence to sell Schedule X drugs wholesale
Form 25 / 28	Licence to manufacture drugs (28 = Schedule C & C(1) biological/special products)
Form 25A / 28A	Loan licence to manufacture
Form 29	Licence to manufacture drugs for examination, test or analysis
Form 13	Report of the Government Analyst on a sample
Form 16	Receipt given by the Inspector for a sample taken
Form 17	Intimation to the person from whom a sample has been taken that it is being sent for test/analysis
Form 18	Order of the Inspector prohibiting the sale/disposal of stock (stock "frozen")

★ **SCHEDULE X TRIAD** — Application in Form 19C → Retail licence in Form 20F → Wholesale licence in Form 20G.

Schedule X drugs are kept in a **separate locked cupboard**, the prescription is required **in duplicate** (one copy retained by the pharmacy), and the records are preserved for **2 years**.

11.2 Schedule H, H1 and X — Record Requirements Compared

Point	Schedule H	Schedule H1	Schedule X
What it covers	General prescription-only drugs (most antibiotics, cardiovascular, hormonal, etc.)	Specified higher-generation antibiotics, anti-TB drugs and certain sedatives/psychotropics	Habit-forming / narcotic-type drugs such as barbiturates and amphetamines
Label marking	"Rx" + the Schedule H warning	"Rx" + H1 warning inside a box with a red vertical line on the left border	"NRx" (in red) + the Schedule X warning
Sale only on prescription of	Registered medical practitioner	Registered medical practitioner	Registered medical practitioner
Separate register	Not separately prescribed (prescription record kept)	Yes — a separate H1 register showing the name & address of the prescriber, the name of the patient, the name of the drug and the quantity supplied, with the date	Yes — a separate Schedule X register
Prescription handling	Retained/recorded as prescribed	Recorded in the H1 register	Prescription in duplicate ; one copy retained by the seller
Storage	With general stock	With general stock (register controlled)	Separate cupboard under lock and key
Special licence	Normal Form 20/20B	Normal Form 20/20B	Form 20F (retail) / 20G (wholesale)
Preservation of records	2 years	3 years	2 years

11.3 Records under the NDPS Act, 1985

- Narcotic drugs and psychotropic substances may be possessed, stored and used only under the **licence/permit/authorisation** prescribed by the NDPS Rules (and the State Rules).
- A **separate, bound narcotic register (Dangerous Drug register)** is maintained with a **separate page for each drug and each strength**, showing date, quantity received (with source and voucher), quantity issued (with the name of the patient/ward and the prescriber), and the **running balance**.
- Entries are made **immediately, in ink, by the responsible person**; corrections are **attested**; no page may be removed.
- The **balance must be physically verified regularly** (daily/at each shift change in a hospital ward) and the register **countersigned** by the designated officer.
- Stock is kept in a **double-locked, fixed steel cupboard/safe**, the keys remaining in the personal custody of the authorised person.
- **Loss, theft or discrepancy must be reported at once** to the prescribed authority and to the police as required.
- **Destruction** of expired/unusable narcotics is done only **before/by the officer authorised** under the Rules, and a **certificate of destruction** is preserved.
- Records and registers must be preserved for the period prescribed by the Rules (commonly **2 years**) and produced for inspection on demand.
- Prescribing of narcotics in hospitals is regulated (e.g. **Recognised Medical Institution (RMI)** provisions for morphine for palliative care), each state prescribing its own formats.

⚠ **ZERO TOLERANCE** — An unexplained shortage of a narcotic drug is not merely an audit objection — it is a **criminal matter** under the NDPS Act, which carries stringent penalties and where the burden of explanation lies on the custodian. This is why the narcotic register is the most rigorously examined record in any drug store.

11.4 Other Statutory Records Relevant to a Drug Store

Law / rule	Record to be maintained
Pharmacy Act, 1948	Registration certificate of the pharmacist (and its renewal); record of the pharmacist on duty
Drugs (Prices Control) Order / NPPA	Price list of scheduled formulations, purchase and sale records proving compliance with the ceiling price
Bio-Medical Waste Management Rules, 2016	Daily record of waste generated category-wise (yellow/red/white/blue), handing-over record to the common facility, annual report , training and accident records
Schedule M (GMP) — manufacturers	Master formula records, batch manufacturing and packaging records , raw-material and finished-product analytical records, distribution records (for recall), validation, calibration and training records; records generally retained for one year beyond the expiry date of the batch
Blood bank (Schedule F Part XII-B)	Donor records, testing records, issue records, preserved for the prescribed period (commonly 5 years)
Legal Metrology / Packaged Commodities Rules	Verified/stamped weighing and measuring instruments with calibration certificates
GST law	Tax invoices, purchase and sale registers, e-way bills; books of account preserved for the statutory period (72 months / 6 years)
Income-tax & financial rules / GFR	Vouchers, cash book, stock registers, physical verification certificates, condemnation records
AERB (radiation)	Licence, radiation survey log, personnel dosimetry (TLD) records, source inventory
Factories Act / Explosives & Petroleum Rules	Licence and register for storage of inflammables and compressed gases
Excise rules (rectified spirit)	Bonded store register, permits, wastage statements
NABH / accreditation standards (voluntary)	SOPs, temperature logs, LASA & high-alert drug lists, emergency-trolley checklists, expiry-check records, staff training records

11.5 Retention (Preservation) Periods of Records

Record	Usual retention period	Governed by
Prescription / records of sale of Schedule H and C(1) drugs	2 years	Drugs & Cosmetics Rules
Schedule H1 register	3 years	Drugs & Cosmetics Rules
Schedule X records, registers and prescriptions	2 years	Drugs & Cosmetics Rules
Narcotic / psychotropic registers and vouchers	2 years (or as the State NDPS Rules prescribe)	NDPS Rules
Purchase bills / invoices and sale bills (cash & credit memo counterfoils)	Commonly 2–3 years for drug-law purposes, but 6 years (72 months) under GST law — so keep for the longer period	Drugs & Cosmetics Rules + GST law
Batch manufacturing & analytical records (manufacturers)	One year beyond the expiry date of the batch (Schedule M); some records longer	Schedule M / GMP
Blood bank records	Commonly 5 years	Drugs & Cosmetics Rules, Sch. F
Stock registers of consumables, GRNs, issue vouchers, indents	Generally 5 years (or until audit is completed)	GFR / State store & financial rules
Dead stock (non-consumable) register	Permanent — until the item is condemned and written off	GFR / State rules
Condemnation, write-off and auction records	Commonly 5–10 years	Financial rules
Bio-medical waste records & annual returns	Commonly 5 years	BMW Rules, 2016
Temperature / cold-chain logs	Commonly 2–3 years (accreditation bodies often ask for the current + previous year)	Institutional SOP / accreditation
Licences, agreements, court and audit files	Permanent / till finally settled	Institutional rules

● **HOW TO ANSWER A RETENTION QUESTION** — Quote the **statutory** periods with confidence — *Schedule H/X = 2 years, Schedule H1 = 3 years* — and remember that where two laws prescribe different periods (drug law vs tax law), the record must be kept for the **longer** period. Departmental/financial periods vary from State to State and are prescribed by the local financial code or record-retention schedule.

11.6 Inspection by the Drugs Inspector — What is Examined

1. **Licence** — valid, displayed, covers the categories actually stocked (including Schedule C/C(1) and X).
2. **Registered pharmacist** present and his registration current.
3. **Premises** — area, cleanliness, refrigeration, storage as per the label/Schedule P.
4. **Stock** — presence of expired, unlabelled, spurious, misbranded, physician-sample or "not for sale" drugs; recalled batches.
5. **Purchase records** — bills showing manufacturer, batch and quantity; ability to trace the source of any drug on the shelf.
6. **Sale records** — cash/credit memos, prescription records, **Schedule H1 register**, **Schedule X register**.
7. **Narcotic register** and the double-locked cupboard; balance tallied physically.
8. **Cold-chain records** — refrigerator temperature log, functioning thermometer.
9. **Price compliance** — DPCO ceiling prices, price list displayed.
10. **Samples** — the Inspector may take a sample, give a **receipt in Form 16**, send it for test with **intimation in Form 17**, and may **prohibit sale of the stock by an order in Form 18**; the Government Analyst reports in **Form 13**.
11. **Records of bio-medical waste** disposal of expired medicines.
12. Remarks are recorded in the **inspection book**, and compliance must be reported within the time allowed.

★ **PENALTY PERSPECTIVE** — Contravention of the conditions of a licence (including **failure to maintain the prescribed records**) can lead to **suspension or cancellation of the licence**, apart from prosecution. Sale of a **spurious or adulterated** drug attracts the most severe penalties under the Drugs and Cosmetics Act. Offences relating to narcotics are dealt with under the **NDPS Act**, which is far stricter.

CHAPTER 12 | COMPUTERISATION, BAR CODING AND MODERN STORE MANAGEMENT

Manual vs computerised records • software modules • e-procurement • data integrity • advantages and limitations

Every state health service is now moving to computerised drug distribution (DVDMS, e-Aushadhi and similar systems). Questions increasingly ask what a store software does, and what safeguards an electronic record needs.

12.1 Why Computerise Store Records?

Problem of the manual system	How computerisation solves it
Slow, repetitive writing; posting backlog	One entry updates the ledger, bin balance, batch record and consumption data simultaneously
Arithmetical and posting errors	Balances computed automatically; bar-code capture removes transcription error
Expiry loss	Automatic near-expiry alerts and FEFO-based batch prompting at the time of issue
Stock-outs	Automatic re-order level alerts and draft indent generation
Difficulty of analysis	ABC/VED analysis, consumption trends, expiry-loss and supplier-performance reports at a keystroke
Recall of a defective batch	Instant list of where every unit of a batch went
Duplicate ordering	Dues-in automatically deducted from the requirement
Weak accountability	Audit trail records who made or altered every entry, and when

12.2 Typical Modules of Drug Store / Pharmacy Software

- **Item master (stores vocabulary)** — code, description, unit, storage condition, min/max/re-order level, ABC-VED class.
- **Indent & purchase module** — indent generation, approval workflow, tender/rate-contract data, purchase order printing, dues-in tracking.
- **Goods receipt module** — GRN entry with batch, expiry, rate; inspection status; quarantine flag; discrepancy record.
- **Inventory module** — batch-wise stock, bin location, expiry alerts, stock transfer, physical verification sheets.
- **Issue / distribution module** — ward & patient-wise issue, FEFO prompt, imprest top-up.
- **Narcotic & Schedule X module** — restricted access, mandatory balance tally, printable statutory register.
- **Financial module** — priced ledger, bill passing, budget control, expenditure statements.
- **MIS & analytics** — consumption trend, turnover ratio, stock-out and expiry reports, vendor rating.
- **Bar-code / scanner interface** and, in advanced systems, **RFID and temperature-logger integration**.
- **Integration with the HMIS** — prescription → dispensing → billing → stock deduction in one flow.

12.2.1 Public-sector systems worth knowing by name

System	What it is
DVDMS (Drugs & Vaccines Distribution Management System)	National/state platform for demand, procurement, warehousing and distribution of drugs and vaccines
e-Aushadhi	State-level drug inventory and distribution software used by several state medical services corporations
GeM (Government e-Marketplace)	Online public procurement portal (e-bidding, reverse auction)
e-Tendering / CPP Portal	Electronic publication and submission of tenders
Jan Aushadhi / PMBJP	Generic medicine supply scheme with its own stock and sale software
eVIN (Electronic Vaccine Intelligence Network)	Real-time vaccine stock and cold-chain temperature monitoring on a mobile platform

12.3 Safeguards for Electronic Records

Safeguard	Requirement
Individual user accounts	Unique user ID and password for every user; no shared logins ; periodic password change
Role-based access rights	Only the authorised role can create a code, approve an indent, or post a write-off
Audit trail	Automatic, non-editable log of every entry, change and deletion with user and time stamp
No deletion of transactions	Corrections made only by a reversal entry, with a reason recorded
Back-up	Daily back-up, stored off-site , with periodic restoration testing
Business continuity	UPS/generator; a manual fallback register for use during downtime, posted into the system afterwards
Validation & calibration	The software is validated; bar-code scanners, printers and temperature loggers calibrated
Printing & preservation of statutory registers	Where the law requires a physical record (e.g. narcotics), print, bind, sign and preserve it
Data privacy	Patient-identifiable data (Schedule H1, prescriptions) protected and accessed on a need-to-know basis

● **DATA INTEGRITY — ALCOA+ —** Attributable • Legible • Contemporaneous • Original • Accurate — plus Complete, Consistent, Enduring, Available. Applies equally to paper and to electronic records; increasingly asked in pharmacy examinations.

12.4 Limitations and Precautions

- **Capital cost** of hardware, software, scanners and networking; recurring cost of maintenance and licences.
- **Training and resistance to change** among staff; a half-trained user creates worse data than a careful clerk.
- **Garbage-in, garbage-out** — if the opening balances, codes and batch data are entered wrongly, every report is wrong.
- **Dependence on power and network**; failure halts issue unless a manual fallback exists.
- **Data loss/corruption and cyber-security** risks; hence back-up, antivirus and access control.
- **Legal acceptability** — statutory registers may still be required in physical form; electronic records must satisfy audit-trail requirements.
- **Physical verification is still compulsory** — a computer shows the book balance, never the ground balance.

★ **REMEMBER** — Computerisation improves **speed, accuracy and information** — it does **not** replace the pharmacist's duties of **physical verification, storage-condition monitoring, expiry checking and statutory record keeping**.

PART VIII • REVISION

CHAPTER 13 | QUICK REVISION CAPSULE

Everything examinable from Chapters 1–12, compressed — read this the night before the examination

13.1 One-Line Definitions

Term	Definition in one line
Store	A specified secure place where materials are received, kept and issued
Records keeping	Systematic writing, posting, preserving and retrieving of all documents evidencing receipt, storage, issue and disposal of stores
Goods inward section	The receiving section where consignments are unloaded, checked against the PO, inspected and documented on a GRN
GRN	Internal note certifying the quantity and condition of material actually received — authority to take on charge and to pay
Delivery challan	Supplier's document of <i>movement</i> of goods, travelling with the consignment
Invoice	Supplier's <i>demand for payment</i> showing rate, tax and total value
Quarantine	Segregation of material awaiting a decision on its quality (yellow label)
Classification	Grouping of similar items on the basis of a common characteristic
Codification	Allotting a unique symbol (number/letter) to each item of stores
Standardisation	Fixing a definite specification for each item stocked
Variety reduction	Reducing the number of varieties/brands/pack sizes of the same item
Stores vocabulary / item master	Master list of all code numbers with descriptions
Stock book / stock register	Item-wise permanent account of receipts, issues and balance
Bin card	Card at the bin showing quantity received, issued and balance, posted by the store keeper
Stores ledger	Office record of quantity (and value) posted from vouchers by the ledger clerk
Indent	Formal written demand for supply of specified materials
Purchase order	The external contract placed on a supplier
Lead time	Time from raising the indent to the material being available for issue
Re-order level	Balance at which a fresh order must be placed
Safety (buffer) stock	Reserve held to absorb unexpected demand or delay
EOQ	Order size at which total ordering cost = total carrying cost, so total cost is minimum
Inventory turnover ratio	Annual consumption value ÷ average inventory value
FIFO / FEFO	First-in-first-out / first-expiry-first-out method of issue
Imprest system	Fixed scale of stock at a ward, topped up to full each cycle
Cold chain	System keeping thermolabile products continuously at +2 to +8 °C from manufacturer to patient
Shelf life	Period for which a drug retains its specified potency ($\geq 90\%$) under stated storage conditions
Condemnation	Formal declaration by a board that stores are unserviceable and may be disposed of

13.2 All Formulae on One Page

Quantity	Formula
Closing balance	Opening balance + receipts – issues – losses + returns
EOQ	$\sqrt{(2AO / C)}$ where A = annual demand, O = ordering cost/order, C = carrying cost/unit/year ($= i \times p$)
Number of orders per year	Annual consumption ÷ EOQ
Re-order level	(Average consumption × lead time) + safety stock
Re-order level (simplified)	Maximum consumption × maximum lead time

Quantity	Formula
Safety stock	$(\text{Maximum consumption} - \text{average consumption}) \times \text{lead time}$
Minimum level	$\text{ROL} - (\text{average consumption} \times \text{average lead time})$
Maximum level	$\text{ROL} + \text{order quantity} - (\text{minimum consumption} \times \text{minimum lead time})$
Danger level	$\text{Average consumption} \times \text{emergency lead time}$
Average stock	$\text{Minimum level} + \frac{1}{2} \text{ order quantity}$ or $(\text{maximum} + \text{minimum}) \div 2$
Indent quantity	$\text{AMC} \times (\text{period of cover} + \text{lead time}) + \text{buffer} - \text{stock in hand} - \text{dues-in}$
Average monthly consumption (AMC)	$\text{Total issues of n months} \div n$
Inventory turnover ratio	$\text{Annual consumption value} \div \text{average inventory value}$
Average inventory value	$(\text{Opening stock} + \text{closing stock}) \div 2$
Months of stock in hand	$\text{Closing stock} \div \text{average monthly consumption}$
Expiry (wastage) rate %	$(\text{Value expired} \div \text{value consumed}) \times 100$
Carrying cost	Commonly taken as 20–25 % of average inventory value per year

13.3 Selective Inventory Control — Criterion Table (memorise)

Technique	Criterion	Classes
ABC	Annual consumption value	A $\approx$ 10 % items/70 % value • B $\approx$ 20 %/20 % • C $\approx$ 70 %/10 %
VED	Criticality	Vital • Essential • Desirable
VEN	Criticality (WHO)	Vital • Essential • Non-essential
HML	Unit price	High • Medium • Low
XYZ	Value of stock lying in store	X high • Y moderate • Z low
FSN / FSND	Rate of movement	Fast • Slow • Non-moving (• Dead)
SDE	Availability	Scarce • Difficult • Easy
GOLF	Source of supply	Government • Ordinary • Local • Foreign
SOS	Seasonality	Seasonal • Off-seasonal
MUSIC-3D	Cost + criticality + availability together	Multi-dimensional

13.4 Storage Temperatures and Conditions (IP)

Term	Temperature
Deep freezer / store frozen	Below -18 to -20°C
Cold place	Not exceeding 8°C
Refrigerator	$2 - 8^{\circ}\text{C}$
Cool place	$8 - 25^{\circ}\text{C}$
Room temperature	$15 - 30^{\circ}\text{C}$ (WHO $15 - 25^{\circ}\text{C}$)
Controlled room temperature	$20 - 25^{\circ}\text{C}$
Warm	$30 - 40^{\circ}\text{C}$
Excessive heat	Above 40°C
Whole blood / packed cells	$2 - 6^{\circ}\text{C}$
Platelets	$20 - 24^{\circ}\text{C}$ with continuous agitation
Fresh frozen plasma	Below -30°C
Vaccines (cold chain)	$+2$ to $+8^{\circ}\text{C}$; ice packs/OPV in deep freezer at -15 to -25°C
General drug store	Below 25°C , RH below 60 %
Distance of stock from wall / floor	30 cm from the wall • 10 cm above the floor (pallets)

13.5 Schedules, Forms and Retention Periods

Item	Answer
Schedule prescribing life period & storage conditions of drugs	Schedule P
Schedule prescribing minimum equipment for a pharmacy	Schedule N
Schedule prescribing GMP and records for manufacturers	Schedule M
Schedule of prescription-only drugs	Schedule H
Schedule needing a separate register kept for 3 years and a red vertical line on the label	Schedule H1
Schedule of habit-forming drugs needing a separate locked cupboard and duplicate prescription	Schedule X
Schedule of biological & special products (sera, vaccines, insulin)	Schedule C and C(1)
Schedule needing the caution "to be taken under medical supervision"	Schedule G
Schedule listing exemptions	Schedule K
Schedule for clinical trials / new drugs	Schedule Y
Retail licence (ordinary drugs)	Form 20
Wholesale licence (ordinary drugs)	Form 20B
Retail / wholesale licence for Schedule C & C(1)	Form 21 / 21B
Application for Schedule X licence	Form 19C
Retail / wholesale licence for Schedule X	Form 20F / 20G
Government Analyst's report	Form 13
Inspector's receipt for a sample	Form 16
Intimation that a sample is sent for test	Form 17
Order prohibiting sale/disposal of stock	Form 18
Retention: Schedule H / Schedule X records	2 years
Retention: Schedule H1 register	3 years
Retention: narcotic registers	2 years (as per NDPS Rules)
Retention: GST books of account	6 years (72 months)
Retention: batch records of a manufacturer	1 year beyond expiry date (Schedule M)
Retention: dead-stock register	Permanent (till condemnation)

13.6 Colour Codes

Context	Colour → meaning
Consignment status	Yellow = quarantine/under test • Green = approved • Red = rejected
Expiry traffic light	Red ≤ 3 months • Yellow 3–6 months • Green > 6 months
Oxygen cylinder (India)	Black body, white shoulder (USA: green)
Nitrous oxide	Blue
Carbon dioxide	Grey
Medical air (India)	Grey body, black & white quartered shoulders
Nitrogen (India)	Grey body, black shoulder
Helium	Brown
Entonox	Blue body, blue & white quartered shoulders
BMW — expired/discarded medicines, cytotoxic, anatomical waste	Yellow → incineration / deep burial
BMW — contaminated recyclable plastic (tubing, IV sets, gloves)	Red → autoclave + shred + recycle
BMW — sharps (needles, blades)	White translucent puncture-proof → autoclave + shred
BMW — glassware, ampoules, metallic implants	Blue → disinfect + recycle

13.7 Differences You Must Be Able to Write

Pair	Key distinction
Bin card vs stores ledger	At the bin, quantity only, store keeper vs in the office, quantity + value, ledger clerk
Challan vs invoice	Movement of goods vs demand for payment
GRN vs challan	Prepared by the buyer vs prepared by the supplier
Indent vs purchase order	Internal demand vs external contract
FIFO vs FEFO	Oldest received first vs earliest expiring first
FIFO vs LIFO (valuation)	Issues priced at oldest rate vs latest rate
Rate contract vs running contract	Rate fixed, quantity not fixed vs both fixed
EMD vs security deposit	Deposited with the bid vs kept after award of contract
ABC vs VED	Annual consumption value vs criticality
HML vs ABC	Unit price vs annual consumption value
Perpetual vs periodic verification	Few items daily, all year round vs whole stock on one date
Consumable vs dead stock	Used up (drugs) vs permanent (equipment)
Debit note vs credit note	Raised by the buyer vs issued by the supplier
Quarantine vs rejected area	Awaiting decision vs decision taken — rejected
Cold place vs cool place (IP)	$\leq 8^{\circ}\text{C}$ vs $8-25^{\circ}\text{C}$
Freeze-sensitive vs heat-sensitive vaccine	DPT/Hep-B/TT/insulin (never freeze) vs OPV/measles/BCG (tolerate freezing)
Centralised vs decentralised store	Lower stock & better control vs faster supply at the point of use
Brisch vs Kodak code	7 digits, 3 stages, UK vs 10 digits (3+4+3), source-of-supply oriented, USA

13.8 Sequences and Procedures in Order

Process	Correct order
Purchase cycle	Indent → sanction → enquiry/tender → comparative statement → PO → receipt & GRN → stock records → bill passing → payment
Goods inward clerical procedure	Gate entry → documents vs PO → unload & count → unpack → quantity check → quality check → sample to QC → quarantine → goods inward register → GRN → inspection result → take on charge (bin card + ledger) → discrepancy report/claims → bill passing → filing
Issue cycle	Indent → scrutiny → sanction → issue voucher → FEFO picking → receiver's signature → posting of bin card, ledger and ward register → consumption statement
Stores system design	Classification → standardisation/variety reduction → codification → bin location → stock records
Condemnation	Listing → condemnation board → sanction → disposal (auction/destruction) → register entry + certificate
Expiry handling	Monthly near-expiry list → colour tagging → redistribution/return → on expiry segregate & enter register → write-off sanction → BMW disposal → destruction certificate
Tender process	NIT → tender document → two-bid submission with EMD → technical bid opening → price bid opening → comparative statement → L-1 → approval → PO → security deposit/agreement

13.9 Numbers, Percentages and Standards Worth Memorising

Fact	Value
ABC classes	A: $\approx 10\%$ of items, 70% of value • B: $20\%/20\%$ • C: $70\%/10\%$
Inventory carrying cost	$\approx 20-25\%$ of average inventory value per year
Drugs as a share of the hospital recurring budget	$\approx 30-40\%$
Minimum remaining shelf life demanded in drug tenders	Usually 5/6 ($\approx 83\%$) of total shelf life at delivery
Potency limit defining shelf life	Not less than 90% of the labelled amount
Store temperature / humidity	Below 25°C / RH below 60%

Fact	Value
Temperature recording frequency	Twice daily (morning and evening)
Distance of stock from wall / height above floor	30 cm / 10 cm
Multi-dose eye-drop in-use period after opening	Commonly 28 days
VVM usable stages	Stages 1 and 2 only (discard at 3 and 4)
Cytotoxic waste incineration temperature	≥ 1,200 °C
Frost thickness at which an ILR must be defrosted	More than about 5 mm
Number of GRN copies usually prepared	4-5
Members of a condemnation board	Usually 3 (including one officer unconnected with the store)

13.10 Mnemonic Collection

Topic	Mnemonic
5 R's of purchasing	Q-Q-P-T-S — Quality, Quantity, Price, Time, Source
Functions of stores	R-I-S-R-I-S-V-H-D-I — Receive, Inspect, Store, Record, Issue, Stock-control, Verify, Housekeep, Dispose, Inform
Layout zones of a store	R-Q-S-C-N-H-R-I-O-B — Receiving, Quarantine, Storage, Cold, Narcotic, Hazardous, Rejected, Issue, Office, Bulk
Goods inward steps	G-D-P-U-U-Q-Q-S-Q-G-I-T-D-B-F (gate entry ... filing)
Good code	U-B-F-S-M-E — Unique, Brief, Flexible, Simple, Mnemonic, Exhaustive
Inventory criteria	A=Amount(value), V=Vitality, H=High price, X=eXisting stock value, F=Frequency of movement, S=Scarcity, G=Geographic source, S=Season
Data integrity	ALCOA+ — Attributable, Legible, Contemporaneous, Original, Accurate (+ Complete, Consistent, Enduring, Available)
Special storage categories	N-X-T-B-L-M-V-C-C-R-G-L-H — Narcotics, Schedule X, Thermolabile, Blood, Light-sensitive, Moisture-sensitive, Volatile, Corrosives, Cytotoxics, Radiopharmaceuticals, Gases, LASA, High-alert

13.11 Fifty Facts — Final Rapid Fire

1. Records keeping ensures **accountability, legal compliance, financial control and traceability**.
2. A wrong entry is corrected by **scoring out with one line + attestation**; never erased.
3. Registers must be **bound with machine-numbered pages** and carry a page certificate.
4. The **stock verifier must be independent of the custodian**.
5. **Purchase, receipt and payment functions must be separated** (internal check).
6. The **PTC** decides which drugs are stocked; the **purchase committee** decides from whom.
7. The **hospital formulary** is the list of drugs approved for use in that hospital.
8. **Rate contract** fixes the rate, not the quantity.
9. The lowest evaluated bidder is **L-1**; bid security is **EMD**.
10. **Three-way matching** = Purchase Order + GRN + Invoice.
11. The **GRN** is prepared by the receiving section, usually in 4-5 copies.
12. The **delivery challan** accompanies the goods; the **invoice** demands payment.
13. Transit claims need the **lorry receipt/consignment note** and damage noted at delivery.
14. **Quarantine = yellow, approved = green, rejected = red**.
15. Late supply attracts **liquidated damages**.
16. Drug tenders normally demand at least **5/6 of the shelf life** remaining at delivery.
17. Excess or unordered goods are **not taken on charge** without sanction.
18. **Dues-in must be deducted** while preparing the next indent.
19. **Bin card** = store keeper, quantity, at the bin; **ledger** = ledger clerk, quantity + value, in the office.
20. **Dead-stock register** = non-consumable items; removed only on condemnation.

21. Sequence: **classification → standardisation → codification.**
22. **Brisch = 7 digits, 3 stages (UK); Kodak = 10 digits (USA); decimal system = Dewey.**
23. The commonest practical coding system is **alpha-numerical.**
24. The master list of codes is the **stores vocabulary / item master.**
25. **ABC** is based on annual consumption **value** (Pareto 80:20 idea).
26. **VED** is based on **criticality**; all **V** items go into control **Category I.**
27. **HML** = unit price; **XYZ** = value of stock in store; **FSN** = movement; **SDE** = availability; **GOLF** = source; **SOS** = season.
28. **EOQ** = $\sqrt{(2AO/C)}$; at EOQ, ordering cost = carrying cost.
29. **ROL** = **(average consumption × lead time) + safety stock.**
30. **Danger level** → stop normal issues and make an emergency purchase.
31. **Two-bin system** suits cheap **C** items.
32. **Perpetual verification** covers a few items daily throughout the year.
33. The stock equation: **closing = opening + receipts – issues – losses + returns.**
34. **Inventory turnover** = **annual consumption value ÷ average inventory value.**
35. Indent quantity = **AMC × (cover + lead time) + buffer – stock in hand – dues-in.**
36. For a new hospital with no consumption data, use the **morbidity method.**
37. Indent by **generic name + strength + dosage form + pack + IP standard.**
38. Commonest store arrangement: **dosage form-wise, alphabetical within the form.**
39. **FEFO** is superior to **FIFO** for drugs.
40. **IP: cold ≤ 8 °C, refrigerator 2–8 °C, cool 8–25 °C, warm 30–40 °C, excessive heat > 40 °C.**
41. **Schedule P** = life period and storage conditions; **Schedule N** = pharmacy equipment.
42. Cold chain = **+2 to +8 °C; ILR** for vaccines, **deep freezer** for ice packs.
43. **Freeze-sensitive:** DPT, TT/Td, Hep-B, IPV, pentavalent, insulin. **Heat-sensitive:** OPV (most), measles, BCG.
44. A **VVM** at stage 3 or 4 means **discard the vial.**
45. Blood **2–6 °C**, platelets **20–24 °C with agitation**, FFP **below –30 °C.**
46. Narcotics: **double-locked cupboard, daily tally, NDPS register, destruction only before the authorised officer.**
47. Schedule X: **Form 19C → 20F/20G**, separate locked cupboard, duplicate prescription, records **2 years.**
48. **Schedule H1 register is preserved for 3 years; the label carries a red vertical line.**
49. Expired medicines go into the **yellow** bio-medical waste category; cytotoxic waste is **incinerated.**
50. Stock kept **30 cm from the wall, 10 cm above the floor**, store **below 25 °C, RH below 60 %**, temperature logged **twice daily.**

CHAPTER 14 | GLOSSARY AND ABBREVIATIONS

Every technical term used in this book, in one alphabetical list

14.1 Glossary of Terms

Term	Meaning
ABC analysis	Classification of items by annual consumption value
AMC	Average monthly consumption
Beyond-use date (BUD)	Date after which an opened/reconstituted product must not be used
Bin	Compartment, rack or shelf position in which an item is kept
Bin card	Card at the bin recording quantity received, issued and balance
Bonded store	Store for goods on which duty/excise has not yet been paid (e.g. spirit)
Buffer stock	Reserve/safety stock
Carrying cost	Cost of holding inventory (interest, storage, insurance, expiry)
CIF	Cost, Insurance and Freight — seller bears these up to destination
Cold chain	Unbroken +2 to +8 °C chain for thermolabile products
Condemnation	Declaring stores unserviceable and fit for disposal
Credit note	Document by which a supplier reduces the amount payable
Danger level	Stock level at which emergency purchase becomes necessary
Dead stock	Non-consumable, permanent items (equipment, furniture)
Debit note	Document by which the buyer reduces the supplier's claim
Dues-in	Quantity ordered but not yet received
EMD	Earnest Money Deposit (bid security)
EOQ	Economic Order Quantity
FEFO	First Expiry First Out
FIFO	First In First Out
FOB	Free On Board
Gate pass	Authority for material to leave the premises (returnable or not)
GDP / GSP	Good Distribution Practice / Good Storage Practice
GRN	Goods Received Note
GS1 / GTIN	Global standards body / Global Trade Item Number used in bar codes
HML analysis	Classification by unit price (High, Medium, Low)
ILR	Ice-Lined Refrigerator (+2 to +8 °C)
Imprest	Fixed scale of stock held by a ward, replenished to full each cycle
Indent	Written demand for supply of material
Inventory turnover ratio	Annual consumption value ÷ average inventory value
Kardex	Visible-index card system for stock records
LASA	Look-Alike Sound-Alike drugs
L-1	Lowest evaluated (successful) tenderer
LD	Liquidated damages (penalty for late supply)
Lead time	Time from raising the indent to material being available for issue
LR / RR	Lorry Receipt / Railway Receipt (consignment note)
MRN	Material Return Note
NLEM	National List of Essential Medicines
NSQ	Not of Standard Quality
NDPS	Narcotic Drugs and Psychotropic Substances (Act, 1985)
PO	Purchase Order
PTC	Pharmacy and Therapeutics Committee
Quarantine	Segregation of material awaiting a quality decision

Term	Meaning
Rate contract	Contract fixing the rate for a period, quantity not fixed
Re-order level (ROL)	Balance at which a fresh order must be placed
RFID	Radio-Frequency Identification
Salvage store	Area holding expired, damaged or condemned stock
SDE analysis	Classification by availability (Scarce, Difficult, Easy)
Shelf life	Period for which the product retains its specified quality
Stores vocabulary	Master list of code numbers and descriptions
Three-way matching	Agreement of PO, GRN and invoice before payment
Two-bin system	Re-ordering when the first of two bins empties
VED analysis	Classification by criticality (Vital, Essential, Desirable)
VMI	Vendor Managed Inventory
VVM	Vaccine Vial Monitor
XYZ analysis	Classification by the value of stock lying in store

14.2 A Final Word on Method

✓ HOW TO REVISE THIS BOOK IN THREE PASSES

- **Pass 1 (understanding):** read Chapters 1–12 once, slowly, following the flow diagrams. Do not memorise yet.
- **Pass 2 (structure):** re-read only the **tables** and the **coloured boxes**. Write the step-lists of Chapter 3 (goods inward), Chapter 4 (issue procedure) and Chapter 9 (indent preparation) from memory.
- **Pass 3 (recall):** use Chapter 13 alone. Cover the right-hand column of every table and recite the answer.
- Pay special attention to: **definitions, the temperature table, schedules and forms, retention periods, the inventory-criteria table, all formulae, and the colour codes** — these carry the highest density of objective questions.

Prepared as complete study notes for the Records Keeping — Stores Records & Procedures section of the JKSSB Junior Pharmacist syllabus: clerical procedure in the goods inward section, records and procedures in main stores, classification and codification, keeping of stock books, preparation of indents and methods of storing drugs. Content is compiled from standard hospital pharmacy, drug store management and materials management texts and from the Drugs and Cosmetics Act & Rules, the NDPS Act, the Bio-Medical Waste Management Rules and WHO storage/cold-chain guidance. Where State rules or institutional practice vary (financial powers, departmental retention periods, local formats), the position given is the commonly prescribed one.